Develop or Conserve?
How the U.S. Government Allocates Public Lands

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Abstract

In the United States, the federal government owns 30% of the country and half of the Western US. The Bureau of Land Management (BLM), the largest federal land manager, delegates land-use decisions to 175 geographically dispersed field offices. These offices determine where mineral development, which generates billions in public revenue annually, can proceed and where development is restricted for conservation. Federal law dictates that field offices (i) make decisions to maximize federal land's market and non-market resource values, and (ii) publicly justify their choices to stakeholders. However, field offices maintain significant discretion in how to value nonmarket resources and how to reconcile competing demands from conservation and industry stakeholders. I show that field offices' land-use decisions can be modeled as many discrete choices and use this framework to transform 90,000 pages of planning records across 11 field offices into data suitable for discrete-choice econometric methods. I then estimate how much revenue each office would forgo to (i) conserve non-market resources and (ii) follow established conservation standards rather than justify departing from them to stakeholders. I find that the median office is willing to forgo $60,000 in annual revenue to conserve one acre. These valuations vary substantially across offices and non-market resources. Suggestive evidence indicates that differences in local preferences better explain this heterogeneity than differences in environmental damages from development. In seven out of ten decisions, the median office passes up higher net resource benefits to remain closer to an established standard. Counterfactual analyses show that heterogeneous conservation valuations and preferences for established standards jointly shape the relative cost-effectiveness of federal conservation policy instruments and the spatial distribution of the induced conservation.

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1 Introduction

The U.S. federal government owns 30% of the United States and half of the Western U.S. (Hanson & Vincent 2020). The largest federal land manager is the Bureau of Land Management (BLM), which delegates its management authority to 175 field offices. These offices establish where mineral development, which generates billions in public revenue annually, can proceed and where development is restricted for conservation (Office of Natural Resources Revenue 2025). Despite the substantial value at stake, little is known about how field offices make these decisions.[1] This knowledge gap is particularly salient amid ongoing public debate over the objectivity and effectiveness of federal land management (Tobias 2024, O'Donoghue 2023). Observed outcomes also offer grounds for criticism. Fossil fuel production on federal lands falls far below production on adjacent non-federal lands, and BLM lands consistently have poor ecosystem health.[2] This paper takes a first step toward evaluating federal land management by recovering how BLM field offices trade off conservation and development.

A fundamental challenge in studying land-use decisions is their high-dimensional nature.[3] To protect non-market resources (bald eagles, prairie dogs, caves, scenic landscapes, soil health, and others), offices choose which market activities (oil, gas, coal, wind, solar, grazing, and others) to restrict and how far those restrictions extend around non-market resources. I address this challenge by modeling jurisdiction-wide land-use allocations as combinations of separate decisions for individual non-market resources. I model each of these decisions as a discrete choice. Specifically, each decision amounts to choosing (i) a *restriction* (open to development, available subject to restrictions on timing or location, or closed to development) and (ii) a *buffer* specifying how far from the resource the *restriction* applies. Field office planning records support this representation; offices document separate decisions for different non-market resources and consider a discrete set of restriction-buffer alternatives.

In this paper, I focus on offices' decisions to restrict oil, gas, and geothermal drilling to protect water, soil, and wildlife. Federal regulations require offices to maximize the value of public resources, weighing the costs of conservation (revenue forgone) against the benefits (non-market resources preserved). But regulations do not specify what value to place on non-market resources. As a result, each office has considerable discretion over whether and how to protect these resources. An office might follow a state wildlife agency's recommendation to restrict drilling within a quarter-mile of a nest or instead choose a half-mile restriction based on local conditions. Whatever the office chooses, federal regulations also require the office to publicly justify its decision to stakeholders, including both fossil fuel companies and conservation groups.[4] Research on public bureaucracies suggests that this accountability requirement can make offices care not only about maximizing resource values, but also about how easily they can defend their choices (Prendergast 2003, Dixit 2002). An office may therefore follow an *established standard*, e.g., the state agency's quarter-mile recommendation, even when local benefits and costs favor a different restriction, because citing the standard is easier than defending its own assessment. This paper examines how offices navigate these institutional demands by estimating how much they value (i) conservation and (ii) following established standards.

[1] See Libecap (2026) for a literature review.

[2] See Lewis (2019) and Edwards & O'Grady (2026) for evidence on fossil fuel production and Public Employees for Environmental Responsibility (2024) for evidence on ecosystem health.

[3] See Irwin & Geoghegan (2001) and Redding & Rossi-Hansberg (2017) for discussions of the challenges of spatial economic modeling.

[4] See appendix A and D.1.

Recovering field office valuations through discrete-choice estimation requires data on each office's choices, the feasible choice set for each decision, and the expected consequences of each alternative (McFadden 1974). Federal regulations mandate disclosure of this information, but it is spread across thousands of pages of text and maps in each office's planning record.[5] I therefore transform these unstructured records into a dataset suitable for discrete-choice estimation in *two* steps.

First, I use large language models (LLMs) to construct a structured database from 90,000 pages of planning documents. Database construction proceeds in three stages, each performed by a different set of LLM agents. First-stage agents read the full record to locate relevant passages. Second-stage agents reread and label these passages according to what each establishes. Finally, third-stage agents use these labels to select and reread relevant context, then code the chosen and considered restriction-buffer pairs, along with the BLM's projections of development, revenue, and disturbance under different land-use scenarios. I verify every coded value ultimately used to construct the estimation sample against the original planning records.

Second, I combine this text database with national and state-level geospatial databases to reconstruct chosen allocations, feasible alternatives, and their expected consequences. The text database specifies the restriction-buffer pairs chosen and considered for different non-market resources, such as streams within a watershed or nests in a particular region. I use geospatial data identifying individual resources, such as stream centerlines and nest locations, to geolocate these descriptions. I then reconstruct the chosen and feasible alternatives, such as buffers extending 25-, 50-,... 500-feet around all streams in a given watershed. Finally, I intersect these reconstructed allocations with each office's development projections to measure the expected revenue forgone (costs) and disturbance avoided (benefits) under each alternative. The resulting dataset contains 12,509 choice situations across 11 field offices in Colorado, Idaho, New Mexico, and Wyoming.

With these data, I estimate a mixed logit model to recover each office's value for avoiding disturbance to non-market resources and following established conservation standards. I identify each office's conservation valuation from variation in the marginal cost (revenue forgone) and marginal benefit (disturbance avoided) of extending protection farther from a non-market resource.[6] I do not systematically observe each office's established standards, so, for each non-market resource type, I construct a *proxy* for established standards using the median protection level chosen by other offices, leaving out the office whose decisions I am modeling. This proxy captures common practice but need not equal the standard relevant to any particular decision. To allow the office's standard to differ from this proxy, I model it as the sum of the proxy and an unobserved *correction*. The correction is drawn from a discrete distribution with office-specific mixing shares, reflecting the recurring nature of standards. The model's mixing distribution thus captures unobserved variation in the standards relevant to offices' decisions, rather than variation in offices' valuations of conservation or adherence to standards. I identify the mixing distribution (discrete support and office-specific probabilities) from clustering at particular protection levels in locations with little or no expected development, where both expected disturbance avoided and revenue forgone are negligible across alternatives. I identify each office's valuation of following standards from how far it is willing to depart from a standard to increase net benefits. An office that remains close to a standard despite substantial gains from departing reveals a stronger preference for following standards than an office that departs for comparable gains. The recovered valuations are reduced-form objects that rationalize decisions emerging from a planning process shaped by

[5] See appendix A.

[6] This identification argument parallels Hendel (1999).

agency personnel, stakeholders, and the regulatory environment.[7]

I find substantial variation in how offices value conservation. Estimated valuations range from \$12k to \$1.6m in annual development revenue that offices are willing to forgo to avoid disturbing an acre of land. This variation occurs both across non-market resource types within an office and across offices for the same resource type. For soil, estimated valuations range from \$20k to \$400k per acre, for water, from \$12k to \$1.6m, and for wildlife, from \$30k to \$900k. This heterogeneity may reflect differences in the environmental damages avoided or differences in local preferences for conservation. Suggestive evidence indicates that the latter better explains the variation; offices' valuations of soil conservation are unrelated to expected soil loss from development but fall with the local Republican vote share.

The fitted model recovers six unobserved *corrections*, each with substantial estimated mixing weights in multiple offices. The estimated corrections are unevenly spaced, and their locations remain largely stable as additional corrections are introduced. These patterns support the interpretation that the corrections locate offices' established standards rather than simply approximate smooth variation in choices.

I find that established standards play a substantial role in shaping offices' choices. Offices are willing to sacrifice net benefits (conservation value minus revenue forgone) to remain close to established standards. In seven out of ten decisions, the model's most likely protective alternative (among those that restrict development) yields lower net benefits than applying the same restriction at an adjacent buffer distance. In a third of these cases, the net-benefit difference exceeds \$1,000 annually. In other words, offices' predicted protective choices are frequently inefficient relative to the conventional benchmark of maximizing conservation net benefits.

In one application of the model, I use counterfactual analysis to illustrate how heterogeneous valuations of conservation and preferences for following established standards jointly shape the effects of policy. I compare two ways BLM headquarters can influence field offices' conservation decisions. The first is a quantity-like instrument that recommends a specific conservation practice, such as a one-mile protective buffer around birds' nests, without requiring offices to adopt it. The second is a price-like instrument that sets a common value for bird conservation across offices. I calibrate the common value to match the aggregate conservation gains of the quantity-like instrument. I find that the price-like instrument is no more cost-effective than the quantity-like instrument in increasing the number of fledglings (baby birds). But the instruments differ in where they achieve gains. Half of the quantity-like instrument's gains occur in Buffalo, WY, whereas half of the price-like instrument's gains occur in Casper, WY and none in Buffalo, WY. Buffalo largely drives the price-like instrument's failure to outperform the quantity-like instrument. In Buffalo, adherence to established standards keeps the office from protecting many birds at relatively low cost, despite its high valuation of conservation. That same high valuation exceeds the common value set by the price-like instrument, so the instrument cannot induce additional protection there. The quantity-like instrument, however, does change Buffalo's choices by giving it a new standard to follow, without requiring headquarters to raise the valuation applied to every office.

Relevant Literature Little is known about how the Bureau of Land Management makes land-use decisions, even though these choices determine the allocation of vast public resources across competing users and

[7] For theories of how political pressure and interest-group competition shape regulatory outcomes, see Stigler (1971), Peltzman (1976), and Becker (1983). For discussions more specific to environmental regulation and public-land management, see Oates & Portney (2003) and Martin et al. (1996).

generations (Libecap 2026). This absence mirrors a broader gap in empirical studies of the regulatory process despite longstanding recognition that such work provides "important, useful insights into what is actually happening" (Joskow & Noll 1981, Magat et al. 1986, Oates & Portney 2003). In the empirical literature on environmental regulatory behavior only two papers—Magat et al. (1986) and Cropper et al. (1992)—study how environmental agencies make decisions in the US. These papers apply revealed preference techniques pioneered by McFadden (1976, 1975) to examine how agency discretion is applied in actual decision making. Magat et al. (1986) develop a revealed-preference approach to recover how EPA regulators make tradeoffs among various factors, including information and input supplied by regulated industries. Cropper et al. (1992) assess the extent to which the EPA meets its congressional mandate when regulating pesticides. In a similar vein, this paper uses land-use decisions to assess the extent to which the economic benefits of development are weighed against the costs of development in terms of the non-market resources lost.

An existing literature studies tradeoffs between conservation and alternative economic uses of land. A central concern in this literature is additionality: whether conservation policies protect land that would otherwise have been converted for market uses. Conservation creates real opportunity costs when it displaces development (Barbier & Burgess 2001), motivating policies that compensate landowners or governments for forgone development opportunities (Angelsen et al. 2012). Economists have studied whether conservation investments actually deliver environmental benefits (Ferraro & Pattanayak 2006), how selection into conservation contracts can limit additionality (Aspelund & Russo 2026), and why protected areas may fail to prevent land conversion when protections are weak or targeted toward low-threat locations (Reynaert et al. 2024). Evidence from federal biodiversity regulation similarly suggests that conservation boundaries may be drawn in part to avoid large economic costs (Frank et al. 2025). This paper extends the additionality concern to discretionary public-land planning. Rather than asking whether conservation programs change private land-use behavior, I ask whether public-land managers choose conservation where doing so requires forgoing meaningful development value.

This question connects to the literature on decentralized public decision-making. A central insight of fiscal federalism is that decentralized decisions can better accommodate heterogeneous local preferences, but may fail to internalize spillovers across jurisdictions (Oates 1972, Oates 1999, Besley & Coate 2003). Costello & Kaffine (2018) formalize this tension in the context of natural resources, where differences in preferences for resource extraction versus conservation favor different allocation decisions across locations, while ecological connectivity can make uncoordinated decisions costly for conservation outcomes. My findings document substantial heterogeneity in conservation valuations across field offices, highlighting the relevance of this less-studied[8] form of spatial heterogeneity for optimal natural resource management.

Recent empirical work increasingly uses structural and revealed-preference approaches to recover the objective functions and constraints of local decision makers from observed public decisions in environmental, infrastructure, tax, and land-use settings (Bodéré et al. 2026, Fajgelbaum et al. 2023, Kashner 2026, Parkhomenko 2023, Slattery 2025, Wehrle et al. 2024). My paper shares this focus on public decisions as endogenous determinants of outcomes, but differs by focusing specifically on the administrative planning process. I treat the planning record itself as data, using it to reconstruct the feasible choices, information environment, regulatory constraints, and perceived payoffs that structure public-land managers' decisions.

[8]Costello & Kaffine (2018) note that the literature on optimal natural resource management primarily considers how the spatial organization of management affects ecological outcomes, with comparatively little attention to heterogeneity in resource users' preferences.

Rather than evaluating these decisions using only ex post estimates of benefits and costs, I focus on the benefits and costs documented at the time decisions were made. Because ex ante assessments may differ substantially from realized outcomes, this approach lets me estimate the tradeoffs as offices percieved them.[9]

Finally, my findings extend research on how external scrutiny shapes decision-making, including incentives to follow established practices and avoid criticism. Scharfstein & Stein (1990) show that reputational concerns can lead managers to imitate others' investment decisions despite their private information. Bernheim (1994) shows that individuals with different preferences may conform to a common standard of behavior when others judge them by their actions. Frakes (2013) finds that physicians' treatment decisions move toward national practice patterns when malpractice law shifts from local to national standards of care. More closely related to my setting, Leaver (2009) and Prendergast (2003) examine how the prospect of criticism and complaints shapes bureaucratic decision-making, showing that bureaucrats have incentives to make decisions that reduce their exposure to criticism. BLM field offices face scrutiny from both conservation groups and industry, whose competing objectives make it difficult for field offices to avoid criticism altogether. Consistent with this literature, my findings suggest that offices may choose to follow established standards to minimize their exposure to criticism. While adhering to a standard cannot eliminate criticism of the resulting allocation, it may redirect criticism away from an office's discretionary judgment and toward the standard it follows.

2 Modeling Framework

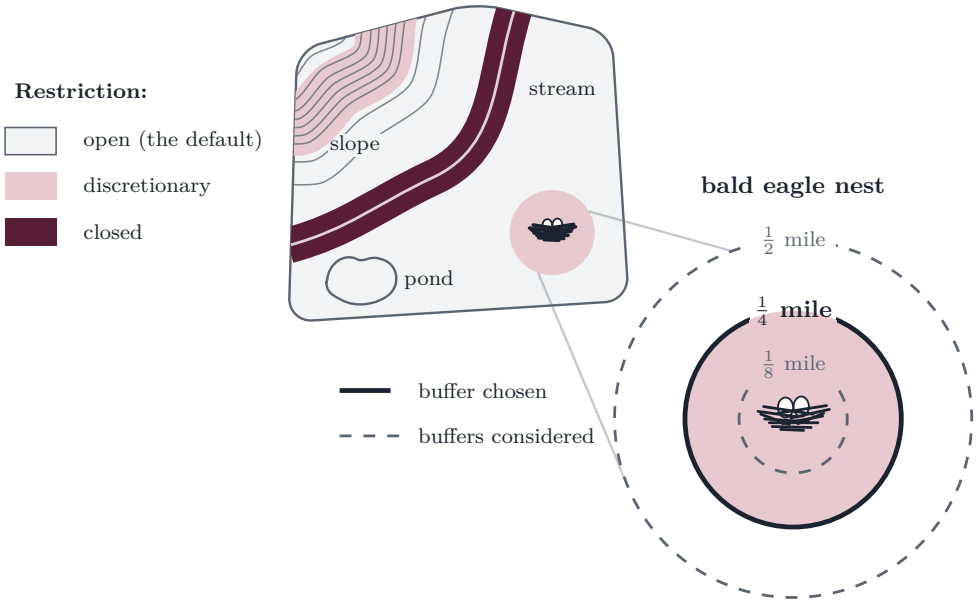
2.1 An Allocation Decision

This section defines a single allocation decision. Each field office decides where to withhold land from development. Each office makes that decision for each *natural feature* f, e.g., a bird's nest, stream, or steep hillside. I represent the allocation decision for each feature f as two parts: a restriction r, which determines whether and how strictly development is restricted, and a scalar buffer b, which determines how far that restriction extends around f. For all features, the feasible set for the restriction is {open, discretionary, closed}. In contrast, the feasible set for a buffer varies by feature type. For example, for nests the feasible set might be $\{1/8 \text{ mi},\ 1/4,\ 1/2,\ 3/4,\ 1 \text{ mi}\}$ and for streams $\{25 \text{ ft},\ 50,\ 100,\ \ldots,\ 450,\ 500 \text{ ft}\}$.

Diagram 1 maps a *set* of allocation decisions (individual allocation decisions for multiple features). In the schematic, the field office chose r = discretionary (pink) on steep hillslopes, restricting drilling activities when the soil is wet to mitigate risk of landslides. The office also chose r = closed (brown) around the stream to protect the riparian habitat on either side and chose r = open (white) for the pond. Finally, the office chose r = discretionary (pink) around a bald eagle's nest, limiting drilling during the nesting season. The spatial extent of each restriction is governed by its buffer b. The zoom for the nest highlights this specifically. Diagram 1 maps the chosen buffer (b) for the eagle's nest at a 1/4 mile (solid line) and feasible alternatives at 1/8 and 1/2 mile (dashed lines).

[9] Rollet (2025) makes a related point in the context of urban redevelopment, arguing that New York City's zoning code may have been less costly, and potentially beneficial, under the economic and industrial conditions prevailing when it was adopted. The policy became increasingly costly as demand, land values, and the externalities from different land uses changed. Because zoning adjusts slowly, observed outcomes today can reflect institutional persistence rather than contemporaneous policymaker preferences. In my setting, using realized outcomes alone to infer public-land managers' objective function could similarly conflate the tradeoffs embedded in the original planning decision with later constraints, shocks, and implementation dynamics.

Diagram 1: A Set of Allocation Decisions



2.2 The Optimization Problem

This section distills the optimization problem each field office solves when making an allocation decision. Field office i, for every feature f of non-market resource class k picks $\boldsymbol{y} = (r, b)$.

office i — **feature** f (class k: soil, water, wildlife) — **choice** $\boldsymbol{y} = (r, b)$ (r: open, discretionary, closed; b: $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$ mile, ...)

Field office i makes this choice for a given feature f to maximize the payoff function:

$$\overbrace{\underbrace{\beta_k \cdot \mathbb{E}_i K(\mathbf{y})}_{\text{Expected conservation value}} - \underbrace{\mathbb{E}_i M(\mathbf{y})}_{\text{Expected revenue forgone}}}^{\textbf{Net benefits of conservation}} - \overbrace{\underbrace{\kappa \cdot D(\mathbf{y})}_{\text{Cost of departing from standards}}}^{\textbf{Administrative burden}}$$

β_k reflects an individual field office's revealed value for conserving f of class k. Under $\boldsymbol{y}$, $\mathbb{E}_i K$ is the expected amount of conservation achieved around f and $\mathbb{E}_i M$, the expected revenue forgone. The expectation $\mathbb{E}_i$ is office i's expectation of revenue forgone and conservation achieved. κ is the office's cost of departing from an *established standard*, which I define as a buffer recommended by an entity outside the office, such as a state wildlife agency. D is the distance of $\boldsymbol{y}$ from the standard office i follows. The first two terms together are the net benefits of conservation. The third term is the administrative burden of justifying a choice that departs from an established standard. Estimation recovers the β and κ that best rationalize the observed choice.

2.3 Motivating Evidence

This section motivates the modeling framework outlined above with evidence from planning records and legal requirements.

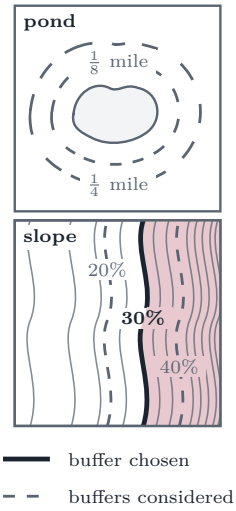
2.3.1 Planning Record Evidence Motivating the Choice Representation

As mentioned in Section 1, federal regulation requires field offices to distill both the allocation decisions they chose and what feasible alternatives they could have chosen, but did not. Offices comply with this requirement by outlining at least four feasible sets of allocation decisions that together span all possible allocation decisions (see Appendix A.2). The maps of restrictions corresponding to each set and the accompanying text describing each show that decisions are (i) made feature by feature and (ii) discrete.

The choice situation: one natural feature. Figure 1 maps one region of Four Rivers, Idaho, under three considered sets of allocation decisions. Panel 1a maps all restrictions in the region. Each considered set anchors its restrictions on four features: a historic trail, a critical groundwater area, elk winter range, and sage-grouse habitat. Panels 1b and 1c map three considered allocation decisions each for one feature, the trail and the groundwater area, respectively. Along the historic trail, the considered restrictions form a constant-width band around the trail, where the width varies across alternatives. When the choice protects the critical groundwater area, restrictions stop exactly at the area's surveyed boundary. In each case, the geometry of the underlying feature dictates the choice's geometry. The planning text has the same structure, discussing restrictions one feature at a time. The resulting map of restrictions is therefore composed of overlapping feature-level footprints rather than drawn to a landscape-scale design.

The alternatives: restriction–buffer pairs. In the considered sets of allocation decisions, every individual allocation decision is a restriction paired with a buffer, and the two are set separately. Across considered decisions, choices can differ in both the restriction and buffer, or in just one. For example, along Figure 1's historic trail, the protective band widens from an eighth of a mile to a quarter to a half across the three considered options, and the restriction tightens from no surface disturbance to closure in the most restrictive. The set of alternatives at a feature is therefore a cross product: every restriction paired with every buffer in the feature's range.[10]

Diagram 2



This interpretation of each feasible choice as a restriction–buffer pair requires thinking about buffers differently for different kinds of features. Along the trail the buffer is a distance. At a pond, it is the pond's mapped boundary buffered by a given distance. On a hillside, it is a threshold on the slope's grade above which drilling is restricted. Diagram 1 draws the first form; Diagram 2 draws the other two. Each form is a single number that grows the protected footprint outward from the feature, so each can be cast as a buffer b on its own scale.

[10]The open restriction protects no ground and so pairs with no buffer; Section 3.2.4 states the set exactly.

2.3.2 Legal Requirements Motivating the Payoff Function

Field offices make allocation decisions while navigating significant legal requirements. These requirements motivate the payoff function above.

Agency Mandate The Federal Land Policy and Management Act (FLPMA) mandates the BLM to manage public lands to balance competing uses, such as drilling and conservation, while sustaining public benefits over time. Federal regulations make this mandate more concrete, establishing that the objective of allocation decisions is to "maximize resource values for the public through a rational, consistently applied set of regulations and procedures" (see Appendix A.1). While more concrete than the statute, regulations do not specify how offices should weigh competing resource values against one another. For example, if a site is valuable for both drilling and bald eagle habitat, regulations do not specify which use an office should prioritize or to what extent. One interpretation of the legal requirements is that, holding conservation value fixed, offices should conserve less where development is more valuable and more where development is less valuable. Whether offices actually respond to resource values in this way is an empirical question. Ultimately, the mandate amounts to requiring field offices to consider *both* market and non-market resource values when making decisions. The payoff function above reflects this requirement by allowing *both* to influence allocation decisions.

Process Federal statutes and regulations place significant structure on the process field offices must follow when making allocation decisions. FLPMA and the National Environmental Policy Act (NEPA) together require offices to **(1)** solicit public input to identify the issues allocation decisions should address, **(2)** publish draft decisions addressing these issues and make them available for public comment, **(3)** revise decisions in response to substantive comments or explain why no revision is necessary, and **(4)** publish revised allocation decisions that eligible participants may protest, raising issues previously submitted during the planning process (see Appendix A and D.1).

Put simply, this process requires field offices to publicly justify their choices to stakeholders, including pro-conservation and pro-industry groups with competing interests. Field offices frequently handle this requirement by choosing and citing an established standard, such as a state wildlife agency's recommended buffer (see Appendix D.2-D.4). Following an established standard provides a readily available justification for a choice, reducing the administrative burden of defending it relative to a buffer of the office's own design (see Appendix D.5). To capture this incentive, the payoff function above includes a cost of departing from established standards.

2.3.3 Empirical Evidence Motivating the Payoff Function

This section provides motivation in the data for the payoff function's two components.

Offices consider the net resource benefits of conservation. Figure 2 plots the share of each township[11] closed to development against each township's development potential. Eight of the eleven offices in the estimation sample protect a steadily smaller share as projected revenue rises; the share is flat or rises in

[11] Townships are 6-by-6 mile squares defined by the Public Land Survey System grid.

Buffalo, WY, Colorado River Valley, CO and Uncompahgre, CO. Pooling all eleven offices, the share closed falls from 41% at about $260 a year to 18% at about $18 million.

Two aspects of the figure help motivate the payoff function. *First*, the cost of conservation matters; as the revenue at stake rises, most offices protect less. A model that omitted forgone revenue would therefore miss a central decision-making margin. *Second*, the tradeoff differs across offices. The office curves fall at visibly different rates over the same tenfold increase in revenue.[12] The revealed preference for conservation must therefore be allowed to vary by field office.

Offices follow several common standards. Figure 3 shows that offices repeatedly choose the same few protection extents. Two distances, a quarter mile and a half mile, account for 85% of protective decisions at bird nests. Two setbacks, 100 and 500 feet, account for 71% at streams and rivers. Two slope grades, 25% and 30%, account for 64% at steep slopes. This clustering is not simply the result of just a handful of offices consistently choosing the same buffers. For example, of the ten offices with bird nests in the sample, eight use quarter-mile buffers and seven use half-mile buffers to protect at least some nests. Two conclusions follow. *First*, the clustering suggests that offices consider more than local benefits and costs when choosing conservation policies. If each choice were tailored to the local benefits and costs, chosen buffers would vary across features, because development potential varies across features. The model captures this phenomenon with a cost of departing from established standards. *Second*, the multi-modal clustering suggests that offices may follow different standards for the same type of feature. This motivates allowing offices to have different established standards, an aspect of the empirical model that I will distill in detail later in the paper.

3 Data

This section outlines how I convert 90,000 pages of planning records into a dataset suitable for discrete-choice estimation. This section proceeds in two parts. First, I describe how I create a structured text database from the planning records. Second, I describe how I convert that text database into a dataset suitable for discrete choice estimation.

3.1 Structuring Data in Planning Records

In processing the planning record, the central challenge is finding *every* passage containing relevant information. For example, information about buffers to protect bird nests may be scattered across three appendices: the buffer an office adopted in one, the buffers it considered in another, and a wider buffer proposed by a commenter in a third. The median office's record runs to about six thousand pages, the longest to more than twenty thousand, and discussion of any one feature may appear throughout. Diagram 3 illustrates how I convert these scattered records into reliably coded values, representing feasible choice sets and the inputs needed to construct their attributes. Its left column shows how the record passes through *three* readings, each covering fewer pages than the last and each converting the record's text into a more structured form. Its right column shows how the output of each reading is checked. The first reading is checked by a literal

[12] This heterogeneity is consistent with the institutional setting; each office makes its own allocation decisions with plausibly its own interpretation of the BLM's mandate.

text search, the second and third by having each agent reread the pages cited in the previous reading rather than trust the earlier agents' conclusions, and the final coded values by hand.

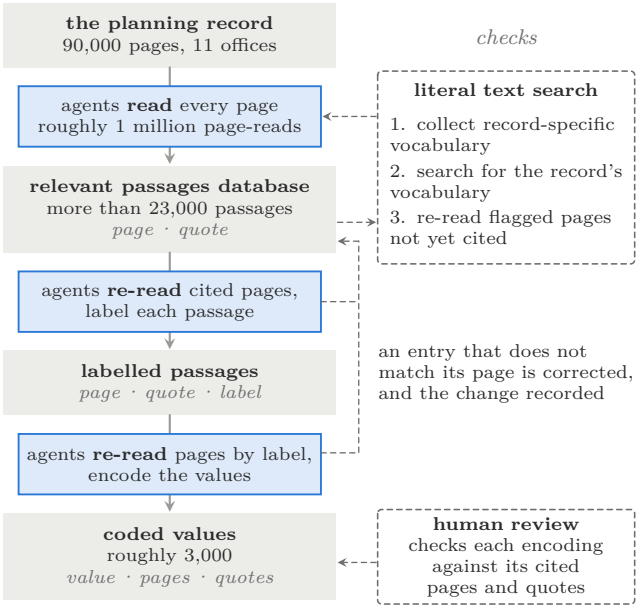
This subsection describes the *three*-stage process I use to convert unstructured planning records into coded values.

3.1.1 Stage 1: Populate Database

The first reading covers every page of the record. I divide this work among many *agents* (LLMs assigned a narrowly defined task). Each agent reads approximately thirty pages, searching for discussion of a single topic, either a group of related feature types (e.g., water and riparian features) or one resource (e.g., geothermal). When an agent finds a relevant passage, it records the source document, page number, and a quote or description explaining the passage's relevance. I assign agents both a narrow topic and a short page range because agents with more focused tasks tend to identify relevant passages more reliably. This approach requires multiple agents to read the same pages. Each page is therefore read approximately eight times, once for each topic. In total, this first reading involves roughly one million agent-page reads.

The primary risk in this first reading is that an agent misses a relevant passage. If a passage is never logged, agents in later stages cannot read it, and the information it contains may be permanently omitted from the database. I therefore complement the first reading with a literal text search that matches strings exactly and does not involve an LLM. Once the first reading is complete, I collect relevant vocabulary for each topic (designation names, stipulation codes, table labels) from the passages agents cited and search for these terms across the entire record. The first reading for a topic is not accepted until every page flagged by this literal text search has either been cited or explicitly recorded as irrelevant.

Diagram 3: From Planning Record to Structured Data



Notes: The grey boxes document the input and output data of each pipeline stage. Every downward arrow marks a LLM processing step. At each step, agents process fewer pages in aggregate. The side column documents how LLM work is checked after each stage. The dashed boxes are the steps where non-LLM text processing enters the pipeline.

A reader familiar with text-processing tools may wonder why I have agents read every page rather than begin with semantic retrieval. Retrieval would be cheaper because agents would need to read only the passages returned by the search. However, at the outset I do not know much of the vocabulary an office uses to discuss a given topic, so queries constructed without that vocabulary risk missing relevant passages. I could iteratively refine the queries, adding terms discovered in passages returned by earlier searches. But this approach would still depend on what those earlier searches happened to retrieve. I instead incur the larger upfront cost of reading the full record because omissions directly affect the estimation sample. A passage that is never surfaced might contain a restriction–buffer pair an office ultimately chose, causing that choice to be incorrectly coded or omitted from the database.

3.1.2 Stage 2: Label Relevant Passages

The first reading produces a database of passages, each linked to a feature type (e.g., bird nests) or a market resource (e.g., geothermal). The completed database contains more than twenty-three thousand passages. The second reading covers only passages in this database. I again divide the work among many agents, each assigned a single passage and topic. Each agent rereads its assigned passage and labels the database entry according to what the passage establishes about that topic, such as a restriction–buffer pair protecting a feature or a discussion of why protecting that feature is important.

The labeling step is necessary because the first reading is deliberately permissive. To minimize the risk of omissions, first-stage agents log any passage that might bear on a feature or market resource, postponing judgments about whether it contains information needed for estimation. This reduces the likelihood of having to reread the full record to recover missing information, but means a large collection of passages in the database have varying relevance to downstream tasks. The second reading labels what each passage establishes so that agents in later stages can identify the information they need without reading irrelevant passages. Separating passage identification (first-read) from labeling (second-read) also improves reliability, because, as previously discussed, agents are more reliable when assigned narrowly defined tasks.

3.1.3 Stage 3: Code Values

The third reading codes the values needed to construct the estimation sample. As in the earlier readings, I assign each agent a narrowly defined task, such as identifying the restriction–buffer pairs an office chose to protect nests in its jurisdiction. The agent locates the relevant passages for its task by querying the database using a topic (e.g., nests) and second-stage labels (e.g., chosen or considered alternative). The agent then rereads the returned passages and codes the requested values. For features, agents code each office's chosen and considered restriction–buffer pairs from passages labeled as describing restrictions. For market resources, agents code offices' projections of development, revenue, and disturbance from passages labeled as containing those projections.

3.1.4 Checks and Verification

The three-stage reading procedure includes several checks designed to prevent an agent's misinterpretation of a passage from propagating to later stages (see right-hand column of Diagram 3). At each stage after the first, a new agent rereads cited pages. As a result, multiple agents examine a passage for information relevant to a given topic before it is used to code a value. When a later agent finds that an earlier agent's description does not match the cited passage, it corrects the database entry and records the change for review. When the underlying record is ambiguous, the agent stops and refers the question to me. For example, if statements within a plan conflict, the agent flags the discrepancy and asks me how to proceed.

The three-stage procedure is standardized and can be rerun. Two runs may not identify exactly the same passages, because LLMs are not deterministic. However, both runs must meet the same verification requirements. The instructions are recorded in plain-text files that an orchestrating agent follows step by step, assigning reading, labeling, and coding tasks to other agents. This written procedure serves two purposes. First, the same instructions are applied to every office, so all eleven records are read and coded using the same procedure. Second, the procedure can be rerun on previously processed offices as an audit.

Audits allow me to compare the passages identified across runs, helping to reveal passages missed during earlier first readings.

I personally verify every final coded value against the passages supporting it. This verification establishes two things. First, the cited pages support the coded value. Second, when a restriction–buffer pair is coded as chosen, the cited pages establish that the office actually adopted it, rather than merely considering it in a draft analysis or alternative. Ludwig et al. (2025) show that when an LLM's output serves as a measurement in a downstream estimate, valid inference requires checking that output against a validation sample. Here, I hand-validate the full set of final coded values rather than use a validation sample. However, verifying coded values does not establish that the database is complete. The literal text search and audits described above provide additional safeguards against omissions.

3.2 Converting Text Database to a Discrete Choice Dataset

The LLM data pipeline outlined above produces a text database of field offices' allocation decisions, with each entry pertaining to a particular type of feature. For example, one entry might specify that a field office chose to prohibit drilling within half a mile of all bald eagle nests in its jurisdiction during nesting season, and within three-quarters of a mile of nests in a particular region year-round. This section describes how I translate these text entries into a feature-level dataset of feasible choices.

3.2.1 Defining a feature

Section 2.1 defines a natural feature f as the choice situation, where a feature is *a* nest or *a* stream. This discussion treats individual features as though they are easily delineated. In practice, however, the boundaries between features are not always obvious. For example, when one stream flows into another, are the two connected streams two features or one? Constructing the estimation sample therefore requires rules for delineating individual features. I define a feature at the finest spatial scale at which offices make distinct allocation decisions. For example, I treat all streams within one watershed as one feature. Offices often make allocation decisions at a coarser scale, adopting a single restriction for an entire feature type. For example, an office might choose how to protect all streams in its jurisdiction rather than make separate decisions for streams in each watershed. Nonetheless, I retain the finer unit because offices have the discretion to differentiate protections across these features.[13]

3.2.2 The universe of features

The preceding section defines a single choice situation, a feature. This section describes how I identify the universe of features each field office could have protected and which features I ultimately include in the estimation sample.

Field offices select which features to protect from mapped inventories of resources in their jurisdictions, but I do not observe each office's complete inventory. I therefore construct a common list of feature types from the union, across all offices, of every type protected by a considered allocation or raised in a public comment. For each office, I then include every feature of these types present in its jurisdiction, regardless of whether the office considered protecting it. This ensures that a feature's inclusion in an office's sample

[13] See Appendix C.2 for examples.

does not depend on the allocation that office chose. An alternative would be to include a feature type in an office's sample only if that office's planning record mentioned it. In practice, this would produce nearly the same sample. Offices discuss almost every feature type present in their jurisdictions, if not in a considered allocation then in a public comment. Nonetheless, defining the sample with the common feature type list rather than mentions guards against missed mentions, a particular concern given that planning records average more than 8,000 pages per office.

Next, I restrict the set of feature types included in the estimation sample to those I can consistently geolocate. Maps locating features are not always publicly available for every type discussed in the planning records. I therefore retain feature types whose locations can generally be reconstructed and verified using publicly available data. The resulting sample covers fifteen feature types across eleven field offices (Figure A1): four *soil* feature types (slope, water erodibility, wind erodibility, saline soils), eight *water* types (major rivers, perennial streams, intermittent streams, waterbodies, riparian wetlands, floodplains, drinking-water sources, groundwater), and three *wildlife* types (bald eagle roosts, prairie dog towns, and raptor and other nests). The planning records contain decisions concerning many other feature types, including big-game seasonal ranges, cultural and historic sites, and the historic trail shown in Figure 1, which are (currently) excluded from the estimation sample.

3.2.3 Georeferencing features

Field offices frequently publish geodatabases containing partial inventories of the features in their jurisdictions. I use these geodatabases, along with the underlying state and national inventories from which they are derived, to locate individual features. For example, I use the national hydrography database to locate stream features. Once a feature is georeferenced, I can combine its location with the text description of an allocation decision to reconstruct each choice. For example, I can extend the mapped streams by 500 feet on either side to reconstruct a 500-foot buffer restriction. When possible, I verify these reconstructions against field offices' published geodatabases of allocation decisions. When these geodatabases are incomplete, I compare the acreage covered by my reconstructions against acreage figures reported in the planning records.

3.2.4 Constructing georeferenced alternatives

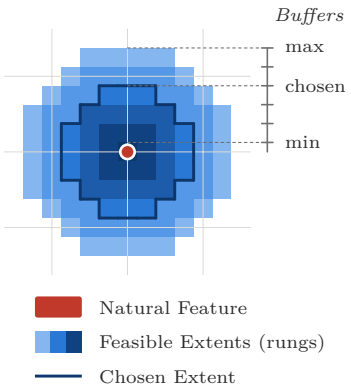
For each feature type, I define the same feasible choice set across field offices. As previously discussed, each choice is a combination of a restriction r and a buffer b, denoted $\boldsymbol{y} = (r, b)$. I construct the feasible choice set as the Cartesian product of feasible restrictions and buffers. This section first explains how I define the feasible restrictions and buffers separately, then describes the resulting set of feasible alternatives.

Feasible restrictions I allow every office to choose any restriction for any feature. The planning records contain five restrictions: open, timing limitations, controlled surface use, no surface disturbance, and closed. Timing limitations restrict when activities may take place over the year, while controlled surface use restricts where operations may occur. No surface disturbance prohibits drilling on the surface but allows directional drilling from adjacent land, whereas closed removes the land from leasing altogether. I collapse these five restrictions into three. Open remains unchanged. I group timing limitations and controlled surface use into a discretionary restriction, as both restrict surface disturbance without prohibiting it. I group no surface disturbance and closed into a closed restriction, as both prohibit surface disturbance within the protected

area. The feasible restriction set is therefore $\mathcal{R} = \{\text{Open}, \text{Discretionary}, \text{Closed}\}$, with $r \in \mathcal{R}$ for every feature.

Feasible buffers The smallest and largest feasible buffers (e.g., 1m and 1000m for streams) defines the feasible set of buffers and is specific to each feature type. The smallest feasible buffer is the smallest increment that protects any ground at all. This increment depends on the feature's geometry. For areal features such as ponds, the minimum buffer is zero, meaning the restriction's footprint is the feature itself. For point and line features, such as nests and streams, the minimum buffer is just above zero, because 0 would protect no ground. For features such as slopes, which are defined by raster data not vector data, the minimum buffer is the least restrictive threshold that protects any ground, for example the grade threshold that covers only the steepest slopes. The largest feasible buffer is the maximum buffer considered for a given feature type in any field office's considered allocation decisions. Between these endpoints, I construct the feasible policy footprints as a discrete ladder (Diagram 4). Each rung is a dilation of the one below it, and the chosen footprint is always included as one rung on the ladder.[14] This ladder preserves the coarseness of the policy choice rather than treating the buffer choice as continuous.

Diagram 4: The Buffer Ladder



Notes: Each shaded band is one rung of the buffer ladder for a nest, drawn on the grid. The ladder starts at the cell holding the nest, grows one cell per step in every direction, reaches the chosen buffer, itself a rung, and continues to the largest feasible buffer.

Feasible alternatives The feasible choice set is $\mathcal{Y}_{i,f} = \{\text{Open}\} \cup (\mathcal{R} \setminus \{\text{Open}\}) \times \mathcal{B}(f)$, where $\mathcal{R}$ is common to every office and feature, and $\mathcal{B}(f)$ is the ladder just described. The ladder's range is determined by the feature type, while its rungs are constructed for each individual feature. One concern with this construction is that it may include infeasible alternatives. An alternative feasible at one office may not have been feasible at another. For example, a legally required minimum buffers for a species listed under the Endangered Species Act may have applied when one office made its decision but not when another did. I limit this concern by checking the planning records for evidence that offices had discretion over whether and how extensively to protect each feature type. For example, I verify that offices considered leaving feature types unprotected (open) or protecting them substantially more extensively than they ultimately chose.

[14] The ladder starts at the minimum footprint (setting aside the open alternative), dilates step by step until it reaches the chosen footprint, includes the chosen footprint as a rung, and then continues dilating until it reaches the maximum buffer, unless the chosen footprint is already the maximum. The chosen footprint enters at the distance stated in the plan; only the intermediate rungs between it and the endpoints are generated by grid dilations rather than directly from stated distances. Each step consists of a fixed number of 8-adjacency dilations on a grid of roughly 30×30 m cells. Threshold-based feature types step through successive attribute thresholds instead of geometric dilations. For these feature types, the chosen threshold is the cutoff stated in the plan.

4 Empirical model, identification and estimation

Field office i, for every feature f of non-market resource class k, chooses $\boldsymbol{y} = (r, b)$ to maximize $V_{i,f}(\boldsymbol{y}; \eta_{i,f}) + \sigma_{i,k}\, \varepsilon_{i,f,\boldsymbol{y}}$, where

$$V_{i,f}(\boldsymbol{y}; \eta_{i,f}) = -\overbrace{\mathbb{E}_i\, M_{i,f}(\boldsymbol{y})}^{\text{expected revenue forgone}} + \underbrace{\beta_{i,k}}_{\text{WTP}} \overbrace{\mathbb{E}_i\, K_{i,f}(\boldsymbol{y})}^{\text{expected disturbance avoided}} - \underbrace{\kappa_{i,k}}_{\substack{\text{departure} \\ \text{cost}}} \overbrace{D_{i,f}(\boldsymbol{y}; \eta_{i,f})}^{\substack{\text{distance from} \\ \text{the standard}}} + \underbrace{\gamma_i}_{\substack{\text{habit to} \\ \text{close}}} \underbrace{\mathbf{1}[r = \text{closed}]}_{\substack{\text{closure} \\ \text{fixed effect}}} \quad (1)$$

Equation 1 is the payoff function Section 2.2 outlines with two changes.[15] First, the equation writes the distance from the standard as conditional on the correction $\eta_{i,f}$, the latent type known to the office but not the econometrician. This correction helps locate each office's standard. I will outline its role in full below. Second, the empirical payoff function includes a closure fixed effect γ_i, which absorbs office i's propensity to close land regardless of the net-benefits.

The remainder of this section precisely defines each component of the payoff function, how each coefficient is identified, assumptions on unobserved portions of the payoff function, the resulting choice probabilities and the estimation procedure.

4.1 Expected Net Benefits

4.1.1 Measuring Expected Revenue Forgone and Disturbance Avoided

When an office chooses how to protect a feature (e.g., a nest), the cost is revenue forgone and the conservation benefit is disturbance avoided. Both depend on the wells the choice prevents. Closing an area to development ($r = \text{closed}$) prevents every well that would otherwise have been drilled there, whereas imposing discretionary restrictions ($r = \text{discretionary}$) prevents only a share of the wells. The revenue these wells would have generated is the revenue forgone, and the disturbance they would have caused near the feature is the disturbance avoided.

At the time an allocation decision is made, the office does not know where wells will be drilled. To estimate offices' preferences for conservation relative to development, I therefore need the expectations of revenue forgone and disturbance avoided that informed their decisions, whether accurate or not. I recover these expectations using the offices' own development forecasts.

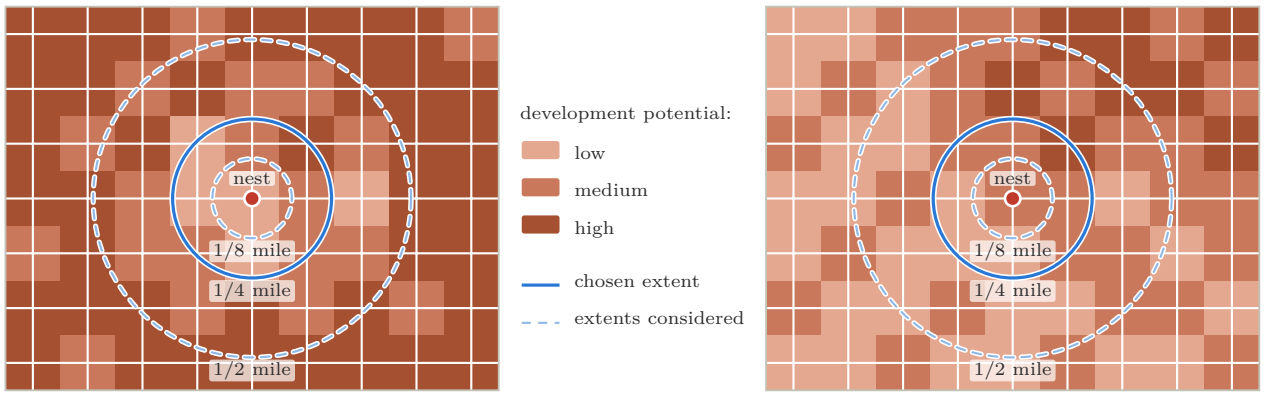
Federal regulations require field offices to evaluate alternative *sets of allocation decisions* (as described in Section 2.3.1) and to estimate each set's consequences for non-market and market resources (Appendix A). To do so, each office maps expected development in its jurisdiction, assuming that all land over which the office has discretion remains open for development.[16] These development potential maps partition the planning area into categories c (low, medium, and high in Diagram 5). Within each category, the office expects the same number of wells to be drilled per acre. Multiplying this expected number of wells per acre by the office's

[15] As previously discussed, $\mathbb{E}_i$ denotes office i's expectations of development.
[16] See Figure A2a for an example.

expected revenue per well gives each category c an expected revenue per acre. Multiplying it by expected surface disturbance per well gives the probability an acre in that category is disturbed.[17]

Offices intersect these maps with the alternative sets of allocation decisions they consider to project production, revenue, and surface disturbance under each (Figure A2 reproduces Rock Springs, Wyoming's accounting).[18] To measure expected revenue forgone and disturbance avoided at the level of choices (features), I intersect each feasible choice for protecting an individual feature with these same development potential maps. Diagram 5 illustrates this procedure for two nests, each representing a separate allocation decision.

Diagram 5: Development potential map intersected with alternatives



a nest on higher potential land

a nest on lower potential land

For both nests, the office considers the same alternatives and chooses the same extent of protection. The first nest, however, lies on higher-potential land than the second. Consequently, the same choice entails greater expected revenue forgone at the first nest because more of the acres it protects lie on land with high development potential. Specifically, I compute expected revenue forgone as:

$$\underbrace{\mathbb{E}_i M(\boldsymbol{y})}_{\substack{\text{expected revenue} \\ \text{forgone}}} = \sum_c \begin{array}{c}\text{share of wells} \\ \text{restriction } r \text{ removes}\end{array} \times \begin{array}{c}\text{revenue per} \\ \text{acre in } c\end{array} \times \begin{array}{c}\text{protected acres under } b \\ \text{in category } c\end{array} \tag{2}$$

When r = open, the share of wells removed is zero, so $M(\boldsymbol{y}) = 0$. When r = closed the share is one, so $M(\boldsymbol{y})$ is the sum of the expected revenue per acre over all acres the choice protects. When r = discretionary the share is between zero and one, so $M(\boldsymbol{y})$ is a fraction of the sum when r = closed.[19]

When estimating consequences for non-market resources, offices measure conservation benefits in acres.[20] I therefore construct expected disturbance avoided much as I construct revenue forgone, but with one important difference. This difference reflects offices' judgment that an acre of avoided disturbance matters more for protecting a feature when it is nearby than when it is farther away, whereas a dollar of revenue

[17] Appendix F details how expected revenue and the probability of disturbance are recovered from the planning records.

[18] One concern is that these tabulations could themselves be constructed to justify a choice already made. Appendix D.1 discusses how the public planning process constrains this possibility.

[19] Appendix F details how this share is recovered from the planning records.

[20] See Figure A2c for an example.

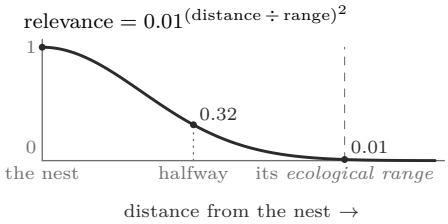
has the same value regardless of where it is earned.[21] The record, however, does not quantify how the importance of avoided disturbance declines with distance. I therefore model this relationship using what I call the *relevance scale*.

The relevance scale assigns an acre at distance l from feature f a value $w_f(l)$; which has full weight at the feature itself, declines as l grows, and is negligible beyond an *effective ecological range* h, i.e.,

$$\underbrace{w_f(0) = 1}_{\text{full weight at the feature}} \quad , \quad \underbrace{w_f'(\cdot) \leq 0,}_{\text{declining}} \quad \underbrace{w_f(h) \approx 0}_{\text{negligible by the range}} \quad .$$

Any function with these three properties would work. I use a Gaussian kernel $w_f(l) = 0.01^{(l/h)^2}$. I test the sensitivity of the results to this functional form assumption. Diagram 6, at right, draws the relevance as an acre's distance from a nest increases. An acre halfway from the nest to the ecological range h has a relevance of about a third of the relevance at the nest, and an acre at the range a hundredth. The planning record supplies the underlying judgment that relevance declines with distance, but not its exact functional form. For each type of feature, I calibrate h from external sources, e.g., perennial streams and intermittent streams each get their own h.[22] Estimating h would confound quantity of conservation achieved with preference for conservation.

Diagram 6



I leverage the relevance scale to measure the amount of expected disturbance a choice avoids as it relates to the feature the choice protects. Specifically, I compute expected disturbance avoided as:

$$\underbrace{\mathbb{E}_i K(\boldsymbol{y})}_{\substack{\text{expected disturbance} \\ \text{avoided}}} = \sum_c \underbrace{\begin{matrix}\text{share of wells} \\ \text{restriction } r \text{ removes}\end{matrix}}_{} \times \underbrace{\begin{matrix}\text{probability of disturbance} \\ \text{on an acre in } c\end{matrix}}_{} \times \sum_{\substack{\text{acre } j \text{ in category } c \\ \text{protected under } b}} \underbrace{\begin{matrix}\text{acre } j\text{'s} \\ \text{relevance}\end{matrix}}_{} \tag{3}$$

This construction differs from that of expected revenue forgone (equation 2) in two ways: i) this sum applies the probability of disturbance instead of the expected revenue per acre, and ii) it weights each protected acre by its relevance to the feature.

4.1.2 Identifying the Willingness to Pay for Conservation (β)

Expected revenue forgone is an acreage-weighted sum of the per-acre expected revenue for each potential category (see equation 2). Expected disturbance avoided without the relevance weighting would be almost perfectly collinear with expected revenue forgone (compare equation 2 and 3), because they are both acreage-weighted sums of scalars derived from the expected number of wells per acre in each potential category. Table A2 shows how collinear the not-relevance weighted disturbance avoided is with the revenue forgone. With all acres weighted equally, the median within-feature correlation between revenue forgone and disturbance avoided is 1.000, and the 5th percentile is 0.983. Differences across restrictions add little independent variation. Among alternatives that share a restriction and differ only in the buffer, the 5th

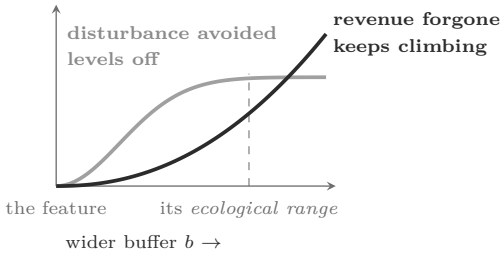
[21] See appendix C.5 for an example.

[22] Appendix E lists the range for the different types of features and how the range is calibrated from external sources.

percentile is 0.986, barely above the 0.983 across all alternatives.

The relevance weighting reduces this collinearity and is therefore what identifies $\beta_{i,k(f)}$. The relevance weighting leads $\mathbb{E}_i M_{i,f}$ and $\mathbb{E}_i K_{i,f}$ to change at different rates as protection extends outward. Diagram 7 plots expected revenue forgone and disturbance avoided as the buffer around a nest grows to beyond the nest's ecological range h. Revenue forgone climbs at an increasing rate as the buffer grows, while the marginal disturbance avoided by added acres falls as the added acres' relevance decreases with b. As the buffer approaches and passes the range h, those added acres carry almost no relevance, so disturbance avoided all but stops increasing with b. The difference in the curves is what identifies β

Diagram 7



This identification argument parallels Hendel (1999), who notes that in a standard discrete-choice model estimated on a single cross-section, brand-specific dummies would absorb the variation in price, because prices vary at the brand level, leaving the price coefficient unidentified. His multiple-unit demand model avoids that problem because firms choose both the quality and the number of computers. Expenditure is linear in the units purchased, while the payoff is a non-linear function of their quantity and quality. Here a single choice, the extent of the buffer, sets both how many acres an office protects and their relevance to the feature, the counterparts of Hendel's quantity and quality. Revenue forgone is roughly linear in the acres protected, while disturbance avoided is non-linear in them.

4.2 Cost of Departing from Established Standards

4.2.1 Measuring distance from the established standard.

I measure the distance of a choice from the office's established standard in two steps. *First*, I place all buffers, for all feature types, on the relevance scale. In other words, I normalize every choice b to a value between 0 and 1 with $w_f(b)$. This normalization allows me to compare buffers in different units and with different implications for the features they protect. *Second*, I locate an office's established standard for a given f with (i) a proxy for the established standard, the median $w_f(b)$ chosen for features of the same type as f across *other* field offices and (ii) an unobserved correction $\eta_{i,f}$, allowing offices to follow standards that differ from the proxy, and hence other offices.[23] I then measure the distance from the established standard as:

$$D_{i,f}(\boldsymbol{y};\eta_{i,f}) = \begin{matrix}\text{distance from the}\\ \text{standard under } \boldsymbol{y}\end{matrix} = \Big[\ \overbrace{w_f(b)}^{\substack{\text{last acre's}\\ \text{relevance}}} - \underbrace{(\overbrace{\overline{w}_{-i,f}}^{\text{proxy}} + \overbrace{\eta_{i,f}}^{\text{correction}})}_{\text{standard the office follows}}\ \Big]^2, \tag{4}$$

I choose this functional form for tractability and because it is standard. Squaring the distance form a reference point follows spatial theories of choice (Davis et al. 1970); centering that reference point on a social norm while allowing offices to follow different standards is motivated by Bernheim (1994). At $\eta_{i,f} = 0$ the office follows the proxy itself; a negative $\eta_{i,f}$ puts the office's standard past the proxy, i.e., at a wider more protective buffer, and a positive $\eta_{i,f}$ puts it short of the proxy, i.e., at a narrower less protective buffer. $\eta_{i,f}$

[23] As discussed in Sections 2.3.2 and 2.3.3, offices follow different standards. $\eta_{i,f}$ allows for this behavior in the model.

is known to the office but not to the econometrician. The office knows which standard they are anchoring their choice to, but I do not.

I assume that $\eta_{i,f}$ takes one of a fixed set of values, common to every field office and feature type, i.e., $\eta_{i,f} \in \{\eta_1, \ldots, \eta_Q\}$. A standard is by definition a protection extent shared across many features, so a discrete distribution for η is the natural choice. While all features share the same support, a given η_q implies a different standard for each feature type because each office's standard for a given feature type is the type's proxy plus the correction. Assuming common support across offices and feature types disciplines the fit by preventing the model from assigning a separate standard to nearly every choice. This assumption implies that across types of features standards are more or less protective than their proxies by similar amounts. This is plausible if offices draw on guidance from state agencies or previous decisions that prescribe similarly stricter or more permissive protections across different kinds of features.

4.2.2 Identifying the Correction (η) and the Departure Cost (κ)

The correction η is identified by choices at features with little or no development potential. At such a feature, the office expects no wells on the land any buffer would protect, so every alternative in expectations forgoes roughly no revenue and avoids roughly no disturbance. In other words, the net-benefit terms of equation 1 drop out of the comparison between buffers, and differences in payoffs across choices arise only through differences in their buffers' distance from the standard. An office that protects such a feature chooses the buffer closest to the standard it follows, assuming $\kappa_{i,k} > 0$, so the relevance of the last acre it protects, $w_f(b)$, reveals $\overline{w}_{-i,f} + \eta_{i,f}$. Clustering of choices at low-potential features across offices and feature types identifies the support values, and clustering within an office reveals which of them that office draws on.

The departure cost κ is identified by how far choices, for features with development potential, stray from the standard. Setting the departure cost aside, the office would choose the buffer that maximizes expected net benefits, $-\mathbb{E}_i M_{i,f} + \beta_{i,k}\, \mathbb{E}_i K_{i,f}$. At a feature on high-potential land, the net-benefit-maximizing buffer likely does not sit on the standard. How far the office's choice moves from the standard toward the net-benefit-maximizing buffer reveals $\kappa_{i,k}$. An office whose choices stay at the standard as $\mathbb{E}_i M_{i,f}$ and $\mathbb{E}_i K_{i,f}$ change across features reveals a high κ. In contrast, an office whose choices track the net-benefits more closely reveals a lower κ.

What keeps the departure cost κ separate from the willingness to pay β is that the attributes they multiply depend on different things. The distance from the standard depends on the buffer only through $w_f(b)$. Revenue forgone and disturbance avoided depend on the ground beneath the restriction. Two features of the same type with buffers at the same point on their relevance curves incur the same distance from the standard, regardless of the development potential beneath them (see Diagram 5). If one lies on high-potential land and the other on low, however, they differ in both revenue forgone and disturbance avoided. The distance from the standard therefore does not move in lockstep with the net-benefit terms across features, and that independent variation identifies $\kappa_{i,k}$ separately from $\beta_{i,k}$.

4.3 Estimation

4.3.1 Additional assumptions

First, I assume that the idiosyncratic shock $\varepsilon_{i,f,\boldsymbol{y}}$ is a type I extreme value draw, independent across the alternatives in a feature's choice set and across features. $\sigma_{i,k(f)}$ sets the shock's scale and is identified in estimation because the coefficient in front of revenue forgone is normalized to one.

Second, I assume, beyond the Q corrections (latent types), there is also an open type. This additional latent type leaves a feature unprotected (open) regardless of the payoff of other feasible protective choices. Without the open type, open choices driven by an office's disposition not to protect would also influence the parameters intended to describe the office's standard conditional on protecting.

Third, I assume the *mixing shares are office-specific*, and run over all $Q+1$ types, i.e.,

$$\Pr(\eta_{i,f} = \eta_q \mid i) = s_{iq}, \quad q = 1, \ldots, Q, \qquad s_{i0} + \sum_{q=1}^{Q} s_{iq} = 1,$$

where s_{i0} is the open-type share.

4.3.2 Choice probabilities and likelihood.

Given type q, the choice probability takes the multinomial logit form whose payoff is equation 1 evaluated at $\eta_{i,f} = \eta_q$:

$$P_{i,f}(\boldsymbol{y} \mid \eta_q) = \frac{\exp\{V_{i,f}(\boldsymbol{y}; \eta_q)/\sigma_{i,k(f)}\}}{\sum_{\boldsymbol{y}' \in \mathcal{Y}_{i,f}} \exp\{V_{i,f}(\boldsymbol{y}'; \eta_q)/\sigma_{i,k(f)}\}}$$

where $\mathcal{Y}_{i,f}$ is the set of feasible choices for feature f in office i. The unconditional probability mixes those Q logits with a point mass on the open type:

$$P_{i,f}(\boldsymbol{y}) = s_{i0}\, \mathbf{1}[r = \text{open}] + \sum_{q=1}^{Q} s_{iq}\, P_{i,f}(\boldsymbol{y} \mid \eta_q)\,,$$

where the open type leaves the feature open with probability one. The Q types also leave features open when the net-benefits and the departure make that the best alternative; s_{i0} therefore captures an additional propensity to leave features unprotected beyond what that payoff predicts.

To estimate the parameters ($\beta_{i,k}$, $\kappa_{i,k}$, $\sigma_{i,k}$, γ_i, $\{\eta_q\}$, s_{iq}), I maximize the log likelihood of the observed choices:

$$\sum_i \sum_{f \in \mathcal{F}_i} \log P_{i,f}\big(\boldsymbol{y}^*_{i,f}\big)$$

where $\mathcal{F}_i$ is the set of features in office i. The likelihood is evaluated exactly rather than by simulation, because the mixture has finite support. The number of types Q is set by adding support points until the next one carries a negligible share.

5 Results

This section reports what the fitted model shows. Every office is willing to give up revenue to conserve, though how much varies widely. Offices also tie their choices to established standards that do not depend on the net-benefits of conservation. They pay to stay near the standard; at the median office, the buffer the model predicts forgoes a higher net benefit at an adjacent buffer in seven of ten decisions. Finally, offices in Wyoming are habitually against closing a feature to development and in Colorado habitually inclined toward closure, whatever the net-benefits.

5.1 Willingness to pay

In the raw data, Figure 2 shows offices conserve less as the revenue at stake rises, at visibly different rates. The fitted model puts a price on this pattern: each office's willingness to pay (WTP) for conservation, the revenue it gives up to avoid disturbing a feature, by non-market resource. Figure 4 plots the WTPs, in dollars of forgone revenue per acre of avoided disturbance at the feature, with one estimate for each office and non-market resource. Table 1 reports the same estimates and their standard errors, alongside the closure fixed effect and the cost of departure. Twenty-eight of the thirty-two willingness-to-pay estimates are positive and statistically significant at the ten percent level. At the median office, WTP is \$58k per acre for soil, \$66k for water, and about \$38k for wildlife. The spread across offices is wide. Specifically, WTP for soil ranges from \$23k per acre at Uncompahgre, CO to \$407k at Royal Gorge, CO at ten of the eleven offices; at the eleventh, Little Snake, CO, the estimate is indistinguishable from zero ($-$\$4k, standard error \$7k). For water, WTP ranges from \$12k at Rio Puerco, NM to \$1.6m at Royal Gorge, CO, and it is positive and significant at all eleven offices. For wildlife, WTP ranges from \$27k at Casper, WY to \$900k at Royal Gorge, CO at seven of the ten offices that carry wildlife feature types; at Four Rivers, ID (\$1k, s.e. \$5k), Grand Junction, CO (\$4k, s.e. \$27k) and Lander, WY (\$65k, s.e. \$66k) the estimate is smaller than its own standard error.

The pattern of positive and statistically significant WTP estimates survives two specification checks. *First*, Figure A5 plots estimates from the same mixed logit fit with a relevance functional form of $0.01^{(l/h)^{1.5}}$ instead of $0.01^{(l/h)^2}$ (the main specification). Across the two functional forms, all thirty-two WTP estimates keep their sign, the same twenty-eight are statistically significant, and magnitudes are similar: every sensitivity check estimate lies within a factor of 1.9 of its main specification counterpart. *Second*, Table A1 and Figure A4 report estimates from a conditional logit; the same model with the standard departure cost and the open type removed. This simplified model assumes all offices are pure benefit-cost optimizers. The two models agree on the sign of thirty-one of the thirty-two WTP estimates; the exception is Little Snake soil, one of the four mixed-logit estimates indistinguishable from zero. Magnitudes differ more than signs, and mostly in one direction; the mixed logit's estimates are larger. For soil, the two models' estimates are within 30% of each other at eight of the eleven offices. For water, the mixed logit produces the larger estimate at nine of the eleven offices, and for wildlife at seven of the ten, with differences as large as a factor of seven (Buffalo wildlife, \$321k versus \$44k). Omitting the departure cost explains part of the difference. In the mixed logit, departing from a standard is allowed to be costly; the conditional logit sees no standard, let alone a cost of departing from it. It reads a buffer at a standard as if the benefits and costs put it there; a standard short of the buffer the office's true value would pick on benefits and costs alone is read as a low

value on conservation, and one past it as a high value. The mixed logit instead credits that buffer to the standard. And when a buffer does move past the standard, the office pays the departure cost as well as the revenue, so the mixed logit estimates a higher WTP. This mechanism explains why the wildlife estimates differ most; there, standards sit short of where the benefits and costs of conservation would put the buffer.

To assess whether the substantial heterogeneity in estimated WTP reflects systematic differences across offices, I next examine its relationship with (i) the environmental benefits of avoiding disturbance and (ii) local political preferences. An office's willingness to pay to avoid disturbance reflects both how much environmental damage the disturbance would cause and how much the office values avoiding that damage. As Muller & Mendelsohn (2009) illustrate in the context of air pollution, the damage caused by a given quantity of emissions depends on where the emissions occur. Similarly, differences in willingness to pay to avoid disturbance need not imply that offices value the same environmental benefit differently; they may instead reflect differences in the damage that disturbance would cause. If differences in environmental damages help explain the differences in willingness to pay, I would expect offices where disturbance causes greater damage to have higher willingness to pay to avoid it. I can examine this possibility for soil because the primary environmental benefit of avoiding disturbance identified in planning documents is reducing erosion, which can be quantified in terms of expected annual soil loss per acre. In contrast, planning documents describe numerous benefits from protecting water and wildlife that are more difficult to capture in a single comparable measure. Figure 5, lower left panel, plots willingness to pay to avoid disturbing fragile soils against expected annual soil loss per acre, a proxy for the environmental damage from such disturbance.[24] The plot reveals essentially no relationship between the two. Rather than the positive correlation one might expect if differences in environmental damages were a major source of heterogeneity in WTP estimates, the estimated slope is modestly negative.

Next, I examine whether differences in willingness to pay are associated with local preferences for conservation versus development. Direct measures of these preferences are difficult to obtain at the geographic scale of BLM offices. I therefore use the Republican share of the 2016 presidential vote as a proxy for local preferences, drawing on precinct-level vote shares and polling evidence that Republican voters are substantially less likely than Democratic voters to prefer conservation over drilling on public lands (Colorado College State of the Rockies Project 2017, Colorado College State of the Rockies Project 2020).[25] Figure 5, lower right panel, plots each office's willingness to pay (the same ten estimates as the left panel) against the Republican vote share in the 2016 presidential election. This relationship runs the expected way; higher Republican vote share is negatively correlated with willingness to pay. This negative relationship is consistent with offices in communities that place greater relative value on development having lower willingness to pay to avoid disturbance.

Figure A6 plots the same relationship for all three resource classes, adding water and wildlife to the soil panel of Figure 5. The negative relationship between WTP and Trump vote share also appears for water and wildlife. However, Rio Puerco, NM stands out as an exception. Its planning area gave Trump 43% of the vote, the lowest share in the sample, yet its willingness to pay for water, \$12k per weighted acre, is also the lowest. Weighting precinct vote shares by their acreage of BLM land, rather than their number

[24] See Section G for a detailed description of how I construct this metric.

[25] The 2020 Conservation in the West survey of registered voters in eight Mountain West states finds 47% of Republican voters prefer that Congress emphasize protecting the water, air and wildlife on public lands, while 40% prefer maximizing the public lands available for oil and gas drilling and mining. In contrast, 86% of Democratic voters choose protection over drilling (Colorado College State of the Rockies Project 2020, p. 3).

of votes, makes this contrast more pronounced (Figure A7). The other ten offices' Trump vote shares all rise by 8 to 22 percentage points, steepening the negative relationship between WTP and vote share. Rio Puerco's share, meanwhile, falls by 6 percentage points, making it an even more pronounced outlier. One possible explanation is New Mexico's unusual political economy; although the state leans Democratic, its public finances depend heavily on revenue from federal mineral development (New Mexico Legislative Finance Committee 2023).[26] Rio Puerco's low Trump vote share may therefore overstate local support for restricting development.

5.2 The standard and the cost of departing from it

The data support including a cost for departing from standards. Adding it to the conditional logit discussed above raises the log-likelihood by about 8,000 points for 93 parameters. If these ninety-three parameters explained nothing, they would raise the log-likelihood by about 45 points, so this gain is more than a hundred times what chance alone would produce. The rest of this section outlines where the fitted model locates the standards and the implied costs an office faces for departing from them.

The fitted model recovers six corrections. Figure A8's top section shows they straddle the proxy, from $\eta_1 = -0.691$ to $\eta_6 = +0.430$, with one at -0.002 sitting on the proxy itself. A positive correction puts the office's standard short of the proxy, i.e., at a less protective buffer; a negative correction puts it past the proxy. The same section prints, under each correction, the buffer it implies for bird nests. The proxy protects a nest to 760 m. The two positive corrections put the standard at 430 m ($\eta = +0.29$) and 230 m ($+0.43$). The correction at -0.16 puts it at 940 m; and the two most negative corrections (-0.52 and -0.69) put it beyond 1.5 miles (2,414 m), the end of the nest's ecologically relevant range.

If the corrections locate standards, which exist outside any one office and are available to all offices, two predictions follow, and both hold in the fitted model. *First*, there are only so many standards to cite, so a few corrections should be enough. The fit finds six suffice. The number of corrections is not set in advance. As Section 4.3 describes, I add corrections one at a time until the next one carries a negligible share. The search stops at six, for eleven offices and 135 office-by-feature type pairs, because a seventh would carry only 2% of the mixing shares, all of it at one office, Grand Junction.[27] *Second*, offices should rely on the same corrections. If a correction instead captures an office's own idiosyncratic taste for conservation, each office would have a correction of its own, and nothing would make those tastes fall on a few shared points instead of spreading over a continuum. Consistent with this prediction, offices do rely on the same corrections. Figure A8's rows plot each office's mixing shares over the six corrections: the estimated probability that the office anchors a decision to the standard that correction implies.[28] The model estimates that six of the eleven offices anchor a decision to the correction $+0.29$ with a probability of at least 50%. Three offices anchor to the correction -0.16 with a probability of at least 25%. Five anchor to the proxy itself with probabilities ranging from 2% to over 60%. Three offices anchor to the correction $+0.43$ and two to the corrections -0.52

[26] New Mexico received roughly $2.3 billion in federal mineral leasing payments in fiscal year 2022, which fund its general fund and its Early Childhood Trust Fund (New Mexico Legislative Finance Committee 2023). When the Biden administration paused new oil and gas leasing on federal lands in 2021, New Mexico's two Democratic senators asked the White House to compensate the state for the revenue it stood to lose (Volcovici & Renshaw 2021).

[27] Going from five corrections to six raises the log-likelihood by 89 points, and going from six to seven by 39 points, each for 12 added parameters: one η and eleven office shares.

[28] The shares are estimated through a softmax, which can approach zero or one but never reach them. The shares the figure leaves blank are below 0.1%, and all but one of them are 10^{-11} or smaller; a circle marked 100% is one to the same precision.

and -0.69. Averaged across the eleven offices (Figure A8's bottom row), the mixing share is 12% on the proxy itself, 60% on the two corrections short of it, 27% on the three past it, and 2% on the open type.[29]

One might object that the corrections are few and discrete by construction. I assume a discrete distribution, a shared support forces offices onto common points, and the stopping rule caps how many corrections there are. The model fit contests this objection. If decisions were spread smoothly across the range, as they would be if there were no standards to cluster around, a handful of corrections would approximate the spread by spacing themselves evenly across it, and each added correction would move the others to cover the range more finely. The fit shows neither. First, the corrections are not evenly spaced: three sit past the proxy, at -0.69, -0.52 and -0.16, one sits on it, and two sit short of it, at $+0.29$ and $+0.43$, with gaps between neighbors from 0.14 to 0.36. Second, the corrections do not move as new ones are added. Going from five corrections to six, the new correction sits at -0.69, beyond the others; the corrections from the five-correction fit move by at most 0.01. Similarly, going from six to seven, five of the six stay within 0.003 and the largest move is 0.04. In summary, the fit confirms what the planning record suggests: offices anchor their choices on standards, and this confirmation is not by construction.

The cost of departure, κ, is the revenue an office gives up to stay close to those standards. The fit finds offices do pay to stay close, in two senses: the costs are not zero, and they are large enough that predicted choices often forgo an alternative with a higher net benefit to stay closer to the standard. *First*, the cost cannot be negative in estimation but it can be zero; only one of the twenty-one estimates sits at zero, and eighteen are distinguishable from zero at the ten percent level. Table 1's final four columns report the cost of departure (κ), for soil and water together and for wildlife. The first and third columns report κ itself, for the two non-market resource groups respectively. Like β, κ is reported in dollars of forgone revenue. Specifically, κ's units are revenue forgone per unit of the squared distance between the chosen buffer's relevance and the office's standard, $\overline{w}_{-i,f} + \eta_{i,f}$. The second and fourth columns price the departure cost each office actually pays, the annual dollar-equivalent cost on an average protective decision. Priced at the departure each office actually makes, the soil-and-water penalty runs from $\$4k$ at Rio Puerco and $\$31k$ at Four Rivers to $\$1m$ at Little Snake and $\$2m$ at Grand Junction. On the wildlife side, it runs from $\$6k$ at Uncompahgre and Grand Junction to $\$86k$ at Casper and $\$2m$ at Buffalo. Lander's wildlife penalty is larger still but imprecise: its standard error exceeds the estimate. At the other extreme, Four Rivers bears no cost of departure for wildlife; the estimate sits at $\kappa = 0$, the edge of the parameter space, and with the other parameters held at their estimates the likelihood falls for every positive value, with a one-sided 95% bound of $36 per unit of squared gap.[30]

Second, predicted choices often forgo an alternative with a higher net benefit. Figure A9 plots the share of each field office's decisions in which a buffer adjacent to the *predicted* buffer has a higher net benefit. The figure includes the decisions for which the fitted model predicts protection. The predicted buffer is then the protective alternative with the highest predicted probability. Its adjacent buffers are the alternatives under the same restriction one step narrower and one step wider. Adjacent buffers differ only in net benefit and in the cost of departure, so without a cost of departure no adjacent buffer would have a higher net benefit than the predicted one, and every share in Figure A9 would be zero. Instead, at the median office an adjacent

[29] The open-type share s_{i0} is below one percent at eight of the eleven offices; Colorado River Valley carries the most, at 8%.

[30] The bound is the value of κ at which the log-likelihood, with every other parameter held at its estimate, has fallen 1.353 points below its value at $\kappa = 0$, the one-sided 95% likelihood-ratio cutoff ($\chi^2_1(0.90)/2$); the bound is one-sided because κ cannot be negative.

buffer has a higher net benefit in 69% of decisions: 71% for soil, 66% for water and 75% for wildlife. Across offices the share runs from 44% at Royal Gorge to 94% at Buffalo. In each of these decisions, the cost of departure holds the predicted buffer in place; moving to the adjacent buffer would raise the cost of departure by at least the gain in net benefit. Table 2 shows that the gain in net benefit from an adjacent buffer is frequently minimal, but sometimes quite large. At the median office the gain exceeds \$100 a year in 49% of decisions, \$1k in 32% and \$10k in 12%. The large gains are concentrated in a few offices: the gain exceeds \$10k a year in 77% of decisions at Buffalo and 64% at Little Snake, against 3% at Four Rivers and Rio Puerco.

5.3 The restriction margin

The cost of departing from standards captures an office's habit of picking particular buffers. The closure fixed effect γ_i captures a second habit, how much more or less inclined an office is to close a feature outright rather than restrict development with a discretionary stipulation, whatever the net benefits. Table 1 reports the estimated closure fixed effect for each office. It is negative at the four Wyoming offices and at Rio Puerco, NM. It is positive at the five Colorado offices and in Four Rivers, ID. As with willingness to pay, the habit follows state politics. The five Colorado offices, in a state that leans toward conservation, are inclined to close; the four Wyoming offices, in a state that leans toward development, are inclined not to. To show how much this habit matters for predicted choices, Figure A10 plots, for each office, the share of protective decisions that are closed as realized, as the fitted model predicts, and as the model predicts with the office's closure fixed effect set to zero. Without the fixed effect, every office but Little Snake would close between 46% and 56% of the features it protects. The fixed effect is what moves Rock Springs from 52% predicted closure without it to 0.3% with it—three closures among 1,035 protected features—and Colorado River Valley from 49% to 84%.

In conclusion, offices respond to the benefits and costs of conservation, and they tie their decisions to standards, treating a departure from established standards as a cost. BLM headquarters could therefore change what offices protect in two ways: by recommending a specific level of protection (setting a standard), or by changing the value offices place on avoided disturbance (setting a value for conservation). Section 6 compares the implications of these two policy instruments.

6 Policy Implications

In this section, I bring the estimates together to examine how economic tradeoffs and established conservation standards jointly shape allocation decisions. I first consider a counterfactual in which every BLM office responds to the Fish and Wildlife Service's highly protective one-mile buffer recommendation for birds' nests, holding all other model primitives fixed. This exercise quantifies how adopting a more protective recommendation at BLM headquarters could change conservation outcomes and costs, measured by additional birds protected and revenue forgone. I then ask whether a more traditional price-like instrument, which sets a common value for avoiding disturbance around nests, could achieve the same conservation gains at lower cost. Finally, I compare the cost-effectiveness of both policy instruments to a benchmark: the outside option cost of saving a bird through retrofitting power poles.

6.1 Revising the standard

The first counterfactual moves the nesting standard to the Fish and Wildlife Service's (FWS) one-mile recommendation. The FWS's primary responsibility is wildlife conservation, unlike the BLM, which must balance development and conservation under its multiple-use mandate. Its recommendation therefore provides a more conservation-oriented standard for BLM offices. This counterfactual asks what would happen if BLM headquarters encouraged offices to use the FWS recommendation as their established standard. This contrasts with existing BLM guidance to use "the least restrictive [policy] that effectively accomplishes the resource objectives" (Bureau of Land Management 2013).

I consider two implementations of this counterfactual, *heterogeneous adoption* and *uniform adoption*, reflecting different assumptions about how offices would respond to new guidance from BLM headquarters. Under *heterogeneous adoption*, I assume $\eta_{i,f}$ reflects persistent differences due to local practices or ecological conditions, so offices may continue to follow different conservation standards despite receiving the same recommendation. I therefore set the nesting proxy to one mile, i.e., $\overline{w}_{-i,f} = w_f(1 \text{ mile})$, while retaining the office- and nest-specific correction $\eta_{i,f}$. Under *uniform adoption*, I assume $\eta_{i,f}$ reflects differences in the guidance available to offices at the time of decision-making, so the recommendation eliminates these differences and all offices respond to the same standard. I therefore set the entire nesting standard to one mile, i.e., $\overline{w}_{-i,f} + \eta_{i,f} = w_f(1 \text{ mile})$ for all offices and nests. Because I cannot empirically determine which assumption is more accurate, I report counterfactual results under both.

I measure conservation gains in expected baby birds (fledglings) protected, rather than acres restricted. Following FWS guidance, my accounting assumes a nest produces two baby birds in the absence of nearby drilling. I use the plans' forecasts of drilling near each nest to calculate the expected number of fledglings under each alternative, assuming independent well placements. Figure A11 plots the changes in baby birds protected and revenue forgone under the two implementations of the revised standard, relative to the baseline. Under heterogeneous adoption, approximately 1,300 additional fledglings are protected at a cost of $337 million in revenue forgone, or $254,000 per additional fledgling. Under uniform adoption, approximately 240 more fledglings are protected for less than $10 million in additional revenue forgone, reducing the average cost to $220,000 per fledgling. The difference between the two implementations illustrates how the cost of departing from standard practice can leave relatively inexpensive birds unprotected while keeping relatively expensive birds protected. Under heterogeneous adoption, the office- and nest-specific correction $\eta_{i,f}$ shifts the standard offices anchor on relative to the common one-mile recommendation. In Buffalo, WY, this correction makes the standard less protective than the recommendation, while in Casper and Lander, WY, it generally makes the standard more protective (see Figure A8). Under uniform adoption, these corrections are eliminated, making Buffalo's standard more protective and those of Casper and Lander less protective than under heterogeneous adoption. Consequently, Buffalo protects more birds, while Casper and Lander protect fewer. The additional birds protected in Buffalo are relatively inexpensive to protect. Its less protective standard, rather than high revenue costs, had discouraged their protection. Conversely, the birds no longer protected in Casper and Lander were relatively expensive to protect. These offices had protected them despite their high revenue costs because reducing protection would have incurred large departure costs. Uniform adoption therefore redirects conservation toward birds that are less costly to protect, increasing conservation while reducing its average cost.

This counterfactual is a back-of-the-envelope exercise intended to illustrate how revising conservation

standards could change allocation decisions. I use a one-mile buffer as a deliberately protective interpretation of FWS guidance for nests in the estimation sample. The FWS recommends one mile for nests of unidentified raptors, ferruginous hawks, and bald eagles, but one-half mile for golden eagles, falcons, and goshawks and one-quarter mile for Swainson's and red-tailed hawks (U.S. Fish and Wildlife Service 2022). Many nests in the planning record lack species identification, and ferruginous hawks predominate among identified nests. Thus, the one-mile buffer matches FWS guidance for much of the sample but exceeds it for some identified species. The results therefore represent the conservation gains and revenue costs of applying this protective one-mile standard uniformly, rather than following FWS guidance species by species.

6.2 Setting the value of conservation

The second policy counterfactual changes the value offices place on avoiding wildlife disturbance, holding all else fixed, including the nesting standard. This counterfactual considers a more traditional price-like policy instrument. Rather than recommending a buffer distance, BLM headquarters would prescribe a dollar value for avoided wildlife disturbance, much as the federal government has historically prescribed a social cost of carbon for agencies to use in their analyses. In the model, this intervention changes $\beta_{i,\text{wildlife}}$, the annual revenue office i is willing to forgo to avoid one relevance-weighted acre of wildlife disturbance.

I consider two implementations of this counterfactual, a *minimum value* and a *common value*, reflecting different assumptions about how headquarters would treat offices that already value wildlife more than it recommends. Under a *minimum value*, headquarters sets a floor $\underline{\beta}$; offices with valuations below this floor must raise them to $\underline{\beta}$, while offices already above it retain their valuations, i.e., $\beta^{\text{cf}}_{i,\text{wildlife}} = \max\{\beta_{i,\text{wildlife}}, \underline{\beta}\}$. Under a *common value*, headquarters requires all offices to use the same valuation, raising some offices' valuations and lowering others.

I calibrate both implementations to protect 1,327 additional baby birds; the number of birds protected under the heterogeneous adoptions revised standard counterfactual. This calibration allows me to compare the costs of achieving the same conservation gains under the two types of policy instruments. Calibrating the valuations is straightforward because the number of additional baby birds protected increases monotonically with the prescribed valuation. Under the *minimum value* implementation, setting the lower bound on offices' valuations at \$212,000 per relevance-weighted acre protects 1,327 additional baby birds, the conservation gains under the revised standard. Under the *common value* implementation, setting all offices' valuations to \$240,000 per relevance-weighted acre achieves the same conservation gains. As expected, the calibrated common value is higher than the minimum value because it also reduces valuations in offices initially above the prescribed value, causing them to protect fewer birds. Therefore, to achieve the same aggregate conservation gains requires greater protection in offices with lower initial valuations.

Figure 6 plots the changes in fledglings protected and revenue forgone under the two price-like instruments alongside the revised standard. Both price-like instruments achieve the same aggregate conservation gains at approximately the same revenue cost as the revised standard, but differ substantially in where birds are protected. Under the standard revision, Buffalo accounts for 680 of the 1,327 additional fledglings protected. Under the minimum value counterfactual, Buffalo protects no additional birds, while Rock Springs, Casper, and Lander, WY, protect far more than under the standard revision. This difference arises because Buffalo already values avoiding wildlife disturbance at more than \$321,000 per relevance-weighted acre, substantially higher than every office except Royal Gorge, CO (see Figure 4). A price-like instrument would therefore

need to prescribe a value above $321,000 to induce additional protection in Buffalo. At that value, however, offices like Rock Springs, Casper, and Lander would protect far more fledglings than needed to achieve the aggregate conservation gains of the revised standard. Under the common value counterfactual, the contrast is even starker. Buffalo protects even fewer birds, while Rock Springs, Casper, and Lander protect slightly more than under the minimum value counterfactual. These differences in where birds are protected explain why the revised standard is as cost-effective as the price-like instruments. As discussed above, Buffalo has many birds that could be protected at relatively little revenue cost but remain unprotected because of its less protective standard. Revising that standard induces protection of these relatively inexpensive birds. In contrast, a common valuation cannot induce protection of these birds without also inducing substantially more protection in other offices. Thus, the cost of departing from standard practice can make a revised standard as cost-effective as a price-like instrument, despite the latter's theoretical ability to direct conservation toward offices where it is less costly. This finding holds under both assumptions about how offices respond to headquarters' recommendation. When calibrated to match the 1,566 additional fledglings protected under uniform adoption, both price-like instruments again concentrate additional protection in Casper, Rock Springs, and Lander rather than Buffalo, making them no more cost-effective than the revised standard (see Figure A12).

6.3 Alternative mitigation as a benchmark

The preceding comparisons establish that revising a conservation standard can be at least as cost-effective as prescribing a social value. However, neither instrument is necessarily the least-cost way to protect birds. To assess this, I compare their costs with an alternative form of mitigation. The FWS estimates that offsetting the expected death of one bird (raptor in this region) costs approximately $120,000, equivalent to retrofitting sixteen power poles to prevent electrocutions (U.S. Fish and Wildlife Service 2024). Applying the FWS's resource-equivalency accounting to disturbed nests implies mitigation costs of $91,000 to $130,000 per baby bird (fledgling) (U.S. Fish and Wildlife Service 2018*b*, U.S. Fish and Wildlife Service 2018*a*). Measured against this benchmark, each policy instrument forgoes approximately twice as much revenue per additional baby bird as the estimated cost of mitigation through power-pole retrofits. For example, protecting 1,327 additional baby birds annually under the revised standard costs $337 million in forgone revenue, compared with approximately $159 million for equivalent mitigation at $120,000 per bird.

I also use this benchmark to examine how conservation would change if BLM headquarters prescribed $120,000 as the value of protecting a baby bird, implementing the common value counterfactual from the previous subsection at the FWS benchmark. To implement this counterfactual, I translate the value per fledgling into the model's units of dollars per relevance-weighted acre. Following the FWS buffer guidance, my accounting treats drilling a well at a nest as costing a brood of two fledglings. Thus, a value of $120,000 per fledgling implies that offices should be willing to forgo $240,000 to prevent a well from being drilled at a nest. The expected surface disturbance per well ranges from 2 to 16 acres across field offices. I therefore divide $240,000 by the expected disturbance per well for each office to obtain the corresponding valuation per relevance-weighted acre.

The left panel of Figure A13 applies the reverse conversion to express offices' estimated willingness to pay in dollars per fledgling, allowing for a more intuitive comparison with the FWS benchmark. Six of the ten offices have estimated willingness to pay above the benchmark, two have estimates near it, and only two

have estimates meaningfully below it. This suggests that setting a common value of $120,000 per fledgling would reduce conservation. The right panel confirms this result, plotting the change in fledglings protected when all offices are assigned this common value, relative to the baseline (gray). Buffalo protects more than 500 fewer fledglings, while the largest increase in protection at any office is just 12 fledglings. Thus, relative to the FWS mitigation benchmark, existing allocation decisions produce more conservation than a common social value of $120,000 would induce. Most strikingly, Buffalo accounts for the largest conservation gains under a revised standard but also the largest losses when offices are assigned the FWS mitigation value.

These comparisons are back-of-the-envelope exercises. They apply the FWS's mitigation cost for a golden eagle to nests of every raptor species in the sample, using the golden eagle as a high-end benchmark. They are intended to provide intuition for how offices' estimated valuations, average costs of protection under the two policy instruments, and mitigation costs compare, not to establish whether offices' allocation decisions are socially efficient. A key concern, particularly when using the mitigation benchmark as a common social value, is whether it understates the social value of protecting a bird. However, other available valuation estimates provide limited grounds for prescribing a substantially higher value.[31] That said, the benchmark-based conclusions are sensitive to assumptions about the effectiveness of alternative mitigation. The FWS's $120,000 benchmark assumes power-pole retrofits remain effective for 30 years. If retrofits instead last 10 years, the estimated mitigation cost rises to approximately $250,000 per fledgling, comparable to the $254,000 per fledgling under the revised standard. Using this higher benchmark as a common social value also leaves conservation nearly unchanged, rather than substantially reducing it. Thus, while the revised standard remains at least as cost-effective as the price-like instruments under both assumptions about adoption, conclusions about their costs relative to alternative mitigation depend on the assumed effectiveness of that mitigation.

7 Conclusion

This paper studies how the Bureau of Land Management (BLM) allocates federal land between development and conservation, a question that has received little attention from economists (Libecap 2026). I estimate how much development revenue local offices are willing to forgo for conservation. I find that their choices depend on both the local benefits and costs of conservation and adherence to standard practice.

The paper builds on work that recovers public decision makers' objectives from their choices (Bodéré et al. 2026, Kashner 2026, Slattery 2025, Parkhomenko 2023). My approach differs in how I construct the alternatives available to each office and the information it had when deciding. Federal regulations require BLM offices to document these alternatives and their forecasts. I use large language models to process roughly 90,000 pages of planning records from 11 local offices. The resulting dataset contains 12,509 allocation decisions. For each decision, I recover the alternatives considered and use the office's own forecasts to calculate the expected revenue forgone and disturbance prevented under each. This allows me to recover the tradeoffs offices faced when deciding, rather than infer them from outcomes years later.

At the median office, willingness to pay for conservation is about $60,000 in annual revenue per acre

[31] O'Brien & Hagerty (2025) provide two alternative per-eagle valuation benchmarks: $127,000 based on costs imposed through criminal enforcement, including fines and compensatory mitigation, and $22,100 obtained by allocating stated willingness to pay for the existence of bald eagles across the population. Neither directly estimates the marginal social value of an additional fledgling.

conserved. This value varies substantially across offices and across non-market resources within each office. Such variation may reflect differences in the environmental damage that disturbance would cause, differences in how offices value avoiding that damage, or both. I provide suggestive evidence that differences in local preferences help explain the heterogeneity across offices; willingness to pay is negatively correlated with Republican vote share, which is positively associated with preferences for drilling over conservation on public lands (Colorado College State of the Rockies Project 2017, Colorado College State of the Rockies Project 2020). In contrast, willingness to pay for protecting fragile soils is essentially uncorrelated with predicted erosion, a proxy for the environmental damages from disturbance. I cannot easily conduct a comparable analysis for water and wildlife because planning documents identify numerous environmental benefits that are difficult to capture in a single comparable measure.

Benefits and costs alone do not explain offices' choices. On the extent margin, offices treat departures from standard practice as costly. At the median office, the buffer with the highest predicted choice probability yields lower net benefits than an adjacent buffer in 70% of decisions. Offices also differ in how willing they are to close land to development, even after accounting for the benefits and costs of conservation. All four Wyoming offices tend to avoid closure, while all five Colorado offices tend to favor it. This follows the same broad political pattern as willingness to pay for conservation, with Republican-leaning Wyoming less inclined to prohibit development than Democratic-leaning Colorado. These tendencies make a large difference. Without them, the model predicts that all but one office would choose closure in roughly half their allocation decisions. With these tendencies, closure rates range from 0.3% in Rock Springs, WY, to 84% in Colorado River Valley, CO.

Offices' adherence to established standards, in addition to their response to the benefits and costs of conservation, has two implications for policy. First, BLM headquarters can steer local decisions by recommending conservation standards. Second, adherence to established standards can make price instruments less cost-effective because some offices stick to less protective standards even when protecting more would be worth the cost. I demonstrate these implications by comparing i) a recommended one-mile no-drilling buffer around bird nests with ii) price instruments that specify a dollar value for avoiding wildlife disturbance, either as a floor under offices' own valuations or as a common value for all offices. Counterfactuals show that the recommended standard can be as cost-effective as either price instrument in achieving the same conservation gains. Revising the standard can encourage inexpensive conservation that a price instrument reaches only by inducing more costly protection elsewhere. That a recommended standard can be as cost-effective as a price instrument is striking, but it does not establish that either policy is the best way to protect birds. The Fish and Wildlife Service can protect a bird for about $120k, roughly half the cost per bird under either land-use policy.

More generally, my findings suggest that how standards develop, spread, and influence future decisions deserves further study, as their influence may be far greater than any one decision. For example, BLM headquarters already uses programmatic plans to steer decisions across local offices. These plans focus on a particular resource or activity, such as protecting sage grouse habitat or facilitating wind development. Debate over programmatic plans tends to focus on the allocations they change when adopted. My findings suggest that their influence extends beyond these initial changes, as offices revising their own plans treat programmatic plan's allocations as standards they must justify departing from.

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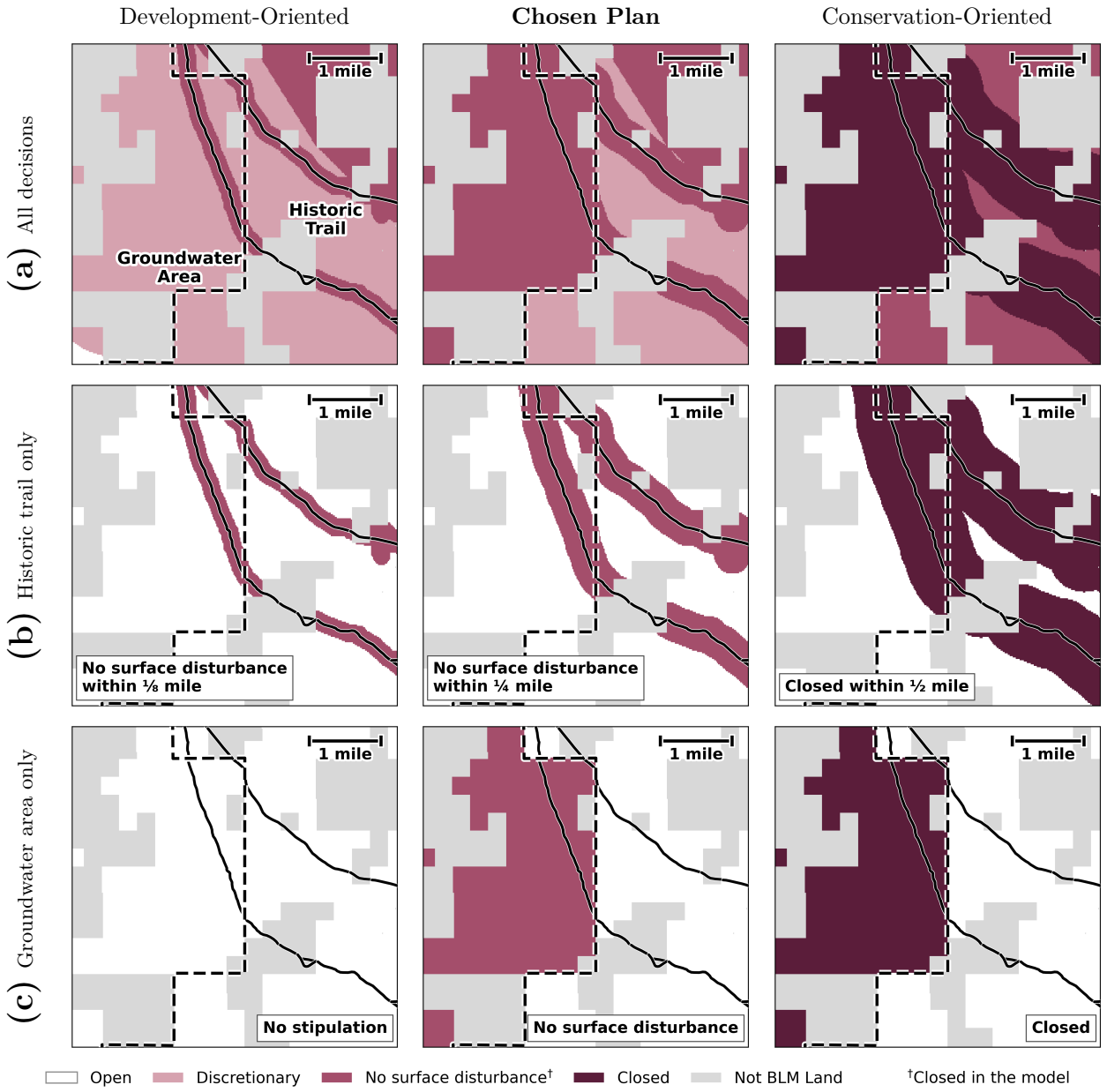
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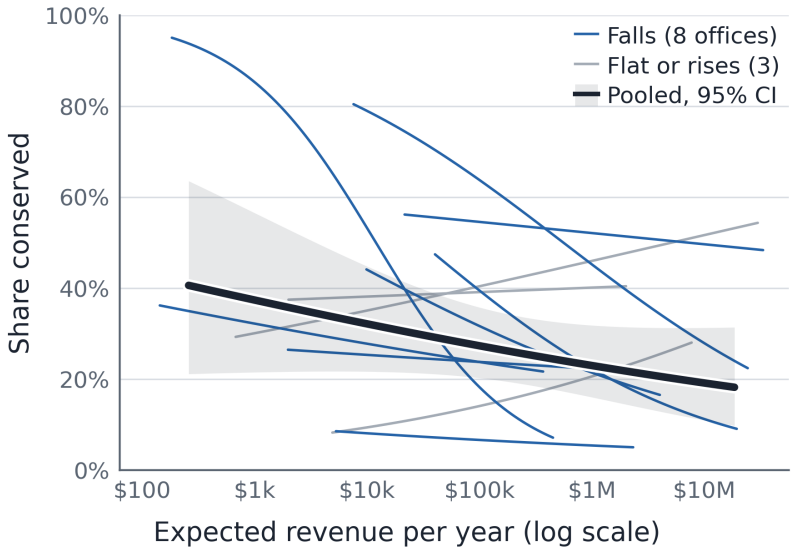
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Figure 1: Example of Considered Allocation Decisions



Notes: The nine maps show one area of the Four Rivers, Idaho planning area. The columns are three sets of allocation decisions the office considered: a development-oriented one (left), the chosen plan (middle), and a conservation-oriented one (right). Row (a) is every decision the plan makes there, which together protect four features: the Oregon National Historic Trail (solid line), a critical groundwater area (dashed line), elk winter range (the discretionary ground covering most of the left column), and sage-grouse habitat (the no-surface-disturbance ground at upper right in the left column). Rows (b) and (c) keep only the allocation decisions tied to the trail and to the groundwater area; tile labels paraphrase the plan. Darker fills are more restrictive: *Discretionary* allows drilling with limits on where and when; *No surface disturbance* bars surface drilling but allows wells angled in from adjacent land (Closed in the model; Section 3.2.4); *Closed* removes the land from leasing. White is land left *Open*—in rows (b) and (c), open with respect to the feature shown; grey is land outside the BLM's jurisdiction.

Figure 2: The Share Conserved vs. the Development Potential

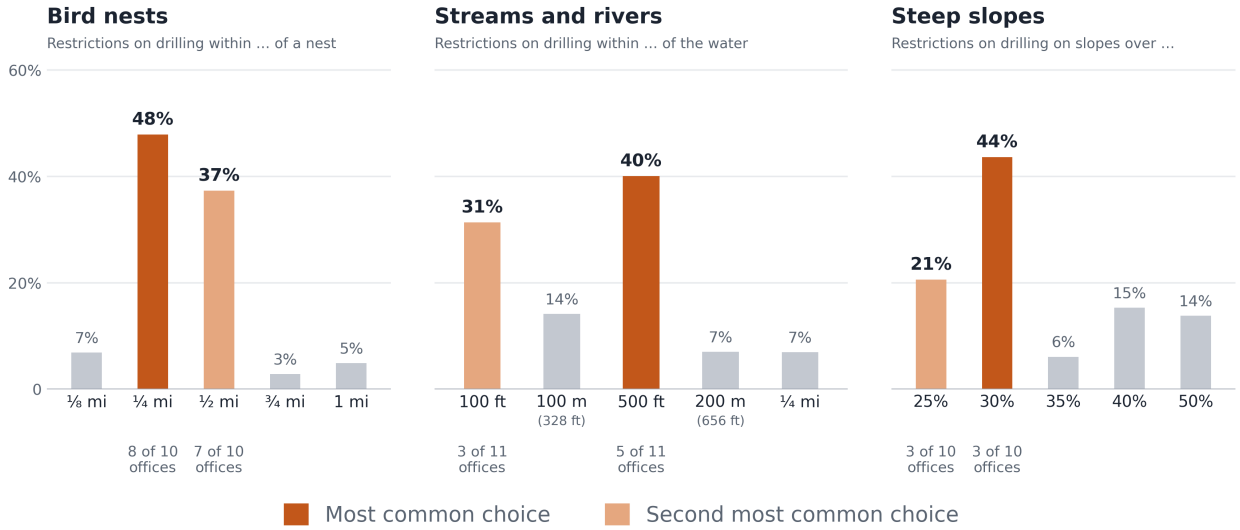


Notes: Figure plots the share of each township[a] conserved against the township's expected revenue from fluid leasable development.[b] Each colored line is the relationship estimated by a logistic regression of the share conserved on log revenue, fit separately for each office. The horizontal axis is development potential: the expected average annual revenue within a township if the township were to be developed, summed across fluid resources, deflated to 2014 dollars and plotted on a log scale. Blue curves fall as expected revenue rises (8 of the 11 offices); gray curves are flat or rise (Buffalo, Wyoming, and Colorado River Valley and Uncompahgre, Colorado). The black curve pools the townships of all eleven offices, weighting each township equally; its shaded area is a 95% confidence interval with standard errors clustered by office.

[a]Townships (6x6 mile squares) are standard land-survey units in the Western United States.

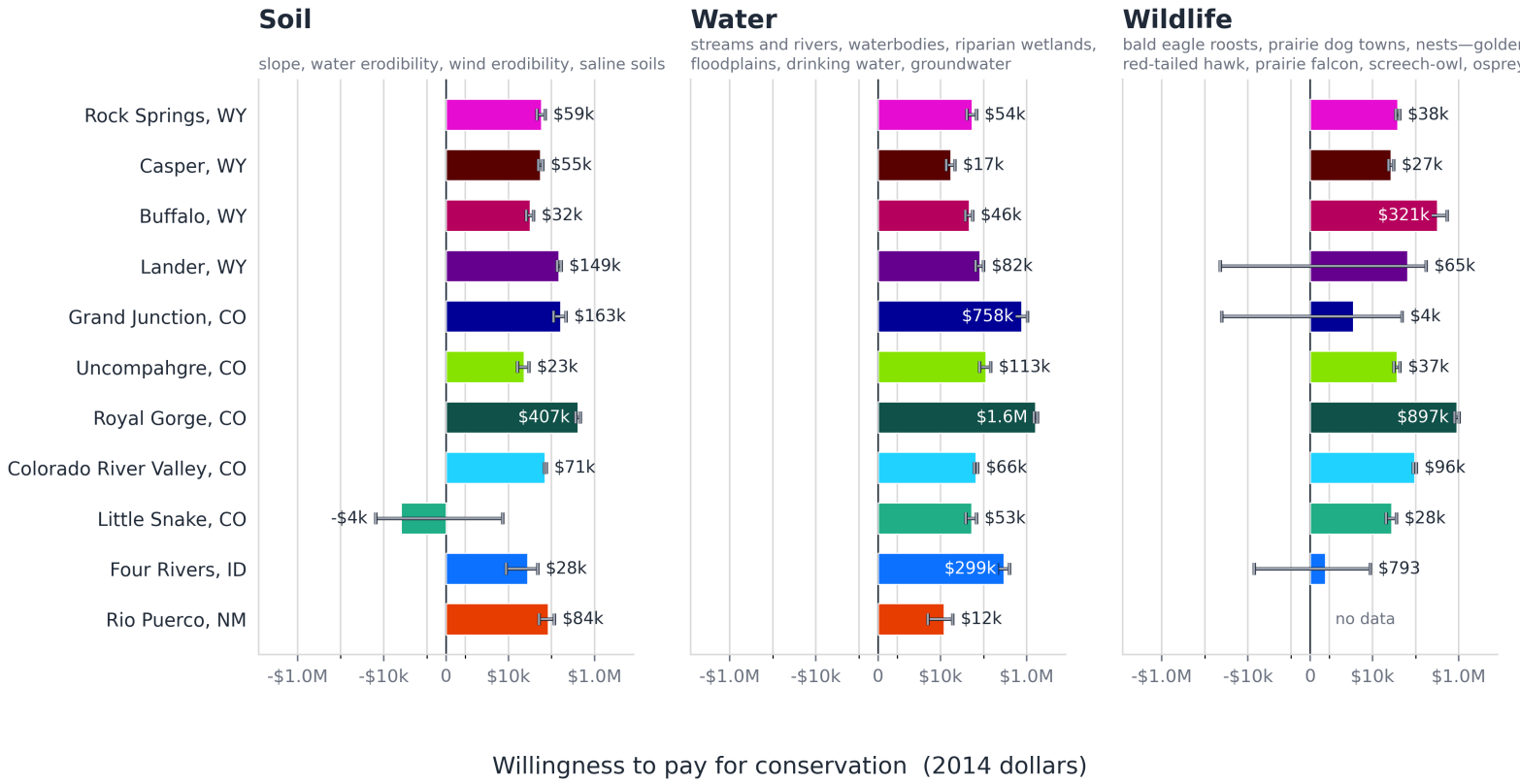
[b]The fluid leasable resources are conventional oil and gas, coalbed natural gas, and geothermal, whichever of the three the plan projects development for.

Figure 3: Distribution of Chosen Protections for Three Kinds of Feature



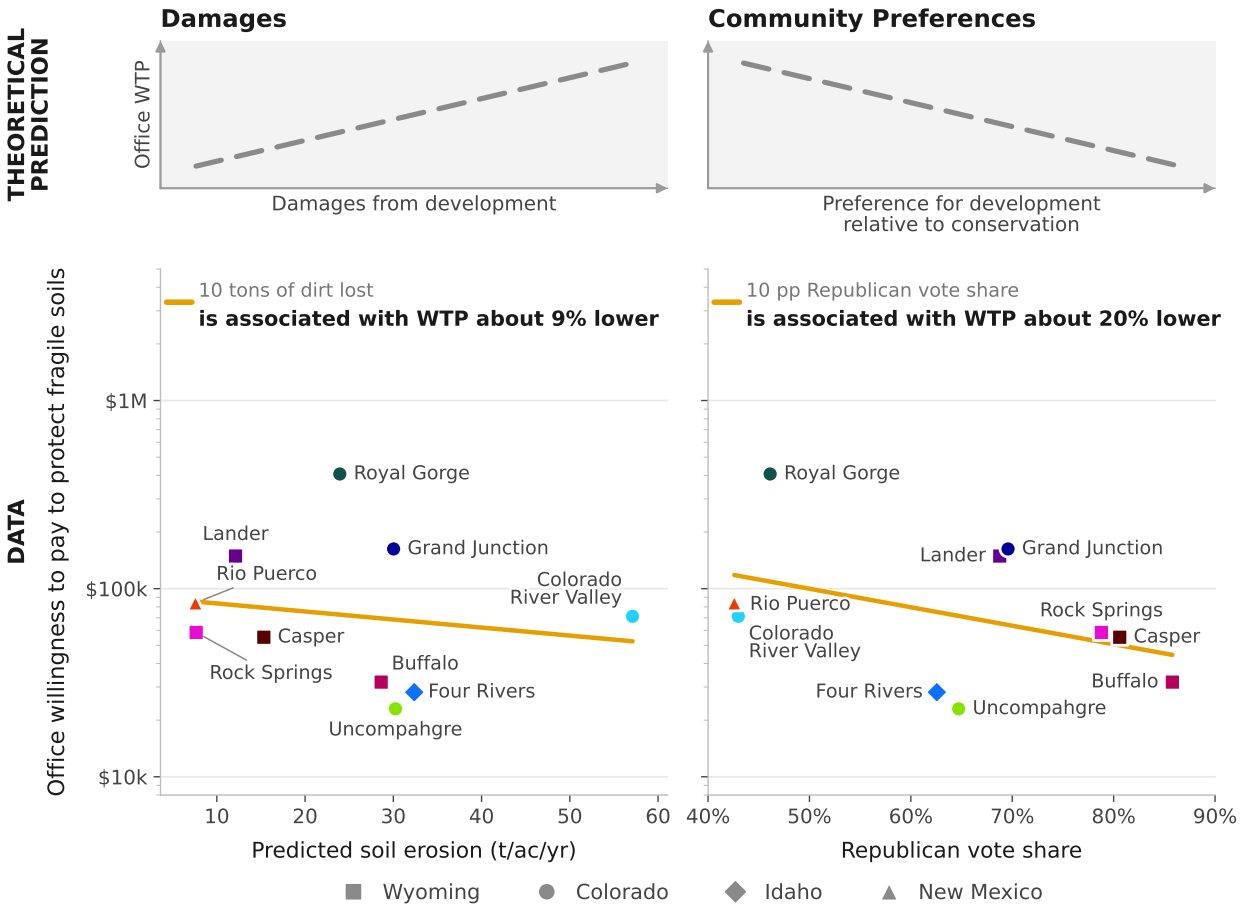
Notes: Figure plots the protections chosen for three types of features across all offices in the estimation sample. From right to left, the panels plot the slope grade above which drilling is restricted, the restricted distance from streams and rivers, and the restricted distance around bird nests. The x-axis shows the chosen extent in each plan's original units, and the y-axis shows the share of protective decisions choosing that extent. Labels below the x-axis show how many offices chose each extent for at least one feature. Features not protected in the chosen plan are excluded from the figure. The displayed bars account for at least 99% of protective decisions in each panel.

Figure 4: Willingness to Pay for Conservation, by Field Office and Feature Group



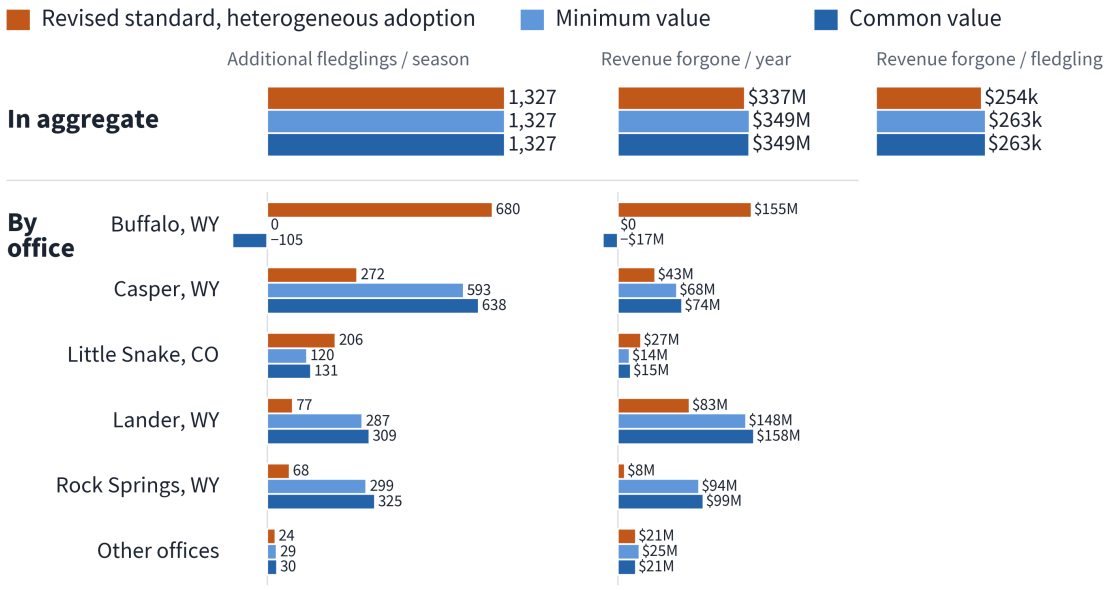
Notes: Figure plots each field office's estimated willingness to pay for conservation, in dollars of forgone revenue. Each bar is the estimate for one office and one non-market resource class; the whiskers are 90% confidence intervals from the Hessian standard errors. The horizontal axis is signed-log, matching Figure A4. "no data" marks an office×class group with no data.

Figure 5: Explanations for Heterogeneity in Office Willingness to Pay



Notes: **Top row** presents schematic relationships (not real data) illustrating the expected signs of the correlations if differences in environmental damages or local preferences contribute to heterogeneity in offices' willingness to pay. On the left, offices should be willing to pay more to avoid disturbance where the environmental damages from disturbance are greater. On the right, offices should be willing to pay less where local residents prefer development over conservation (Section 5.1). **Bottom row** examines these predictions using proxies for environmental damages (left) and local preferences (right). Both panels plot each field office's estimated willingness to pay to avoid disturbing one relevance-weighted acre of fragile soils (Figure 4, soil) against an office characteristic. The left panel's horizontal axis is average erosion potential (Section G; Table A3, column 5). The right panel's horizontal axis is the share of the 2016 two-party presidential vote cast for Donald Trump in the office's planning area (Voting and Election Science Team 2018). Republican vote share serves as a proxy for local preferences regarding drilling versus conservation on public lands (Colorado College State of the Rockies Project 2017, Colorado College State of the Rockies Project 2020). Each marker represents one office, with its shape denoting the state. The panels include the ten soil WTP estimates statistically distinguishable from zero; Little Snake, CO is omitted. Yellow lines show least-squares fits of log willingness to pay on the horizontal variable, with slopes reported in each panel. These relationships are correlational and do not identify causal effects.

Figure 6: Protecting the Same Number of Birds with a Revised Standard or Price-Like Instruments



Notes: Figure plots changes in baby birds (fledglings) protected and revenue forgone under the standard revision and price-like instrument counterfactuals, relative to the baseline. Dark orange bars show the standard revision under *heterogeneous adoption*, in which all offices respond to a common one-mile recommendation but retain office- and nest-specific unobserved heterogeneity in the conservation standards they ultimately follow. Light and dark blue bars show two implementations of the price-like instrument, both calibrated to protect the same aggregate number of additional fledglings as the standard revision. The *minimum value* implementation (light blue) raises each office's value of avoiding wildlife disturbance to at least $212,000 per relevance-weighted acre, while the *common value* implementation (dark blue) sets every office's value to $240,000. The top panel reports aggregate changes, while the bottom panel breaks them out by office. The first column shows changes in fledglings protected, the second changes in revenue forgone, and the third, appearing only in the top panel, revenue forgone per additional fledgling. The "Other offices" category includes Uncompahgre, Colorado River Valley, Royal Gorge, and Grand Junction in Colorado, and Four Rivers in Idaho.

Table 1: Willingness to Pay and Cost of Departure, by Field Office

	Willingness to pay ($/wtd. acre)			Closure FE (logits)	Scale ($\log\sigma$)		Cost of departure (κ)			
							Soil+Water		Wildlife	
Field office	Soil	Water	Wildlife	Any closed label	Soil+Water	Wildlife	$ per unit departure	at the mean departure	$ per unit departure	at the mean departure
Rock Springs, WY	\$58,534***	\$53,944***	\$38,263***	−7.64***	3.73	1.34	\$2,408,524	\$299,692	\$353,696	\$10,918
	(\$7,508)	(\$8,065)	(\$2,513)	(1.29)	(0.223)	(0.336)	(\$652,793)	*at 0.353*	(\$125,802)	*at 0.176*
Casper, WY	\$55,185***	\$17,412***	\$26,966***	−1.92***	3.45	2.91	\$1,960,468	\$52,143	\$1,948,109	\$86,143
	(\$3,783)	(\$2,348)	(\$2,158)	(0.0783)	(0.113)	(0.228)	(\$299,093)	*at 0.163*	(\$466,637)	*at 0.210*
Buffalo, WY	\$31,858***	\$46,315***	\$321,292**	−1.75***	4.12	6.18	\$7,968,144	\$568,432	\$28,681,361	\$2,120,810
	(\$3,625)	(\$5,412)	(\$125,336)	(0.0582)	(0.169)	(0.496)	(\$1,621,067)	*at 0.267*	(\$14,305,417)	*at 0.272*
Lander, WY	\$149,074***	\$82,070***	\$65,302	−0.385***	4.28	6.54	\$3,315,653	\$130,460	\$833,198,799	\$4,285,935
	(\$9,200)	(\$10,380)	(\$66,252)	(0.104)	(0.130)	(1.07)	(\$590,033)	*at 0.198*	(\$909,503,317)	*at 0.072*
Grand Jct., CO	\$162,631***	\$758,014***	\$3,630	1.46***	5.55	0.584	\$37,995,229	\$2,059,150	\$105,624	\$6,450
	(\$31,181)	(\$173,148)	(\$26,853)	(0.164)	(0.234)	(0.813)	(\$9,634,168)	*at 0.233*	(\$93,117)	*at 0.247*
Uncompahgre, CO	\$22,983***	\$112,826***	\$36,994***	0.755***	3.21	0.351	\$2,851,967	\$231,246	\$101,440	\$6,174
	(\$4,110)	(\$19,503)	(\$3,395)	(0.119)	(0.257)	(0.343)	(\$946,023)	*at 0.285*	(\$37,081)	*at 0.247*
Royal Gorge, CO	\$406,979***	\$1,605,124***	\$897,196***	1.64***	2.80	2.39	\$235,181,741	\$38,065	\$1,766,854	\$53,364
	(\$30,271)	(\$97,120)	(\$68,596)	(0.0622)	(0.0840)	(0.385)	(\$22,437,427)	*at 0.013*	(\$699,877)	*at 0.174*
Colo. River	\$71,295***	\$66,425***	\$95,551***	1.93***	3.83	1.29	\$2,207,466	\$331,449	\$7,869,325	\$25,286
Valley, CO	(\$2,054)	(\$3,614)	(\$4,167)	(0.149)	(0.213)	(0.481)	(\$523,061)	*at 0.387*	(\$3,920,763)	*at 0.057*
Little Snake, CO	−\$3,990	\$53,275***	\$27,964***	1.36***	4.16	−0.116	\$23,234,278	\$1,070,296	\$3,964,308	\$13,133
	(\$6,907)	(\$8,891)	(\$4,510)	(0.149)	(0.255)	(0.357)	(\$6,358,336)	*at 0.215*	(\$1,496,850)	*at 0.058*
Four Rivers, ID	\$28,157**	\$299,305***	\$793	0.805***	0.785	−5.28	\$477,808	\$31,266	\$0	\$0
	(\$11,741)	(\$56,004)	(\$4,824)	(0.121)	(0.233)	(0.547)	(\$115,284)	*at 0.256*	($\leq$ \$36)	*at 0.558*
Rio Puerco, NM	\$84,054***	\$12,038***	*no data*	−1.05***	1.39	*no data*	\$104,651,030	\$3,750	*no data*	*no data*
	(\$19,744)	(\$4,233)		(0.138)	(0.270)		(\$31,745,327)	*at 0.006*		

Notes: Table reports the estimates fit on all eleven offices together (12,509 choice situations, log-likelihood −23,208.90). Willingness to pay is the revenue, in 2014 dollars, an office forgoes to protect an acre at the feature from development, for each non-market resource class. The closure effect is the office's residual tendency to close a feature outright rather than leave it under discretionary stipulations, common to all non-market resource classes and measured in log-odds rather than dollars. The scale, reported in logs for soil and water together and for wildlife, is the spread of the idiosyncratic differences among a feature's alternatives. The cost of departure is reported in dollars for soil and water together and for wildlife; it is estimated in logs, so it is positive by construction and carries no significance stars. The *at the mean departure* columns report the penalty an office pays on an average protective decision: the cost of departure times the mean squared gap between its chosen buffer and the standard it follows, the proxy plus its correction, with the gap averaged over the office's latent types by their posterior probabilities given the choice. Open choices carry no departure and are left out. The italic line reads that departure as a distance: how far, on the relevance scale, the office's typical protective decision lands from the buffer its standard called for, where 0 means it stopped exactly at the standard and 0.3 means the last acre it protected mattered about a third less, or more, than its standard would have. *no data* marks a class×office group that I do not currently have data for. A cost printed as \$0 (wildlife at Four Rivers, ID) sits at the boundary of what the fit allows: the office is estimated to pay nothing to depart there, and the parenthesis is the one-sided 95% upper bound. Standard errors in parentheses, by the delta method for the cost of departure; $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 2: How Often an Adjacent Buffer's Net Benefits Exceed the Predicted Buffer's

		Share of decisions with a net benefit gain			
Field office	Decisions	> \$0	> \$100	> \$1k	> \$10k
Rock Springs, WY	1,127	83%	64%	26%	6%
Casper, WY	1,902	67%	49%	33%	12%
Buffalo, WY	2,696	94%	93%	89%	77%
Lander, WY	518	73%	70%	58%	31%
Grand Junction, CO	221	50%	48%	42%	29%
Uncompahgre, CO	386	51%	39%	25%	9%
Royal Gorge, CO	1,427	44%	38%	27%	11%
Colorado River Valley, CO	454	69%	53%	32%	22%
Little Snake, CO	427	89%	87%	84%	64%
Four Rivers, ID	345	47%	18%	8%	3%
Rio Puerco, NM	195	72%	18%	9%	3%
Median office		69%	49%	32%	12%

Notes: Table reports, for the decisions in which the fitted model predicts protection, the share in which a buffer adjacent to the predicted buffer has a higher net benefit—its conservation value net of the revenue given up—by the size of the gain. The predicted buffer is the protective alternative with the highest predicted probability; its adjacent buffers are the alternatives under the same restriction one step narrower and one step wider. Figure A9 plots the first share column by non-market resource class. The Decisions column documents the number of decisions for which the model predicts protection. The first share column counts a gain of any amount; the others count only gains above \$100, \$1,000 and \$10,000 a year (2014 dollars). The last row is the median across offices of each column.

A Economic Interpretation of Land Use Plan Regulation

The Federal Land Policy and Management Act (FLPMA) serves two central functions in BLM land-use planning. First, it grants the agency comprehensive authority to manage public lands under a mandate of multiple use and sustained yield. This mandate requires the BLM to balance competing uses—such as grazing, energy development, recreation, and conservation—while ensuring that public benefits are sustained over time. Second, FLPMA specifies the procedural mechanism through which the agency is to pursue this mandate: Resource Management Plans (RMPs), commonly referred to as land-use plans (LUPs). Throughout the paper I do not refer to LUPs, but rather allocation decisions. The LUP planning documents document allocation decisions. To reduce the use of unnecessary jargon, I do not mention LUPs in the main text, but I do in this appendix to clarify the regulatory context of the allocation decisions.

Under the National Environmental Policy Act (NEPA), LUP development must include all required components of an Environmental Impact Statement (EIS) (Bureau of Land Management 2008). This process requires the BLM to evaluate the environmental and economic impacts of their choices, assess and disclose alternative management strategies, and engage with the public and other jurisdictions throughout plan development (Muhn & Stuart 1988).

Together, NEPA and FLPMA require the BLM to publicly define and solve an optimization problem through land-use planning. FLPMA defines the objective—management under a multiple-use, sustained-yield mandate—while NEPA requires the agency to articulate the feasible set by evaluating a full range of reasonable land-use alternatives. Both statutes require planners to publicly identify the market and non-market resource endowments that enter the objective function. The adopted LUP represents the chosen solution, specifying how each resource is governed over the plan's tenure. This paper's goal is to recover the objective function.

The remainder of this section shows how regulations, implementing NEPA and FLPMA, define the objective and informational content of BLM land-use plans. I also link this regulatory language to the paper's analytical framework.

A.1 The Objective of Land Use Plans

This section outlines the objective of LUPs as stated in regulation.

LUPs' objective is the same as the BLM's mandate: "maximize resource values for the public through a rational, consistently, applied set of regulations and procedures which promote the concept of multiple use management" (CFR 43 Part 1601.0-2).[32] In this context, "multiple use management" is defined in CFR 43 Part 1601.0-5. The following excerpts highlight key elements of that definition, though they do not represent it in full:

1. "making the most judicious use of the lands for some or all of these resources or related services over areas large enough to provide sufficient latitude for periodic adjustments in use to conform to changing needs and conditions; the use of some lands for less than all of the resources"

2. "harmonious and coordinated management of the various resources...with consideration being given to the relative values of the resources and not necessarily to the combination of uses that will give the greatest economic return or the greatest unit output"

To map this legal definition of the LUP objective into a mathematical analogue, I summarize the mandate as follows: land-use plans should maximize resource values, accounting for both market and non-market values, while balancing competing uses. In economic terms, offices choose allocations to maximize a weighted sum of resource endowments, where the weights represent the relative values assigned to each resource.

[32] See Section B for the full subpart.

A.2 Feasible Solution Set

In this section, I outline the federal regulations requiring that land-use plan development specify not only a set of allocation decisions, but also alternative sets of allocation decisions that span the space of feasible options, thereby defining the feasible choice set.

43 CFR Part 1600 requires that the LUP development process "consider all reasonable resource management alternatives and develop several complete alternatives for detailed study" (CFR 43 1610.4-5). The regulation's "alternatives" are each a set of allocation decisions in the vocabulary of this paper, and those developed for detailed study are its considered sets. This legal requirement defines the *full* set of alternatives which address a LUP's goal through different means and are *feasible*.

In economic terms, I interpret the regulatory requirement to consider "reasonable" alternatives, as a requirement to identify *feasible* solutions to the land use planning optimization problem. Specifically, the regulation defines "reasonable" alternatives as those that:

- Respond to the LUP's "purpose and need" and address the "issues" identified during the LUP scoping process.[33]
- Are feasible given estimated costs, logistics, and available technology.
- Contain internally consistent components; for example, an alternative should not both prohibit visual disturbances and allow oil and gas drilling on the same parcel.

I interpret the regulatory requirement to consider "all" alternatives as a requirement to define alternatives which span the feasible set of solutions. This interpretation is supported by guidance in how the BLM planning process should apply NEPA (Bureau of Land Management 2008):

> "For some proposals there may exist a very large or even an infinite number of possible reasonable alternatives. When there are potentially a very large number of alternatives, you must analyze only a reasonable number to cover the full spectrum of alternatives (see Question 1b, CEQ, Forty Most Asked Questions Concerning CEQ's NEPA Regulations, March 23, 1981)."

A.3 Resource Endowments

In this section, I outline the federal regulations requiring land-use plan developers to evaluate the impacts of their decisions, thereby documenting their beliefs about resource endowments across the planning area. I also provide an example of this analysis from a land-use plan.

For each set of allocation decisions, including the final plan—each considered set of allocation decisions and the adopted one, in this paper's language—planners must publicly outline the predicted impact of the given alternative on physical, biological, economic and social resources in the planning area (CFR 43 Part 1610.4-6). This assessment begins with a detailed description of existing resource endowments and then evaluates how each set would affect those resources, identifying both positive and negative impacts. This second step closely resembles a benefit–cost analysis, in which benefits and costs are quantified in terms of the acreage of each resource that would be affected by a given set of allocation decisions. For example, in the Four Rivers Field Office Draft RMP Amendment, the planners quantify the impact of alternate sets on vegetation resources in terms of the acres exposed to "surface disturbing activities". Specifically, the planning documents write:

> "The most acreage of right-of-way (ROW) exclusion, closure or open with a no-surface-occupancy (NSO) stipulation to fluid leasable minerals, and renewable energy exclusion areas would occur under Alternative B, which would reduce surface disturbing impacts from development on vegetation and help maintain

[33] A LUP's "purpose and need" section is a required component that outlines the rationale for the plan. See 40 CFR §1502.13 for the relevant regulatory requirement. In addition, under 43 CFR §1610.4-1 and 40 CFR §1501.7, the scoping process must identify "issues" to be addressed in the LUP. These issues must be within the scope of analysis, describe a relationship between an action and a resource, be scientifically analyzable, and must not be resolved solely by existing law, regulation, or policy.

> intact vegetation communities in those areas. The least amount of ROW exclusion would occur under Alternative C, resulting in the greatest potential impacts from ROW development on vegetation."

The documentation then goes on to detail acreage of permitted surface disturbance by vegetation community under the different sets considered. Figure A3 shows a table from the planning documents summarizing this analysis at the planning area level. In this example, each row of the table details the acres of a given vegetation community where ground disturbance would be allowed under each considered set, labeled as "Alt. A", "Alt. B", "Alt. C" and "Alt. D". Earlier and later discussions in the planning documents highlight that the vegetation communities: Evergreen Forest, Shrub-Steppe & Perrenial Grassland, and Riparian & Wetland are valuable vegetation communities, either for logging, preservation of protected species, wildfire resilience, etc. While the vegetation community "Exotic Annual Grassland" have negative value, reducing habitat for native species and decreasing wildfire resilience.

Mineral and energy resources requirements for data collection and analysis follow a more structured protocol. Bureau of Land Management (1985) requires each plan to rank the planning area's occurrence potential and development potential on a categorical scale (none, low, moderate, high), where development potential combines the geological ranking with historic drilling and recent industry interest to state where the office expects development within the life of the plan. Appendix F describes how these development-potential categories, together with the plan's development forecast, yield the per-acre revenue and disturbance densities of Section 4.1.

B Relevant BLM Land Use Planning Regulation

B.1 CFR 43 Part § 1601.0-2 Objective

> The objective of resource management planning by the Bureau of Land Management is to maximize resource values for the public through a rational, consistently applied set of regulations and procedures which promote the concept of multiple use management and ensure participation by the public, state and local governments, Indian tribes and appropriate Federal agencies. Resource management plans are designed to guide and control future management actions and the development of subsequent, more detailed and limited scope plans for resources and uses.

B.2 CFR 43 Part § 1601.0-5 Definitions

> Multiple use means the management of the public lands and their various resource values so that they are utilized in the combination that will best meet the present and future needs of the American people; making the most judicious use of the lands for some or all of these resources or related services over areas large enough to provide sufficient latitude for periodic adjustments in use to conform to changing needs and conditions; the use of some lands for less than all of the resources; a combination of balanced and diverse resource uses that takes into account the long term needs of future generations for renewable and non-renewable resources, including, but not limited to, recreation, range, timber, minerals, watershed, wildlife and fish, and natural scenic, scientific and historical values; and harmonious and coordinated management of the various resources without permanent impairment of the productivity of the lands and the quality of the environment with consideration being given to the relative values of the resources and not necessarily to the combination of uses that will give the greatest economic return or the greatest unit output.

B.3 CFR 43 Part § 1601.0-6 Environmental impact statement policy

> Approval of a resource management plan is considered a major Federal action significantly affecting the quality of the human environment. The environmental analysis of alternatives and the proposed plan

shall be accomplished as part of the resource management planning process and, wherever possible, the proposed plan and related environmental impact statement shall be published in a single document.

B.4 43 CFR Part § 1610.4-5 Formulation of Alternatives

> "At the direction of the Field Manager, in collaboration with any cooperating agencies, BLM will consider all reasonable resource management alternatives and develop several complete alternatives for detailed study. Nonetheless, the decision to designate alternatives for further development and analysis remains the exclusive responsibility of the BLM. The alternatives developed shall reflect the variety of issues and guidance applicable to the resource uses. In order to limit the total number of alternatives analyzed in detail to a manageable number for presentation and analysis, all reasonable variations shall be treated as sub-alternatives. One alternative shall be for no action, which means continuation of present level or systems of resource use. The plan shall note any alternatives identified and eliminated from detailed study and shall briefly discuss the reasons for their elimination."

NEPA Section 102(2)(E) directs the BLM to "study, develop, and describe appropriate alternatives to recommended courses of action in any proposal that involves unresolved conflicts concerning alternative uses of available resources" (Bureau of Land Management 2008). Specifically, the BLM handbook on NEPA compliance states that (Bureau of Land Management 2008):

> "The range of alternatives [should explore] alternative means of meeting the purpose and need for the action. ...the purpose and need statement[34] helps define the range of alternatives. The broader the purpose and need statement, the broader the range of alternatives that must be analyzed.[35] You must analyze those alternatives necessary to permit a reasoned choice (40 CFR 1502.14). For some proposals there may exist a very large or even an infinite number of possible reasonable alternatives. When there are potentially a very large number of alternatives, you must analyze only a reasonable number to cover the full spectrum of alternatives (see Question 1b, CEQ, Forty Most Asked Questions Concerning CEQ's NEPA Regulations, March 23, 1981). When working with cooperating agencies, your range of alternatives may need to reflect the decision space and authority of other agencies, if decisions are being made by more than one agency."

B.5 CFR 43 Part § 1610.4-6 Estimation of effects of alternatives

> The Field Manager, in collaboration with any cooperating agencies, will estimate and display the physical, biological, economic, and social effects of implementing each alternative considered in detail. The estimation of effects shall be guided by the planning criteria and procedures implementing the National Environmental Policy Act. The estimate may be stated in terms of probable ranges where effects cannot be precisely determined.

C Examples from the Planning Record

This section provides examples from the planning record supporting assertions with regards to how allocation decisions are made. Each subsection names an assertion made in the main text and gives the examples that support it. All examples are from planning documents published by one of the eleven offices in the estimation sample.

[34] The purpose and need statement of an RMP or RMP amendment "shall briefly specify the underlying purpose and need to which the agency is responding in proposing the alternatives including the proposed action" (40 CFR 1502.13); the BLM NEPA handbook quotes the same language in its §6.2 (Bureau of Land Management 2008).

[35] Alternatives are only "reasonable" if they respond to the purpose and need for the RMP or RMP amendment. Only reasonable alternatives need to be considered (Bureau of Land Management 2008).

C.1 Choices Not Fixed by Ecology, Hydrology, or Geology

This section provides examples supporting the assertion that an allocation decision, the restriction an office places on a feature and the distance it extends, is a choice rather than something the feature's ecology, hydrology, or geology determines.

Grand Junction, Colorado. Colorado Parks and Wildlife asked the Grand Junction office to raise its 30 m buffer on ephemeral streams to 100 m. Existing literature, already cited in the plan itself, places that 30 m at the low end of the range of effective riparian buffers (Johnson & Buffler 2008, p. 26). The office declined on the ground that "the minimum buffer of 30 meters will allow flexibility for protecting these systems according to site-specific circumstances" (Grand Junction Field Office 2015, p. 6-19).

Uncompahgre, Colorado. The office's existing plan prohibits surface use within a half mile of an osprey nest, on the ground that "[t]he sensitivity of osprey to human associated disturbance activities requires a half-mile buffer zone to avoid nest abandonment" (Uncompahgre Field Office 2019, p. B-105). The approved plan cuts the radius to a quarter mile, offering no rationale for the revision and no reassessment of the sensitivity the earlier stipulation invoked (Uncompahgre Field Office 2020, p. B-33).

Casper, Wyoming. Casper's approved plan is explicit that development on slopes over 25 percent is hazardous, yet the office does not close those slopes to surface disturbance outright. It prohibits surface disturbance "unless individual site plans are submitted to and approved by the authorized officer" (Casper Field Office, Wyoming State Office 2007, Decision 1022).

C.2 Decisions Made Feature by Feature Rather Than by Feature Type

This section provides examples of offices making different decisions for one feature compared to the rest of its type.

Rock Springs, Wyoming. Rock Springs closes the saline soils of the Red Creek watershed to fluid mineral leasing while comparably saline soils elsewhere in the planning area remain open to leasing; the alternatives considered would have left Red Creek open as well (Rock Springs Field Office, High Desert District 2024*b*). The office likewise closes the aquifer recharge areas around the towns of Superior and McKinnon to drilling but assigns discretionary restrictions to other mapped recharge areas (Rock Springs Field Office, High Desert District 2024*b*).

Rio Puerco, New Mexico. One contiguous portion of the planning area's winter elk habitat is Closed while the rest receives discretionary protection, a seasonal limit on drilling (Rio Puerco Field Office, New Mexico 2024).

C.3 Different Decisions for Different Market Resources at the Same Feature

This section provides examples of offices restricting different market resources differently at the same feature.

Four Rivers, Idaho. The Four Rivers plan amendment separates oil and gas from geothermal management actions, closing areas with low or no oil and gas potential while leaving geothermal management unchanged (Bureau of Land Management 2023).

Rock Springs, Wyoming. Rock Springs restricts oil shale within a quarter mile of a national historic trail while restricting all other resources within five miles of it (Rock Springs Field Office, High Desert District 2024*b*).

C.4 Closures Opposed Whether or Not Development Was Expected

This section provides examples of closure being opposed as a decision in itself, whether or not development was expected on the land closed.

Rock Springs, Wyoming. A trona operator protesting the Rock Springs draft plan objected to the closure of solid mineral leasing on slopes greater than 25%, which under that alternative covered 2,119,920 acres—of which 49,224, or 2.3 percent, fall inside the one area the plan identifies as materially affecting trona development (Rock Springs Field Office, High Desert District 2024*a*, ES-6, 4-84, W-20–W-21). The operator asked why BLM was "prohibiting solid mineral leasing in areas where soils have ... slopes greater than 25%," when "most of the RMP area exhibits at least one of these characteristics, as the area is a high desert" (Rock Springs Field Office, High Desert District 2024*a*, W-20–W-21).

C.5 Disturbance Matters Less the Farther It Is from the Feature

This section provides an example of an office treating disturbance as mattering less the farther it lies from the feature.

Lander, Wyoming. Lander's proposed plan protects riparian areas with a 500-foot buffer. One alternative the office considered would have protected more. The office judged that 500 feet "would likely be adequate" to provide "the necessary attributes for water dissipation, sediment capture, and habitat needs," and that "it is currently undetermined what level of beneficial impact there would be from the additional buffer" under the wider alternative (Lander Field Office, Wind River/Bighorn Basin District 2013, Ch. 4, p. 833).

C.6 Decisions Regarding Prairie Dog Towns Factor in Their Size

This section provides examples of offices factoring prairie dog town size when making protection town protection decisions.

Little Snake, Colorado. Little Snake prohibits disturbance of more than one acre in active white-tailed prairie dog towns smaller than ten acres and permits it in larger ones (Little Snake Field Office 2010, p. Q-237). When a commenter objected that the rule "would conserve relatively small colonies while allowing for development in larger colonies even though these larger colonies have greater biological value" and "promotes habitat fragmentation." The office justified their choice by arguing it "would preserve smaller colonies that are more sensitive to surface disturbance while allowing for some disturbance within larger colonies that could more easily adapt to, recover from, or absorb impacts associated with surface disturbances" (Little Snake Field Office 2010, p. Q-67).

Uncompahgre, Colorado. Uncompahgre considered two designs for prairie dog towns, each in a different alternative. One protects every active town the same way, with a 150-foot ring around it in which surface use is prohibited. The other protects towns by size: in towns smaller than ten acres no activity may disturb more than one acre, while larger towns carry no such limit (Uncompahgre Field Office 2019, p. B-38).

Lander, Wyoming. Lander weighed prohibiting disturbance in all colonies against avoiding it only in "complexes larger than 100 acres" (Lander Field Office, Wind River/Bighorn Basin District 2013, Ch. 2, p. 131).

D Standards in the Planning Record

This section provides examples from the planning record supporting assertions with regards to how allocation decisions are publicly justified. The first subsection sets out the legal requirements and agency guidance that

oblige an office to justify each restriction in writing when it responds to public comment. The remaining subsections give examples of a common way of meeting that obligation, citing a standard set elsewhere and stating that the choice is consistent with it, as well as of the departures from standards that offices defend on their own and of the limits on which standards an office will cite. Each subsection names an assertion made in the main text and gives the examples that support it. All examples are from planning documents published by one of the eleven offices in the estimation sample or from the agency guidance that governs them.

D.1 The Public Comment Process

This section sets out the regulatory requirements and agency guidance that underpin the formal public review every land-use plan goes through. Regulation requires a written response to every substantive comment, followed by a protest review and the Governor's consistency review. The agency's handbook directs an office to adopt the least restrictive measure that meets a resource objective and to justify each restriction in the plan. The commenters press in opposite directions, and the office owes each of them a written answer.

Regulatory requirements. A plan is published in draft for public comment and again, as a proposed plan, for protest, before a final decision is made. The regulation in force when most of the sample's plans were written says the agency "shall assess and consider comments both individually and collectively, and shall respond" in the final document. One permitted response is to explain why a comment warrants no change, "citing the sources, authorities, or reasons which support the agency's position" (40 CFR 1503.4). The BLM's NEPA handbook states the duty without qualification: "[w]hen preparing a final EIS, you must respond to all substantive written comments" (Bureau of Land Management 2008, p. 65). Once the office publishes its proposed plan, any person who participated in the planning may *protest* it. The Director must then rule on the protest in writing, either upholding the office's decision with reasons or requiring the office to change the plan before it is approved (43 CFR 1610.5-2).

BLM's recommendation for justifying decisions. For fluid mineral leasing, agency guidance recommends offices justify a restriction by showing that less restrictive measures were considered and found inadequate, and that they state that justification in the plan. Specifically, the BLM's handbook directs that "[t]he least restrictive stipulation that effectively accomplishes the resource objectives or uses for a given alternative should be used" (Bureau of Land Management 2013, p. III-11). It asks the plan to show that a milder option was weighed: "[i]f a closure or operating constraint is discretionary with the BLM, evidence that a less restrictive mitigation measure was considered should be reflected in the range of alternatives analyzed in detail" (Bureau of Land Management 2013, p. III-13). For the alternative the office prefers, the plan "should provide evidence that less restrictive measures were considered but found inadequate" (Bureau of Land Management 2013, p. III-14). The plan's text "should summarize the justification for constraints, nondiscretionary and discretionary closures, including references to any applicable laws, executive orders, etc." (Bureau of Land Management 2013, p. IV-1). Grand Junction, Colorado follows this instruction literally. It tells its commenters that "[t]he justification for each stipulation can be found in Appendix B under the specific stipulation" (Grand Junction Field Office 2015, p. 6-280), and each stipulation there ends with a line headed "JUSTIFICATION" (Grand Junction Field Office 2015, pp. B-40–B-41).

Who comments. Commenters on the same decision frequently ask for opposite changes. On a single page of Grand Junction's response to comments, the county commissioners say the office "has not justified the need" for the buffer it proposed, and the state wildlife agency asks for a stricter restriction (Grand Junction Field Office 2015, p. 6-45). Elsewhere in the same volume an industry group asks for the "scientific justification" of a river buffer, and a conservation group calls the stream buffers "not adequate" (Grand Junction Field Office 2015, pp. 6-270, 6-276). Little Snake, Colorado's response table carries comments from a gas operator, the county commissioners, two conservation groups, and the U.S. Fish and Wildlife Service (Little Snake Field Office 2010, pp. Q-236–Q-247, Q-377–Q-378). In these records operators, counties, and

industry groups generally press for less protection, and conservation groups and wildlife agencies generally press for more. The office owes each of them a written answer.

Comments that changed a plan: Buffalo and Little Snake. Buffalo's draft drew comments in both directions on its bird (raptor) buffers, and the office revised the raptor actions "to fully conform to the distances and dates recommended by the USFWS Wyoming Ecological Services Office" (Buffalo Field Office, High Plains District 2015*b*, p. 30). Little Snake's draft carried forward an eighth-of-a-mile buffer around raptor nests from its earlier plan; a conservation group asked for a quarter mile, and the approved plan adopted it, finding the increase "necessary to protect the raptor species, as well as to be consistent with state-wide stipulations throughout Colorado" (Little Snake Field Office 2011, p. ROD-2). Both exchanges are given in full in Sections D.4 and D.5.

D.2 Standards That Predate the Plan

This section provides examples supporting the assertion that when offices set a buffer, there are already published distances it can adopt instead of defending one of its own.

Wyoming: the Mitigation Guidelines. The BLM in Wyoming maintains a set of *Mitigation Guidelines for Surface-Disturbing and Disruptive Activities*, updated "for all Field Offices in Wyoming." Subject to exceptions, they prohibit surface disturbance on "[s]lopes in excess of 25 percent," "[w]ithin 500 feet of surface water and/or riparian areas," and "[w]ithin either one-quarter mile or the visual horizon (whichever is closer) of historic trails" (Casper Field Office, Wyoming State Office 2007, Appendix I, p. I-2). The guidelines state their own purpose: they "are primarily for the purpose of attaining statewide consistency in how requirements are determined for avoiding and mitigating environmental impacts" (Casper Field Office, Wyoming State Office 2007, Appendix I, p. I-1). They predate any one plan. "All BLM RMPs have included these guidelines as appendices," and "[p]ublic comment on the guidelines, per se, has not been requested" (Casper Field Office, Wyoming State Office 2007, Appendix I, p. I-1). They are also the reference point for the alternatives a plan considers. Planners use them "to develop a baseline for measuring and comparing impacts among the alternatives" and "to help determine whether more stringent or less stringent mitigations should be considered" (Casper Field Office, Wyoming State Office 2007, Appendix I, p. I-1). Lander, Wyoming's plan reprints the same text (Lander Field Office, Wind River/Bighorn Basin District 2013, pp. 1595–1596). The national handbook, written in 1990, already takes the "Wyoming BLM Standard Stipulations for Surface Disturbing Activities" as its worked example of how to describe an alternative (Bureau of Land Management 2013, Illustration 3, p. 1).

Wildlife agencies. Colorado's state wildlife agency published *Recommended Buffer Zones and Seasonal Restrictions for Colorado Raptors* in 2008. Grand Junction's plan, approved in 2015, names that document as the justification for each of its raptor stipulations. The half-mile buffer around ferruginous hawk nests, for example, "is necessary to protect ferruginous hawk nesting habitat per" the state's recommended buffer zones (Grand Junction Field Office 2015, p. B-40). The U.S. Fish and Wildlife Service's Wyoming office publishes its own table of recommended raptor buffers and dates. For special status raptors, Buffalo's proposed plan prints no distances of its own. It adopts the Service's "recommended spatial buffers and dates for breeding raptors" by reference to the Service's web page (Buffalo Field Office, High Plains District 2015*b*, pp. 217–220).

Statewide stipulations. The BLM's state offices issue stipulations meant to be used by every field office in the state. Grand Junction describes the Colorado effort: "[p]art of BLM leasing reform is to enact minimum stipulations (for the various programs) that could be applied across the state," and the group working on water "reduced the number of stipulations from approximately 70 to approximately 12 in Colorado" (Grand Junction Field Office 2015, p. 6-271). Uncompahgre refers to "BLM statewide stipulations, which are used on BLM-administered lands across Colorado" (Uncompahgre Field Office 2019, p. R-223). The state wildlife

agency supplied its own statewide list. It told Uncompahgre that in 2010 it had given the BLM's State Office "recommendations for fluid mineral lease stipulations relevant to Resource Management Plan (RMP) revisions" (Uncompahgre Field Office 2019, p. R-209).

The office's own earlier plan. Each office also inherits the plan it is revising. The handbook requires that "[e]xisting fluid minerals management, including constraints, is summarized and presented in the RMP/EIS as the 'no action' alternative," which "serves as the baseline for discussing the other alternatives" (Bureau of Land Management 2013, p. III-13). The existing buffers therefore sit in every revision as a named alternative. In Rock Springs, Wyoming's revision that alternative "represents a continuation" of the plans then in force, the older of them adopted in 1997 (Rock Springs Field Office, High Desert District 2024*b*, p. 1-3). Little Snake tells a commenter that "[t]he language in Alternative A was obtained from the 1989 Little Snake RMP and ROD and is not subject to revision within Alternative A" (Little Snake Field Office 2010, p. Q-377).

D.3 Commitments to Consistency Made Before Choosing

This section provides examples supporting the assertion that offices commit themselves to consistency with standards and with neighboring offices before they draft allocation decisions. Rock Spring, WY's decision record explains that the approved stipulations "were proposed to align with similar restrictions in adjacent planning area field offices and to provide clarity and resolve potential conflicts from the Draft" (Rock Springs Field Office, High Desert District 2024*b*, pp. 1-4–1-5). Little Snake states, "Lease stipulations will be reviewed for consistency with neighboring BLM Field Offices and States, and where there are discrepancies efforts will be undertaken to achieve consistency or to explain any differences" (Little Snake Field Office 2011, pp. RMP-6–RMP-7). Colorado River Valley, Colorado states, "Lease stipulations will be reviewed for consistency with neighboring field offices and states. Where there are discrepancies, efforts will be made to achieve consistency" (Colorado River Valley Field Office 2014, p. 1-14). Both offices also commit to a common format for stipulations that a regional committee prepared in 1989. Buffalo states "[t]he planning process will use the Wyoming BLM Mitigation Guidelines to develop management options and alternatives" (Buffalo Field Office, High Plains District 2015*a*, p. 9). Both Little Snake's and Buffalo's decisions state that they "will strive to be compatible with the existing plans and policies of adjacent" agencies (Little Snake Field Office 2011, p. RMP-6), (Buffalo Field Office, High Plains District 2015*a*, p. 7). Colorado River Valley's "will be compatible" with them (Colorado River Valley Field Office 2014, p. 1-13). That said, the Wyoming guidelines add that "[c]onsistency in this sense does not mean that identical requirements would be applied for all similar types of land use activities" (Casper Field Office, Wyoming State Office 2007, Appendix I, p. I-1).

D.4 A Standard as the Answer to Both Sides

This section provides examples supporting the assertion that a buffer taken from an established standard can be defended to public stakeholders (both pro-conservation and pro-development) by citing the standard.

Bird nests: Buffalo, Wyoming. Buffalo's alternatives included a 1.5-mile seasonal buffer around the nests of special status birds (raptors) in the alternative the plan labels "Resource Conservation" and a quarter-mile buffer in the one it labels "Resource Utilization" (Buffalo Field Office, High Plains District 2015*b*, p. 217). The comments split. "Some commenters felt the raptor spatial buffer distances and dates were too restrictive while others commented that the protections were not sufficient" (Buffalo Field Office, High Plains District 2015*b*, p. 30). The office did not defend a distance of its own. It revised the raptor actions "to fully conform to the distances and dates recommended by the USFWS Wyoming Ecological Services Office" (Buffalo Field Office, High Plains District 2015*b*, p. 30). In its formal response, the part that concerns buffers is one sentence long and serves both groups. Commenters who called the nest buffers "unnecessary and not adequately justified" and commenters who "proposed additional restrictions protecting eagles" received the same answer: "Dates and distances for buffers were based on recommendations from the appropriate state and federal agencies" (Buffalo Field Office, High Plains District 2015*b*, p. 2716).

Bird nests: Grand Junction, Colorado. Three commenters pulled Grand Junction's bird (raptor) buffers in three directions. An individual asked the office to go "beyond the 0.33 mile" around ferruginous hawk nests. The county commissioners objected that the Fish and Wildlife Service "recommends only a 200 meter" buffer around eagle nests and that the office "has not justified the need for this additional buffer." The state wildlife agency asked for no-surface-disturbance stipulations in place of seasonal limits. The office gave one answer to all three: "Buffers for raptor stipulations were developed to be consistent with the best available science, including CPW's Recommended Buffer Zones and Seasonal Restrictions for Colorado Raptors (CPW 2008). No change has been made" (Grand Junction Field Office 2015, p. 6-45). The county had cited a standard of its own; the one published by the state's wildlife agency prevailed.

Sensitive plants: Grand Junction, Colorado. Asked to defend a 200-meter buffer around sensitive plant communities, Grand Junction answered with outside references, none of them specific to its planning area. The justification "comes from Onshore Oil and Gas Order No. 1," a national regulation. The distance "is consistent with NSO stipulations in adjacent BLM field offices." It "is supported by" a 2009 set of recommended practices for rare plants and by draft guidance from the Fish and Wildlife Service (Grand Junction Field Office 2015, p. 6-280). NSO, the plans' abbreviation in these quotations, is no surface disturbance: the restriction that bars drilling equipment from the surface.

Historic sites: Buffalo, Wyoming. Commenters told Buffalo that its stipulations around cultural sites "were too restrictive." The office answered that the three-mile distance rested on the state historic preservation office's "experience with historic trails issues in other parts of Wyoming," and concluded: "The distance is not arbitrary since it is consistent with other Wyoming BLM plans" (Buffalo Field Office, High Plains District 2015*b*, p. 2722).

Wildlife stipulations: Uncompahgre, Colorado. Uncompahgre states the practice as a rule and applies it in both directions. TLs are timing limitations, restrictions that apply for part of the year. Commenters asked the office to acknowledge that these "are not adequate to protect big game species from oil and gas development" (Uncompahgre Field Office 2019, p. R-220). The office answered: "The BLM defers to existing federal and state guidelines and policy, as well as agency expertise, such as CPW's and USFWS', to help craft effective land use stipulations." Where the state agency's version does not fit, the plan "ensures TLs are consistent with BLM statewide stipulations" (Uncompahgre Field Office 2019, p. R-223). A rural electric association asked for the opposite, that timing limitations be "the minimum biologically necessary" (Uncompahgre Field Office 2019, pp. R-217, R-221). It was told that they "were developed based on best available biological information in coordination with CPW" (Uncompahgre Field Office 2019, p. R-224).

D.5 Departures from a Standard

This section provides examples supporting the assertion that a buffer not taken from a standard has to be defended on its own. In each case a departure from a standard is what draws the demand for data, or what the office defends with local facts.

Lease stipulations: Uncompahgre, Colorado. Colorado's wildlife agency told Uncompahgre that some stipulations in the preferred alternative were "not consistent with lease stipulations recommended to BLM by CPW." The agency then set the terms for a departure: "If the BLM elects to not incorporate our recommendations in the preferred alternative, we recommend the analysis provide data, with clear rationale supporting that decision" (Uncompahgre Field Office 2019, pp. R-209–R-210).

Fish-bearing waters: Buffalo, Wyoming. Buffalo placed a quarter-mile restriction around fish-bearing waters, wider than the 500-foot standard. Commenters objected. The office's defence names the standard and gives a physical reason for exceeding it: "the buffer from Onshore Order No. 1 is insufficient to protect

the fish habitat from sedimentation since canyon rims are well over 500 feet distance from the water" (Buffalo Field Office, High Plains District 2015*b*, p. 2717).

Major rivers: Grand Junction, Colorado. Grand Junction proposed buffers around major rivers wider than the usual stream buffer. An industry group asked for the "scientific justification," noted that "[h]istoric buffers have been limited to 300 to 500 feet," and argued that "[w]ithout scientific evidence proving its need, this new restriction is unwarranted" (Grand Junction Field Office 2015, pp. 6-270–6-271). The office replied at length. It cited the statewide working group, a statewide risk assessment for drinking water, documented contamination incidents, and "the topography and river channel type in this planning area" (Grand Junction Field Office 2015, p. 6-271).

Raptor nests: Rock Springs, Wyoming. A county objected that a two-mile raptor buffer in Rock Springs' draft "is more restrictive than United States Fish and Wildlife standards," and that "[t]he BLM has not provided information to support a 2-mile buffer zone." Other commenters wrote that the Service's "recommended distances should be used." The office's replies on that page do not defend the distance. Each refers the commenter to the range of alternatives (Rock Springs Field Office, High Desert District 2024*a*, p. W-149). The approved plan sets raptor buffers of a quarter mile to one mile by species (Rock Springs Field Office, High Desert District 2024*b*, p. 2-32), and the decision record gives alignment with adjacent offices as a reason for the final stipulations (Rock Springs Field Office, High Desert District 2024*b*, pp. 1-4–1-5).

D.6 Standards That Are State-Specific and Do Not Bind

This section provides examples supporting the assertion that just because a standard exists does not mean it is followed. A standard, as the paper uses the word, is a published buffer an office can cite: a state agency's guidance, another office's choice, or the office's own earlier plan. The examples in this section show offices declining standards.

Grand Junction, Colorado. A conservation group asked Grand Junction to adopt "the same buffer setback stipulations" that Little Snake had approved in 2011, a quarter mile around all perennial waters (Grand Junction Field Office 2015, p. 6-276). The office kept its own buffers and replied that its analysis "did not show that larger buffers were warranted" (Grand Junction Field Office 2015, p. 6-280). Grand Junction told commenters it "is not obligated to be perfectly consistent with CPW's guidelines" (Grand Junction Field Office 2015, p. 6-348).

Little Snake, Colorado. A conservation group asked Little Snake to adopt raptor guidelines from the Fish and Wildlife Service's Utah office. Little Snake answered that it "is not required to use guidelines prepared by another ecological field office" (Little Snake Field Office 2010, p. Q-247).

Uncompahgre, Colorado. Uncompahgre states "BLM state-specific guidance outside of Colorado are not applicable" (Uncompahgre Field Office 2019, p. R-68). When Colorado's Governor objected that Uncompahgre had not adopted the state agency's stipulations, the BLM replied that "such recommendations do not comprise an officially approved policy or program" under the consistency regulation (Bureau of Land Management, Colorado State Office 2020, pp. 2–3).

E Calibrating the Ecological Ranges

This section provides the evidence behind the effective ecological range h Section 4.1 defines.

E.1 The Range of Each Feature Type

This section lists the range of every feature type in the estimation sample.

Water features. The range is 120 m for riparian wetlands and for floodplains, whose mapped extent is itself the strip that shields the water, so the range covers only the ground that drains into it; 200 m for streams, rivers and waterbodies, and by analogy for drinking-water sources; 400 m for the named major rivers; and 805 m, half a mile, for groundwater. Section E.2 derives the 200 m.

Wildlife features. The range is 400 m for prairie dog towns; 1,609 m, one mile, for bald eagle roosts; and 2,414 m, a mile and a half, for nests. Section E.3 derives the nest range. For prairie dog towns and bald eagle roosts, whose sites are areas rather than points, the range prices only the ring outside the mapped site, since every acre inside it carries full weight whatever the range.

Soil features. The four soil feature types carry no compact geometry to measure distance from, so each is given two anchors in attribute space rather than metres: the level at or above which an avoided acre carries full conservation value, and the level below which it carries essentially none. The shortfall between them is the range h. For slope the anchors are a 50% and a 10% grade; for water erodibility a RUSLE factor of $K = 0.64$ and $K = 0.20$; for wind erodibility an erodibility index of 310 and 56 t/ac/yr; and for saline soils an electrical conductivity of 16 and 8 dS/m. Section E.4 derives the slope anchors.

E.2 Distance Features: Streams

This section derives the 200 m range of streams, rivers and waterbodies. An acre at the water's edge earns full weight, decaying to near zero by about 200 m from the edge. Two distances set this range, and the range takes the upper end of each.

The riparian strip. The first distance is the width of the strip along the stream that keeps its ecosystem healthy: the ground that filters runoff before it reaches the water, holds the bank in place, and supplies streamside habitat for wildlife. Buffer design guidelines written for this region put that strip at up to about 90 m (Johnson & Buffler 2008, p. 26).

The overland path. The second distance is how far upslope of that strip surface disturbance can still affect it. The federal rangeland erosion model puts the distance runoff travels before it soaks into the ground or drops the sediment it carries at no more than about 120 m (U.S. Department of Agriculture, Natural Resources Conservation Service 2021).[36]

Why the upper end of each. What the weighting needs is the distance at which an avoided acre stops mattering *altogether*, so each input is how far its mechanism can plausibly reach, not how far it typically does. Past roughly 200 m, an acre neither belongs to the strip that shields the stream nor drains into it, so disturbance beyond this distance is given almost no weight.

E.3 Distance Features: Nests

This section derives the 2,414 m range of bird nests, a mile and a half. The range is the outer edge of the federal guidance that grades restrictions near an active nest by distance. The Forest Service's conservation assessment for the ferruginous hawk recommends a maximum duration of disturbance by breeding stage and by distance from the nest, in three bands: within half a mile, half a mile to one mile, and one mile to a mile and a half. "Out-of-vehicle" disturbance, which the schedule defines to include "work-related activities (e.g., mineral development, agricultural operations)," is permitted for no time at all inside half a mile during the sensitive breeding stages, for under an hour in the second band, and for under a day in the third. The

[36] The 120 m is the longest slope the model supports, not a distance sediment was observed to move; its standard case is 50 m (U.S. Department of Agriculture, Natural Resources Conservation Service 2021, 647-C.3). Sediment typically travels only a fraction of that distance—monitored plumes at gas well pads extend under 40 m beyond the disturbed area (Williams et al. 2008)—so 200 m is an outer bound rather than a tight estimate.

schedule states no restriction beyond a mile and a half (Collins & Reynolds 2005, Table 7, p. 44). A well pad, with its road and its production life, is permanent, and so exceeds every one of these caps. On this schedule a pad is incompatible with an active nest anywhere inside a mile and a half and unrestricted beyond it. So I set the ecological range at 1.5 miles.

The range is an upper bound. Every other distance the assessment attaches to human disturbance near a nest lies inside this one. It recommends posting wildlife alert signs at least 450 m from active nests (Collins & Reynolds 2005, p. 43), and placing artificial nest platforms at least one mile from any public road and at least a mile and a half from any occupied building (Collins & Reynolds 2005, p. 42). The schedule's own bands end at a mile and a half, and no recommendation in the assessment reaches farther. The range is therefore the farthest distance at which the cited guidance treats disturbance as affecting a nesting pair, an upper bound rather than a typical response distance.

E.4 Threshold Features: Slopes

This section derives the two anchors of the slope range. In the distance case the maximum weight of one sits at the feature and decays with distance until disturbance affects the feature negligibly. In the threshold case it is less obvious where the maximum weight should land, so two points are calibrated: a *lower anchor*, below which disturbance does little lasting harm, and an *upper anchor*, above which almost any disturbance is harmful. For slope the anchors come from the two distinct processes by which disturbing a slope degrades it, surface erosion and mass wasting.

The lower anchor: surface erosion. On gentle slopes the concern from disturbance is surface erosion. Many factors set how fast a disturbed slope sheds soil—rainfall, soil erodibility, ground cover—and steepness adds to that loss at every grade, not only steep ones. What the federal erosion model marks at about 9% is a change in *how fast* steepness adds erosion: below that grade an extra degree of slope adds little, above it each degree adds soil loss more sharply (Renard et al. 1997, ch. 4). A grade near 10% therefore marks where slope begins to drive erosion sharply, the point at which avoiding disturbance begins to meaningfully benefit soil health.

The upper anchor: mass wasting. The upper anchor is where the more damaging process, mass wasting (landslide, slope failure), becomes markedly more likely. The Forest Service's road-failure records show "a significant threshold at 50 percent," with mass-wasting failures per mile almost eight times higher above that grade than below (U.S. Department of Agriculture, Forest Service 1999, App. 2).

F Development Forecasts

Section 4.1 builds the payoff attributes from per-acre revenue and disturbance densities, $m_{i,c}$ and $d_{i,c}$, defined over each resource's development-potential categories c. This appendix details how those densities are recovered from the planning record.

Forecasts. Regulation requires each plan to include a *Reasonably Foreseeable Development* (RFD) report (Bureau of Land Management 1985): projections of commercial development under a reference scenario in which development is permitted everywhere legally allowed, reported at coarse spatial scales—typically by development-potential category or township. For oil and gas, the projections include the expected number of wells, success rates for exploratory ("wildcat") and development wells, expected production per well, and associated surface disturbance. For each category, the RFD expresses expected development as a rate referenced to a township, a fixed areal unit (36 sq. mi.) much smaller than the category's full extent—a category may be labeled "> 150 wells per township." Converting this rate to a per-acre basis and multiplying by the expected disturbance and revenue per well gives each category its disturbance density $d_{i,c}$ (disturbed

acres per acre) and revenue density $m_{i,c}$ (revenue per acre) under the baseline scenario; both incorporate the probability of commercially viable extraction as the RFD reports it. Integrating a density over a footprint treats development as uniform within each category—the township-referenced rate applied evenly to every acre of the category—which is consistent with the coarse resolution at which RFDs report and imposes no further structure on where development falls within a category.

Restriction shares. To evaluate feasible choices, expected outcomes are parameterized as proportional reductions against the baseline: for each restriction r, $d_{i,c}(r) = a_{r,c}\, d_{i,c}$ and $m_{i,c}(r) = a_{r,c}\, m_{i,c}$, where $a_{r,c}$ is the share of the wells the baseline projects in category c that restriction r allows. A single share scales both densities because both derive from the same projected well count: a restriction raises operating costs and so lowers the probability that a projected well is drilled, and expected disturbance and expected revenue fall with the well count together. By construction $a_{\text{Open},c} = 1$, with $a_{r,c}$ weakly decreasing in policy stringency,

$$1 = a_{\text{Open},c} \geq a_{\text{Discretionary},c} \geq a_{\text{Closed},c} = 0.$$

The record's five stipulations collapse onto these three as in Section 3.2.4: timing limitations and controlled surface use share the discretionary factor, and no surface disturbance and Closed both permit no wells on the footprint. Where a plan publishes no reduction for its discretionary stipulations, the share is borrowed from the plans that do, interpolated on the resource's own well density. Where more than one fluid mineral resource is viable at a feature, the feature-level quantities sum the viable resources' expectations—each read off that resource's own potential map under the same pair $\boldsymbol{y}$—so $M_{i,f}$ and $K_{i,f}$ price the one decision against the combined development it restricts.

G Erosion Potential at the Soil Features

Figure 4 documents an eighteen-fold spread across offices in the willingness to pay to avoid disturbing fragile soils, from \$23k per acre at Uncompahgre, Colorado to \$407k at Royal Gorge, Colorado. This difference could reflect a difference in preferences or it could reflect a difference in the environmental benefit of avoiding disturbing fragile soils across offices. To investigate the latter, I build a measure of erosion potential to assess the extent to which the environmental benefits of avoiding disturbing fragile soils differ across offices. This appendix documents how I measure erosion potential; Figure 5's left panel sets the measure beside the soil willingness-to-pay estimates.

G.1 The Erosion Potential Index

I measure erosion potential with the Revised Universal Soil Loss Equation, the Department of Agriculture's standard formula for how much soil rain washes off a piece of ground in a year (Renard et al. 1997). The formula estimates expected soil loss, considering the following mechanisms for erosion. Rain falls on a patch of dirt. Hard rain knocks more soil loose than soft rain. Some soils come loose easily and others hold together. Water running down a steep patch moves faster and carries more soil off the surface than water on a flat one, and water that has farther to run picks up more speed, carrying more soil. A patch covered in plants, or protected by a terrace or a ditch, loses less than bare ground. The formula gives each of these influences a number and multiplies them:

$$A \;=\; R \times K_{\text{w}} \times LS \times C \times P.$$

A is the predicted soil lost, in tons per acre per year. R measures how hard rain falls on average on the patch. K_{w} measures how easily the soil erodes. LS measures how steep the slope is and how far the water runs down the slope before reaching the patch. C measures how much plant cover protects the ground. P measures whether anything has been built to slow the water.

All of the formula's components except R are defined against a standard test strip. To compare soils fairly, the scientists who built the formula laid out many small plots of bare dirt of exactly one size, 72.6 feet

long (about two school buses) on a 9 percent grade, let rain fall on them and weighed the dirt that washed off. R is the erosive energy of the rain itself, in the same units on a test strip as anywhere else. The soil factor K_{w} is the loss from such a strip per unit of R. Each of the remaining factors, LS, C and P, says how much more or less a real patch loses than a test strip would, because it is steeper or flatter, longer or shorter, covered or bare, protected or not. A factor of one means "the same as the test strip."

I hold the two factors that describe how the ground is *managed* at their test-strip values. I let $C = 1$, reflecting the fact that a freshly graded well pad is bare-ground. I assume no erosion control has been built, $P = 1$, having the erosion potential reflect expected erosion under a "baseline open scenario", i.e., where no precautionary measures are required to reduce erosion.

I also hold the slope length at its test-strip value, i.e., $L = 1$, treating every cell as if it were test-strip length. Measuring real hill lengths from a map is hard and fussy, and people argue about how to do it. Holding slope length at the 72.6-foot test-strip value may understate differences in erosion potential across offices. RUSLE predicts that longer slopes increase erosion, with larger proportional effects on steeper terrain. Steeper offices also have higher predicted erosion in the data; the correlation between log mean S and log mean A is 0.93. Thus, if effective slope lengths generally exceed 72.6 feet and are similar across offices, the assumption compresses differences in predicted erosion.

Soil erodibility. K_{w} measures how easily water erodes the hillside. Silty soils wash away easily; clays stick together, and sands absorb the water, i.e., are not easily eroded by water. I use the published factor from the national soil survey (Soil Survey Staff 2025).

Slope steepness. S measures how steep the ground is. Water runs faster down a steeper slope and carries more soil. The steepness factor S comes from McCool et al. (1987), as adopted by the equation (Renard et al. 1997, ch. 4). It rises slowly with grade up to about 9 percent and faster above it: $S = 10.8 \sin\theta + 0.03$ below 9 percent and $S = 16.8 \sin\theta - 0.50$ at or above it, where θ is the slope angle read from the 30 m national elevation model. The factor is one at the test strip's 9 percent grade, about 0.1 on nearly flat ground and about 6 on a 45 percent grade. Grades are capped at 100 percent.

Rainfall erosivity. R measures how hard rain falls on average on the hillslope over a year: the energy of the raindrops, times how fast they arrive, summed over the year's storms. I estimate R from how much rain falls in an average year with the relation of Renard & Freimund (1994): $R = 0.0483\,P^{1.610}$ for annual precipitation $P \leq 850$ mm and $R = 587.8 - 1.219\,P + 0.004105\,P^2$ above it, in the equation's metric units, divided by 17.02 to convert to U.S. customary units. More rain means a higher R, and the relation rises faster than the rainfall itself because wetter places also tend to have harder storms. I derive the annual precipitation data from the 1991–2020 normal of the PRISM climate group at 800 m resolution, using bilinear interpolation (PRISM Climate Group 2022).

G.2 Aggregating the Index to the Office Level

I use the formula discussed in the previous section to compute the expected soil loss A for every $30\ m \times 30\ m$ grid cell an office has discretion over. I then aggregate this cell-level measure to the office level in three steps. *First*, I restrict the cells to those the office considered restricting to development to protect fragile soils. *Second*, I drop the cells that have no soil to lose: bare rock, water and unmapped ground, where the soil survey gives no K_{w}. *Third*, I average A over the remaining cells. The cells are not all the same size, because the map grid is in degrees and a degree of longitude is shorter farther north, so each cell counts in proportion to its true area in acres. This average is the "Mean A" of Table A3; it is taken over 0.6 to 2.6 million acres per office.

Two variants serve as checks. One keeps only the cells inside the buffer the office actually chose at each feature it protected; this is the least ground the office's soil decisions covered ("Mean A, chosen buffer" in Table A3). The other averages A over every cell the office has discretion over (last column of Table A3); the

gap between it and Mean A shows how much steeper and more erodible the ground the plans singled out is than the office's average acre.

G.3 Limitations

The index is an erosion potential under standardized conditions, not the erosion a well pad would cause. The tons are on-site soil loss, not the downstream sediment, water quality or reclamation costs. Erosivity is estimated from annual precipitation rather than from measured storm intensity. The relation is least reliable in the arid West, where a few intense summer storms carry most of a year's erosivity, so two places with the same annual total can differ in how hard their rain falls.

Table A1: Willingness to Pay for Conservation by Field Office and Feature Group: Conditional Logit Specification

	Willingness to pay ($/wtd. acre)			Policy fixed effects (logits)		Scale ($\log \sigma$)		Summary statistics		
Field office	Soil	Water	Wildlife	Leaves all open	Any closed label	Soil+Water	Wildlife	McFadden R^2	$\log L$	Features
Pooled (all offices)	\$76,144***/\$59,511*** (\$7,071)/(\$6,531)	\$34,488***/\$30,777*** (\$8,065)/(\$7,012)	\$44,346***/\$31,815*** (\$1,775)/(\$1,724)	0.927***/0.930*** (0.0254)/(0.0254)	−0.553***/−0.549*** (0.0213)/(0.0214)	5.68/5.69 (0.0806)/(0.0827)	3.74/3.78 (0.0492)/(0.0525)	0.058/0.057	−35,310/−35,352	12,509
Rock Springs, WY	\$57,357***/\$48,716*** (\$16,903)/(\$10,347)	\$47,495***/\$42,168*** (\$6,543)/(\$5,065)	\$36,093***/\$30,572*** (\$509)/(\$286)	0.0535/0.0717 (0.0847)/(0.0845)	−11.1***/−9.25*** (2.19)/(1.43)	3.36/3.29 (0.165)/(0.164)	−0.673/−0.901 (0.0667)/(0.0686)	0.334/0.336	−2,659/−2,654	1,222
Casper, WY	\$93,167***/\$59,380*** (\$7,742)/(\$4,658)	\$4,915/\$6,431* (\$4,198)/(\$3,400)	\$22,058***/\$16,393*** (\$709)/(\$592)	−0.109/−0.136* (0.0735)/(0.0739)	−2.11***/−2.09*** (0.0806)/(0.0802)	3.72/3.66 (0.129)/(0.128)	1.15/1.14 (0.0595)/(0.0592)	0.182/0.180	−5,099/−5,108	2,036
Buffalo, WY	\$36,117***/\$31,417*** (\$3,285)/(\$2,315)	\$5,957/\$10,582** (\$6,847)/(\$5,316)	\$59,229***/\$43,653*** (\$1,386)/(\$929)	−0.514***/−0.530*** (0.0765)/(0.0763)	−1.84***/−1.83*** (0.0562)/(0.0562)	3.75/3.63 (0.128)/(0.129)	2.99/2.86 (0.0459)/(0.0456)	0.142/0.144	−9,411/−9,389	3,300
Lander, WY	\$261,324***/\$172,990*** (\$36,086)/(\$22,341)	\$62,930***/\$65,723*** (\$21,440)/(\$18,086)	\$122,483***/\$99,128*** (\$29,500)/(\$15,151)	1.26***/1.26*** (0.117)/(0.117)	−0.436***/−0.433*** (0.103)/(0.103)	5.01/4.95 (0.181)/(0.182)	6.16/5.80 (0.411)/(0.365)	0.111/0.112	−1,562/−1,560	577
Grand Jct., CO	\$189,666***/\$126,141*** (\$30,382)/(\$21,147)	\$556,687***/\$426,777*** (\$96,757)/(\$78,270)	\$20,170*/\$15,860 (\$11,556)/(\$10,587)	0.214/0.233 (0.258)/(0.258)	0.875***/0.923*** (0.151)/(0.150)	5.19/5.38 (0.194)/(0.205)	−0.0957/−0.0765 (0.457)/(0.489)	0.136/0.126	−628/−635	267
Uncompahgre, CO	\$23,647***/\$24,053*** (\$2,806)/(\$2,157)	\$106,720***/\$78,398*** (\$12,648)/(\$7,278)	\$40,901***/\$30,768*** (\$1,640)/(\$1,675)	1.65***/1.63*** (0.143)/(0.144)	0.668***/0.713*** (0.119)/(0.118)	2.98/2.76 (0.203)/(0.191)	−0.777/−0.275 (0.163)/(0.183)	0.149/0.137	−1,158/−1,173	480
Royal Gorge, CO	\$447,008**/\$365,537*** (\$177,985)/(\$85,039)	\$350,542***/\$347,256*** (\$24,823)/(\$23,568)	\$1,097,580***/\$788,737*** (\$30,252)/(\$17,400)	2.08***/2.08*** (0.0652)/(0.0652)	1.16***/1.18*** (0.0578)/(0.0577)	4.96/4.77 (0.178)/(0.168)	0.643/1.26 (0.143)/(0.169)	0.125/0.119	−5,242/−5,276	2,350
Colo. River Valley, CO	\$75,592***/\$72,405*** (\$4,591)/(\$3,053)	\$65,422***/\$66,153*** (\$6,870)/(\$4,921)	\$121,574***/\$88,481*** (\$1,251)/(\$759)	1.98***/1.99*** (0.169)/(0.170)	1.74***/1.79*** (0.141)/(0.141)	4.46/4.26 (0.250)/(0.254)	−0.940/−0.612 (0.133)/(0.134)	0.261/0.239	−1,077/−1,109	492
Little Snake, CO	\$8,116/\$7,262 (\$6,237)/(\$4,588)	\$18,308**/\$14,433*** (\$7,268)/(\$5,359)	\$22,063***/\$16,435*** (\$150)/(\$82)	2.85***/2.81*** (0.158)/(0.156)	0.745***/1.19*** (0.165)/(0.155)	4.50/4.24 (0.226)/(0.207)	−2.21/−2.07 (0.0816)/(0.0780)	0.542/0.470	−925/−1,071	650
Four Rivers, ID	\$43,674***/\$45,340*** (\$11,394)/(\$8,577)	\$243,355***/\$180,912*** (\$22,300)/(\$15,572)	\$5,507/\$4,145 (\$4,017)/(\$2,954)	1.41***/1.41*** (0.131)/(0.131)	0.395***/0.408*** (0.112)/(0.112)	0.795/0.744 (0.164)/(0.165)	−5.37/−5.41 (0.467)/(0.502)	0.116/0.114	−1,204/−1,207	499
Rio Puerco, NM	\$15,852*/\$12,600* (\$9,620)/(\$7,408)	\$21,565**/\$21,444** (\$9,111)/(\$8,593)	*no data*	2.19***/2.19*** (0.0946)/(0.0946)	−1.10***/−1.10*** (0.137)/(0.137)	2.32/2.28 (0.430)/(0.429)	*no data*	0.290/0.290	−1,162/−1,162	636

Notes: Table reports the estimates from a conditional logit specification—the main specification setting $\kappa = 0$ and dropping the open type. Each office row is fit on that office alone; the pooled row fits the same specification once on all eleven offices together, a single restricted fit rather than an average of the eleven. Each cell gives the estimate under the relevance functional form $0.01^{(l/h)^{1.5}}$ before the slash and under $0.01^{(l/h)^2}$, the main specification, after it; the two differ only in how sharply relevance falls away with distance from the feature or, for soil, in attribute space. Willingness to pay is the revenue, in 2014 dollars, an office forgoes to protect an acre at the feature from development, for each non-market resource class. The policy fixed effects are the office's residual tendencies to leave every resource open and to close a feature outright, each measured against the discretionary stipulations short of closure, common to all non-market resource classes and measured in log-odds rather than dollars. The scale, reported in logs for soil and water together and for wildlife, is the spread of the idiosyncratic differences among a feature's alternatives. The summary statistics are McFadden's R^2, the log-likelihood and the number of features in the fit. *no data* marks a class×office group that I do not currently have data for. Standard errors in parentheses; the conditional logit is written in willingness-to-pay space, so the willingness-to-pay standard errors are ordinary rather than delta-method. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A2: Correlation Between Revenue Forgone and Disturbance Avoided Across a Feature's Alternatives

		Disturbance Avoided with All Acres Weighted Equally						Disturbance Avoided with Relevance-Weighted Acres					
		All alternatives			Within restriction			All alternatives			Within restriction		
	Features	Median	p10	p5	Median	p10	p5	Median	p10	p5	Median	p10	p5
Rock Springs, WY	1,140	0.999	0.986	0.968	1.000	0.989	0.974	0.887	0.734	0.652	0.859	0.670	0.556
Casper, WY	1,973	1.000	0.998	0.995	1.000	0.998	0.996	0.853	0.701	0.652	0.841	0.639	0.578
Buffalo, WY	3,190	1.000	0.998	0.993	1.000	0.998	0.994	0.875	0.751	0.722	0.849	0.645	0.613
Lander, WY	487	0.999	0.990	0.967	1.000	0.990	0.971	0.842	0.638	0.599	0.815	0.481	0.463
Grand Jct., CO	222	1.000	0.985	0.970	1.000	0.994	0.987	0.888	0.669	0.602	0.854	0.519	0.460
Uncompahgre, CO	418	1.000	0.995	0.988	1.000	0.996	0.989	0.864	0.720	0.694	0.841	0.657	0.613
Royal Gorge, CO	1,856	1.000	0.989	0.967	1.000	0.991	0.976	0.829	0.537	0.507	0.797	0.519	0.471
Colo. River Valley, CO	412	1.000	0.996	0.965	1.000	0.996	0.985	0.876	0.677	0.622	0.858	0.549	0.477
Little Snake, CO	602	1.000	0.999	0.998	1.000	1.000	0.998	0.886	0.730	0.701	0.863	0.626	0.571
Four Rivers, ID	391	0.996	0.967	0.944	1.000	0.985	0.955	0.870	0.642	0.603	0.832	0.504	0.458
Rio Puerco, NM	536	1.000	0.994	0.980	1.000	0.995	0.982	0.800	0.603	0.558	0.841	0.511	0.471
Pooled (all offices)	11,227	1.000	0.994	0.983	1.000	0.996	0.986	0.868	0.669	0.597	0.847	0.591	0.515

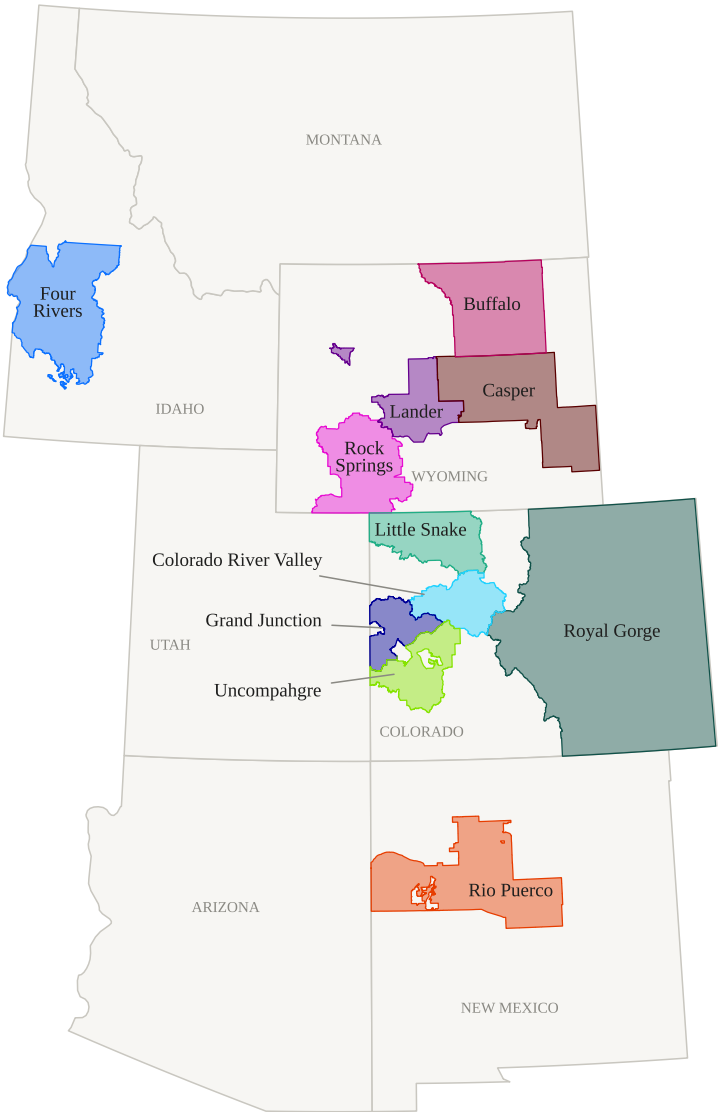
Notes: Entries are the correlation, within each feature's choice set, of revenue forgone M with disturbance avoided K. Columns report the median across features and the 10th and 5th percentiles, the less collinear tail. *All alternatives* uses every alternative in the feature's choice set—the variation the choice model is estimated from. *Within restriction* uses only alternatives that share the same restriction—for example, all alternatives that close the footprint—so they differ in the extent of the footprint alone. *Disturbance avoided with relevance-weighted acres* is K as it enters estimation. *Disturbance avoided with all acres weighted equally* is computed the same way, with every acre's relevance weight set to one. With all acres weighted equally, M and K are nearly proportional by construction: a cell's expected disturbance and its expected revenue are both set by the cell's development-potential level, so over any footprint both are sums of the same acres by level, and they diverge only as the mix of levels changes with extent. The relevance weight enters disturbance avoided and never revenue, and is what separates the two. Features with fewer than 3 distinct extents, or with no revenue at stake, are omitted (1,282 of 12,509).

Table A3: Erosion Potential on the Ground Each Office's Soil Decisions Concern

Office	R	K_w	S	Mean A (t/ac/yr)	Mean A, chosen buffer	Mean A all ground
Rock Springs, WY	21.9	0.305	1.14	7.7	6.6	6.1
Casper, WY	37.7	0.295	1.64	15.3	22.8	9.6
Buffalo, WY	40.8	0.292	2.50	28.6	58.1	17.3
Lander, WY	28.8	0.248	1.81	12.1	30.0	6.7
Grand Junction, CO	43.3	0.229	3.54	30.0	34.7	29.4
Uncompahgre, CO	44.7	0.239	3.08	30.2	56.6	24.6
Royal Gorge, CO	46.6	0.139	3.94	23.9	36.3	10.3
Colorado River Valley, CO	65.8	0.219	4.01	57.1	94.5	51.5
Little Snake, CO	49.3	0.234	2.81	31.1	68.6	19.9
Four Rivers, ID	57.3	0.249	2.99	32.4	38.8	27.9
Rio Puerco, NM	33.0	0.328	0.90	7.6	38.1	5.7

Notes: Table reports erosion potential and its components by field office. Columns R, K_w, and S report acre-weighted mean RUSLE factors within the widest buffer considered for slope and water-erodible-soil features in each office's choice set (see Section G for definitions). The final three columns report predicted annual soil loss (tons per acre) if all land were disturbed. All three are restricted to land where the office has discretion, but cover different areas within this scope. The first covers land within the widest feasible buffers around every slope and water-erodible-soil feature. The second covers only the buffers the office actually chose for features it protected. The third covers all land where the office has discretion, including land without these soil features. Little Snake, CO appears in the table but not in Figure 5, because its soil willingness-to-pay estimate is not distinguishable from zero.

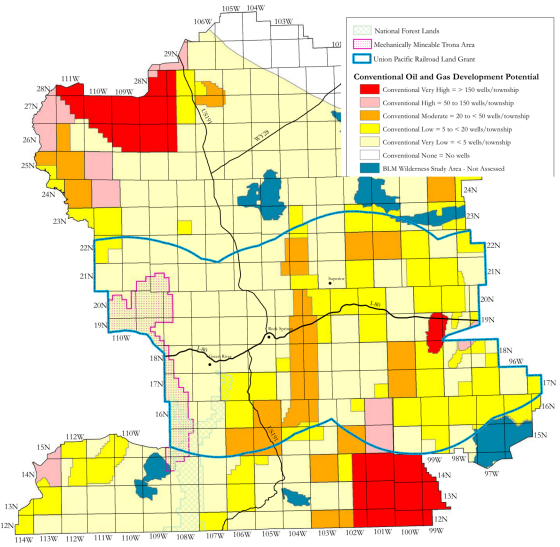
Figure A1: Field Offices in the Estimation Sample



Notes: Figure maps the jurisdiction of each of the eleven field offices in the estimation sample. Within its jurisdiction, each office manages the BLM's surface lands and the federal mineral estate.

Figure A2: Example Benefit-Cost Analysis Inputs from Rock Springs, Wyoming

(a) Oil and Gas Development Potential



(b) Projected Annual Resource Outputs

--	Alt. A	Alt. B	Alt. C	Alt. D
Livestock Grazing (AUMs – Billed Use)[1]	147,631	147,631	147,631	147,631
Livestock Grazing (AUMs – Total Authorized Use)[1,4]	303,238	297,066	160,387	303,238
Conventional Oil and Gas Wells Drilled[1]	232.4	61.5	238.8	230.2
CBNG Wells Drilled[1]	6.2	3.0	7.2	6.7
Crude Oil (bbls – 2016)[2]	2,047,125	542,141	2,104,317	2,026,774
Natural Gas (mcf – 2016)[2]	82,824,048	21,947,512	85,144,702	82,005,381
Coal and Soda Ash (short tons)[1,3]	4,107,267	4,107,267	4,107,267	4,107,267
Recreation (visits – Low Visitation)[1]	426,439	426,439	426,439	426,439
Recreation (visits – High Visitation)[1]	847,318	847,318	847,318	847,318

(c) Big game habitat by Restriction Type

	Alternative A	Alternative B	Alternative C	Alternative D	Proposed RMP
Oil and Gas—Closed					
Elk	561,654	2,713,522	225,782	493,425	220,204
Elk Parturition	92,351	137,100	71	23,422	108,373
Mule Deer	561,656	2,713,530	225,782	493,430	182,445
Mule Deer Parturition	54,722	63,113	4,011	4,011	55,214
Moose	124,815	773,640	24,237	134,474	7,972
Pronghorn	561,656	2,716,103	225,782	496,002	159,583
Oil and Gas—NSO					
Elk	518,268	809,743	15,542	194,648	20,094
Elk Parturition	31,766	6,246	3,094	8,705	4,653
Mule Deer	518,270	809,800	15,542	194,648	11,517
Mule Deer Parturition	3,314	1,568	6,128	27,201	4,395
Moose	176,191	234,198	38,623	126,673	144
Pronghorn	518,270	815,556	15,542	194,648	52,517
Oil and Gas—CSU					
Elk	1,957,092	126,442	1,924,922	2,410,222	70,860
Elk Parturition	17,487	0	47,155	106,998	23,542
Mule Deer	1,957,104	126,426	1,924,917	2,410,233	229,419
Mule Deer Parturition	6,450	0	25,783	33,366	4,473
Moose	502,660	28,580	627,145	581,884	11,190
Pronghorn	1,978,008	127,383	1,929,014	2,418,158	254,216
Oil and Gas—TLS					
Elk	2,600,380	970,235	2,917,087	2,279,364	79,426
Elk Parturition	50,995	6,246	143,275	119,924	30,249
Mule Deer	2,600,433	970,257	2,917,141	2,729,435	184,835
Mule Deer Parturition	9,959	1,568	60,670	60,670	5,024
Moose	792,831	249,618	899,598	801,754	8,018
Pronghorn	2,602,246	982,994	2,918,958	2,731,596	140,448

(d) Projected Surface Disturbance (Alt. C)

Disturbance Associated With All New Drilled Wells and Existing Active Wells (Short-Term Disturbance)					
Wells			**Acres of Surface Disturbance**		
Well Type	**All Lands**	**BLM Managed**	**Well Pad + Access Roads + Pipelines**	**All Lands**	**BLM Managed**
New Conventional Oil and Gas	5,902	4,804	7.1 successful and 5.6 for drilled and abandoned	40,575	33,024
New Coalbed Natural Gas	220	141	7.1 successful and 5.6 for drilled and abandoned	1,545	990
Existing Conventional Oil and Gas	2,259	1,536	2.64	5,964	4,055
Existing Coalbed Natural Gas	52	28	2.64	137	74
Total Wells	**8,433**	**6,509**	**Total Disturbance**	**48,221**	**38,144**

Notes: Each panel reproduces a planning-document table or map projecting development and conservation under different sets of allocation decisions. Panel (a) is an example development-potential map for oil and gas, partitioning the planning area into ordered categories from lower to higher expected development for a set of allocation decisions for which development is allowed everywhere the field office can legally permit it; see Appendix F for more details. Panel (b) shows the projected annual resource outputs tabulated by intersecting each considered set with development-potential maps for each resource (see Panel (a) for an example) (Rock Springs Field Office, High Desert District 2024*a*). Panel (c) shows the big game habitat in closed and stipulated lands, by species, highlighting the primary means conservation is factored into the benefit-cost analysis (acreage) (Rock Springs Field Office, High Desert District 2024*a*). Panel (d) shows the surface disturbance projected under one set of allocation decisions over the plan period, obtained by multiplying the wells that set is expected to permit by an average per-well footprint of well pad, access roads, and pipelines. The report projects those well counts by working out how many well locations each set's restrictions remove from a baseline in which drilling proceeds wherever law allows, and tabulates the resulting disturbance separately for the baseline and for each alternative (Stilwell 2013).

Figure A3: Example Benefit-Cost Analysis Inputs from Four Rivers, Idaho: Permitted Disturbance by Vegetation Community

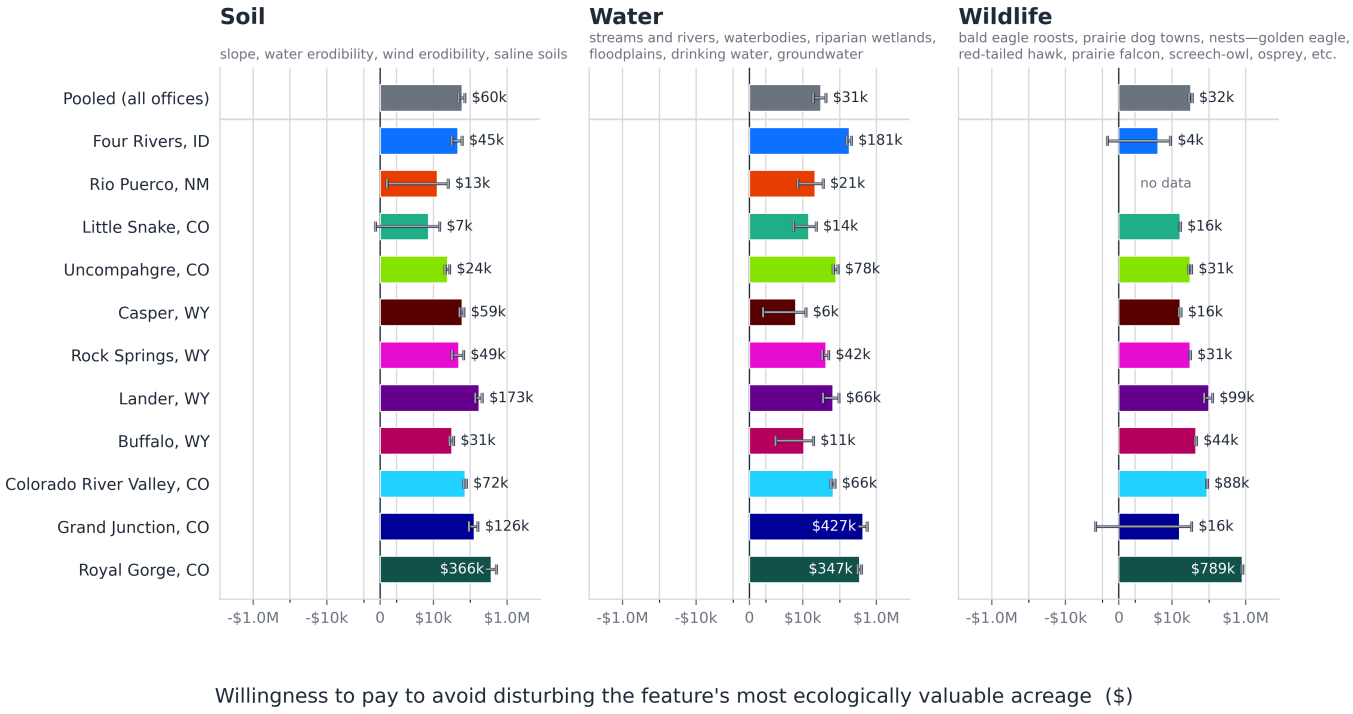
Table 4.4.2 Estimated acres of permitted disturbance for each alternative by vegetation community

Plant Community	Acres (total)	Acres of Human Caused Disturbance Not Including Rehabilitation or Restoration[2]			
		Alt. A	Alt. B	Alt. C	Alt. D
Evergreen Forest	50,830	940	600	1,500	1,000
Aspen & Mountain Shrub	47,150	1,300	1,000	1,300	1,200
Shrub-Steppe & Perennial Grassland	573,110	38,430	32,000	42,000	39,000
Exotic Annual Grassland	162,750	13,700	16,000	15,000	15,000
Riparian & Wetland	38,140	3,010	2,000	3,000	2,500

[2]Surface disturbance from permitted activities (new ROW, new rock & mineral extraction, new roads & trails, commercial logging/firewood extraction, etc.).

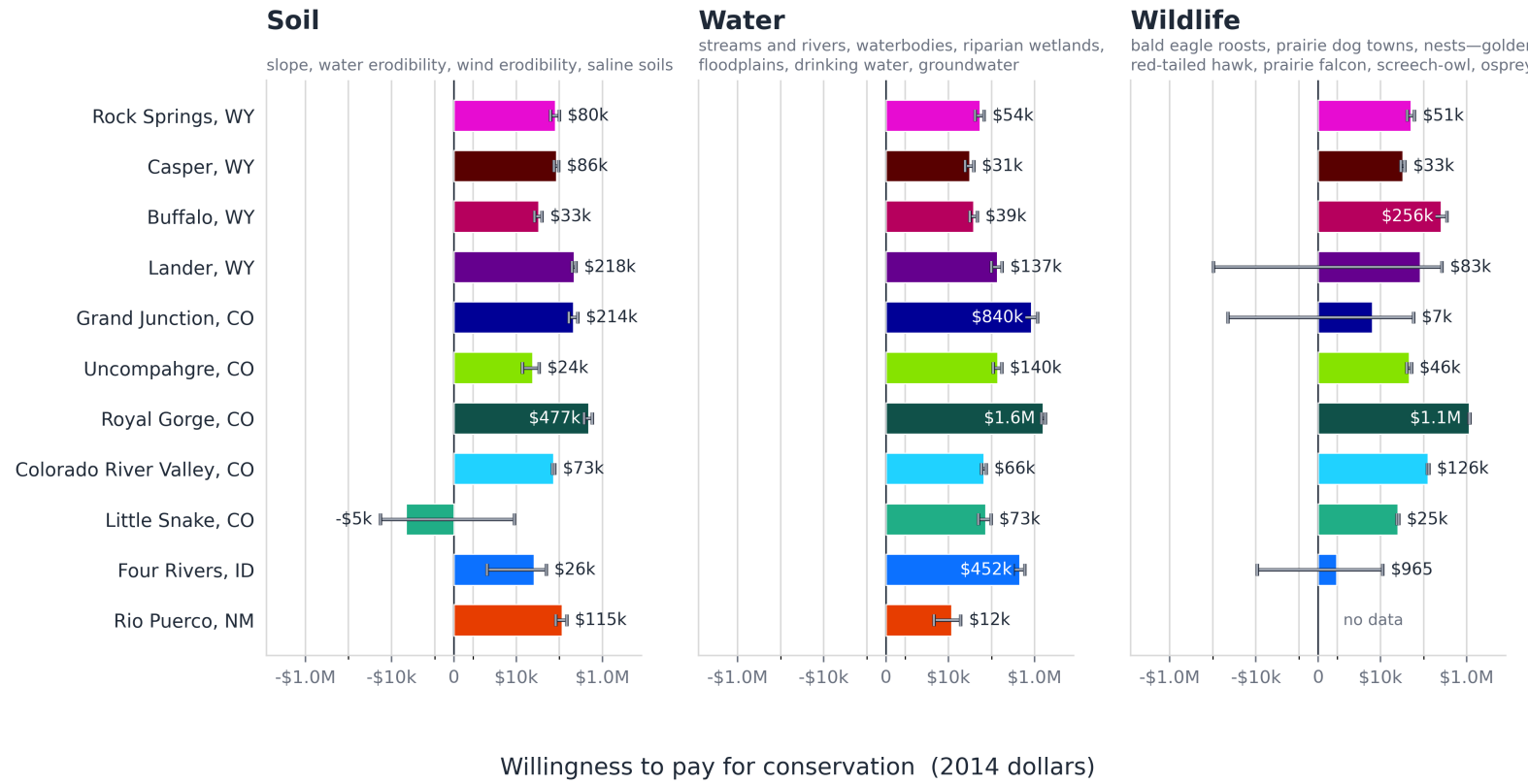
Notes: Reproduces Table 4.4.2 of the Four Rivers Field Office Draft RMP and Draft EIS (Four Rivers Field Office, Boise District 2019). Each row is a vegetation community. The first column gives each community's total acreage in the planning area, and the remaining columns give the acres where the planning documents project human-caused surface disturbance under each set of allocation decisions for each alternative (Alt. A through Alt. D).

Figure A4: Willingness to Pay for Conservation, by Field Office and Feature Group: Conditional Logit Specification



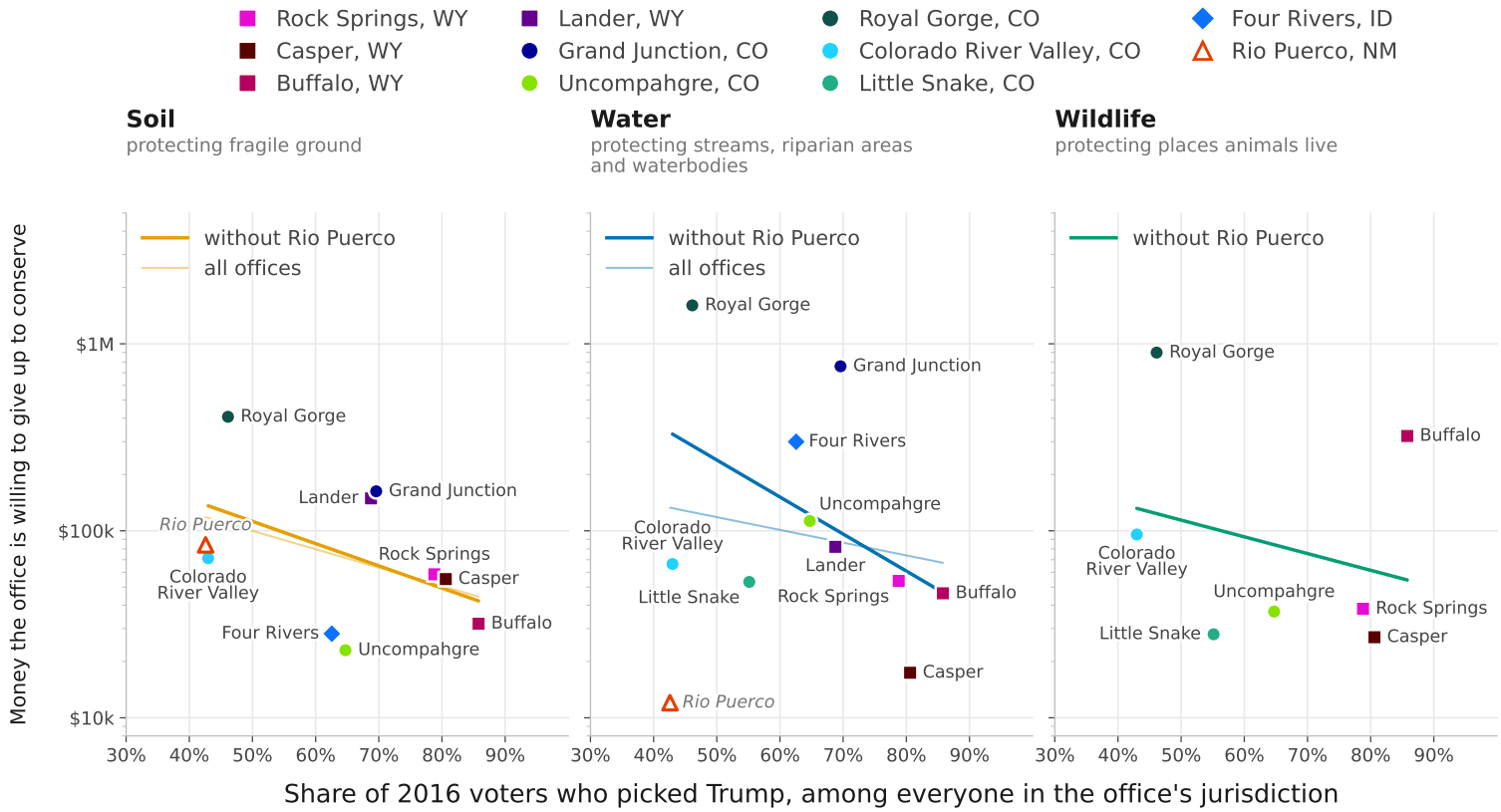
Notes: Figure plots each field office's estimated willingness to pay for conservation, in dollars of forgone revenue, from a conditional logit specification—the main specification setting $\kappa = 0$ and dropping the open type. Each bar is the estimate for one office and one non-market resource class under the main specification's relevance functional form, $0.01^{(l/h)^2}$; the estimates under $0.01^{(l/h)^{1.5}}$ are in Table A1. The whiskers are 90% confidence intervals from the Hessian standard errors. Each office is fit on its own; the *Pooled* bars fit the same specification once on all eleven offices together. The policy fixed effects and the scales are not plotted; Table A1 reports them. The horizontal axis is signed-log, matching Figure 4. "no data" marks an office×class group with no data.

Figure A5: Willingness to Pay for Conservation, by Field Office and Feature Group: Mixed Logit, Alternative Relevance Functional Form



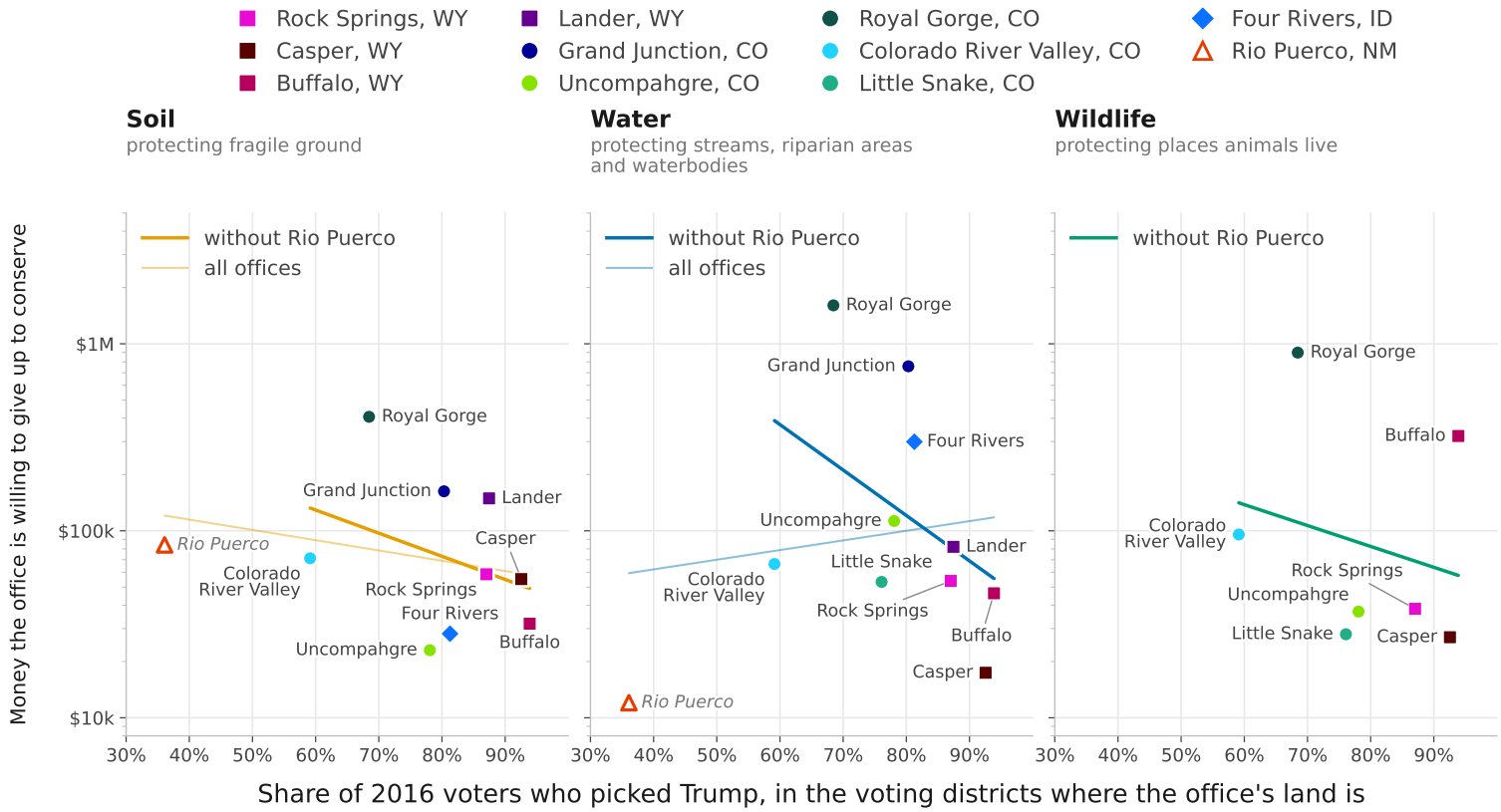
Notes: Figure plots each field office's estimated willingness to pay for conservation, in dollars of forgone revenue. Each bar is the estimate for one office and one non-market resource class under the relevance functional form $0.01^{(l/h)^{1.5}}$; the estimates under $0.01^{(l/h)^2}$, the main specification, are Figure 4. The whiskers are 90% confidence intervals from the Hessian standard errors. The fit reaches a log-likelihood of −23,530.63 on the same 12,509 choice situations. The horizontal axis is signed-log, matching Figure A4. "no data" marks an office×class group with no data.

Figure A6: Willingness to Pay for Conservation and the 2016 Trump Vote Share, by Feature Group



Notes: Figure plots each field office's estimated willingness to pay for conservation (Figure 4) against the share of the 2016 two-party presidential vote cast for Donald Trump in the office's planning area, with one panel for each non-market resource. Each marker is one office, and its shape denotes the office's state. The panels include the twenty-eight estimates that are statistically distinguishable from zero. The vote share sums precinct returns over the planning area; a precinct that straddles the area's boundary contributes in proportion to the share of its 2010 population living inside. The vote share data is from Voting and Election Science Team (2018). Lines are least-squares fits of log willingness to pay on the vote share for each resource type: the dark line leaves out Rio Puerco, New Mexico (hollow marker), and the light line includes it. Rio Puerco is set apart because it is the only New Mexico office, and New Mexico's voters lean Democratic while its state government depends heavily on revenue from federal onshore mineral leasing (New Mexico Legislative Finance Committee 2023, Volcovici & Renshaw 2021). Rio Puerco has no wildlife estimate, so the wildlife panel has one line. The vertical axis is on a log scale.

Figure A7: Willingness to Pay for Conservation and the 2016 Trump Vote Share: Precincts Weighted by BLM Acres



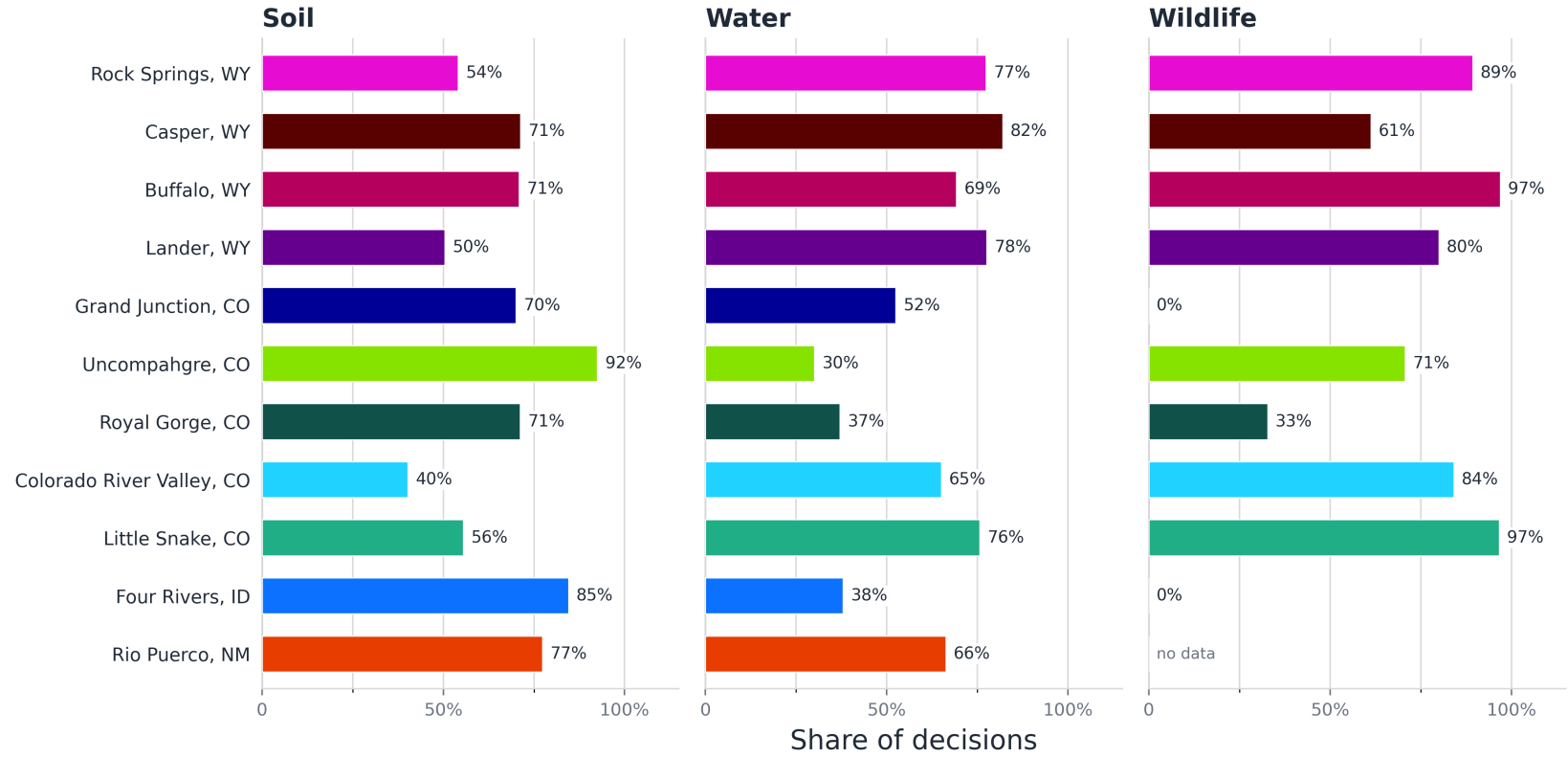
Notes: Figure repeats Figure A6 with a different measure of the vote share. Each marker is one office, and its shape denotes the office's state. The panels include the twenty-eight estimates that are statistically distinguishable from zero. Here each precinct's two-party Trump share is weighted by the acres of BLM-managed surface the precinct holds inside the planning area, so that precincts without BLM land carry no weight. The vote share data is from Voting and Election Science Team (2018). Lines are least-squares fits of log willingness to pay on the vote share for each resource type: the dark line leaves out Rio Puerco, New Mexico (hollow marker), and the light line includes it. Rio Puerco is set apart because it is the only New Mexico office, and New Mexico's voters lean Democratic while its state government depends heavily on revenue from federal onshore mineral leasing (New Mexico Legislative Finance Committee 2023, Volcovici & Renshaw 2021). Rio Puerco has no wildlife estimate, so the wildlife panel has one line. The vertical axis is on a log scale, and the axis limits are those of Figure A6. The weighting moves ten of the eleven offices toward Trump, by 8 to 22 percentage points; Rio Puerco moves 6 points the other way.

Figure A8: The Fitted Discrete Distribution of Corrections for the Standard Proxy



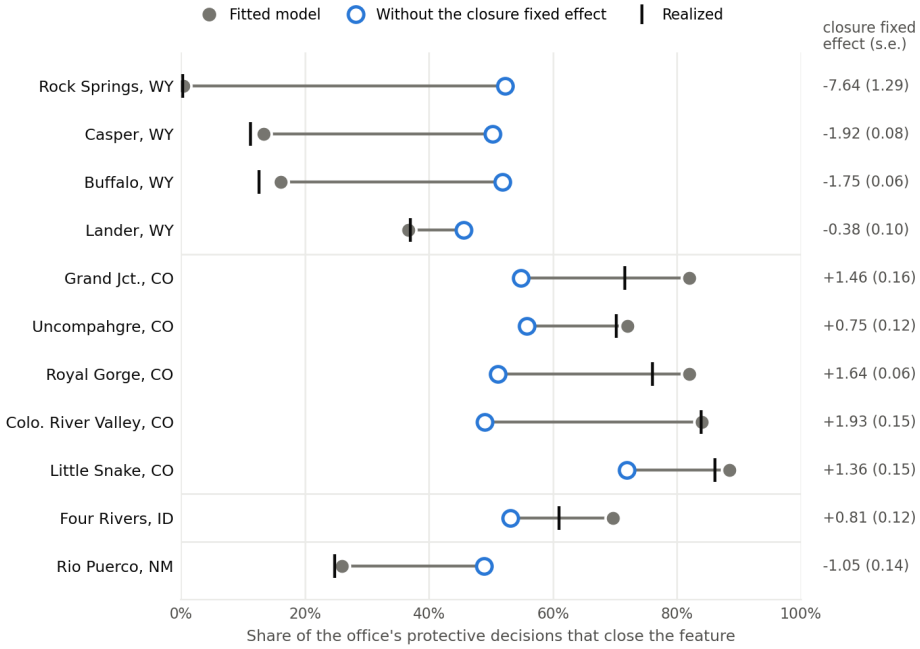
Notes: Figure plots the estimated support of the correction η and each field office's mixing shares over it. The standard is measured with a proxy, the median relevance the other field offices chose for features of the same feature type; the correction allows the standard an office anchors on to sit short of that proxy or past it. The column headed *Proxy* is a correction of zero. Left of it the correction is positive and the office's standard stops short of the proxy, a narrower buffer; right of it the correction is negative and the standard reaches past the proxy. Bubble area is the office's mixing share for that correction; shares sum to one across a row, and shares below 0.1% are not drawn. Under each correction is the buffer the corrected standard implies for one example feature type, bird nests: how far from the nest an office anchored on that standard protects, given the nests' proxy. A correction marked *past their ecologically relevant range* puts the standard beyond where a nest's relevance has fallen to nothing. Other feature types translate the same way, each relative to its own proxy. The left-hand column is the open type, which leaves a feature open regardless of the payoff; it is not a value of the correction and is placed at the left end of the axis. The bottom row averages the eleven offices with equal weight. The corrections describe the sample, not any one office: the effective number of corrections an office carries averages 1.4, and Buffalo, WY and Four Rivers, ID each carry a single correction. The open-type share is below 1% at 8 of the eleven offices; Colo. River Valley, CO is the only office carrying much, at 8%. Standard errors on the fitted corrections run from 0.0012 to 0.0080; they describe how sharply the fitted objective pins down each correction locally and are not confidence intervals for the location of a true latent value.

Figure A9: How Often the Predicted Buffer Yields Less Conservation Value Net of Revenue Forgone than an Adjacent Buffer



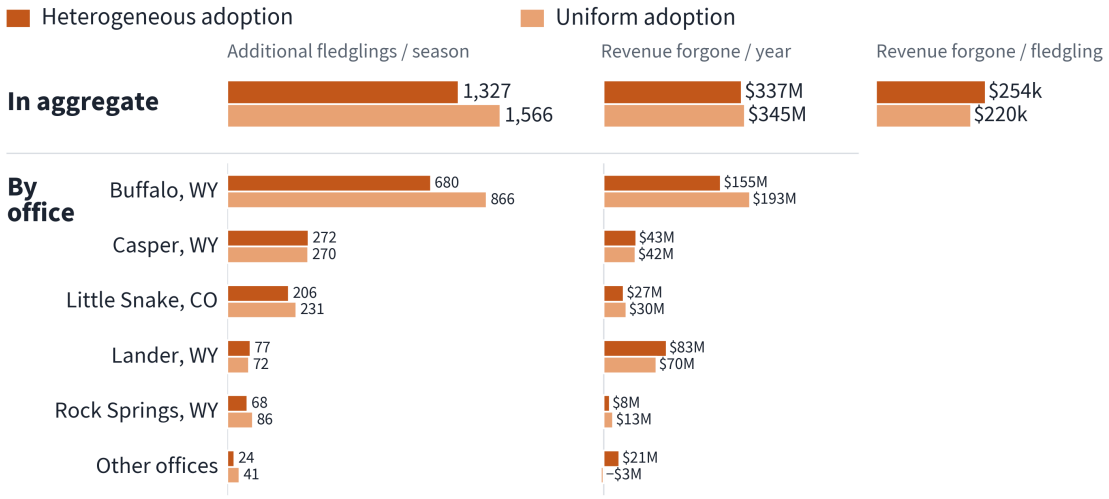
Notes: Figure plots the share of each field office's decisions in which a buffer adjacent to the *predicted* buffer has a higher net benefit—its conservation value net of the revenue given up. Figure includes decisions for which the fitted model predicts protection: it assigns the protective alternatives together a higher probability than leaving the feature open. The *predicted* buffer is the protective alternative with the highest predicted probability. Its adjacent buffers are the alternatives under the same restriction one step narrower and one step wider.

Figure A10: Predicted Closing With and Without the Closure Fixed Effect



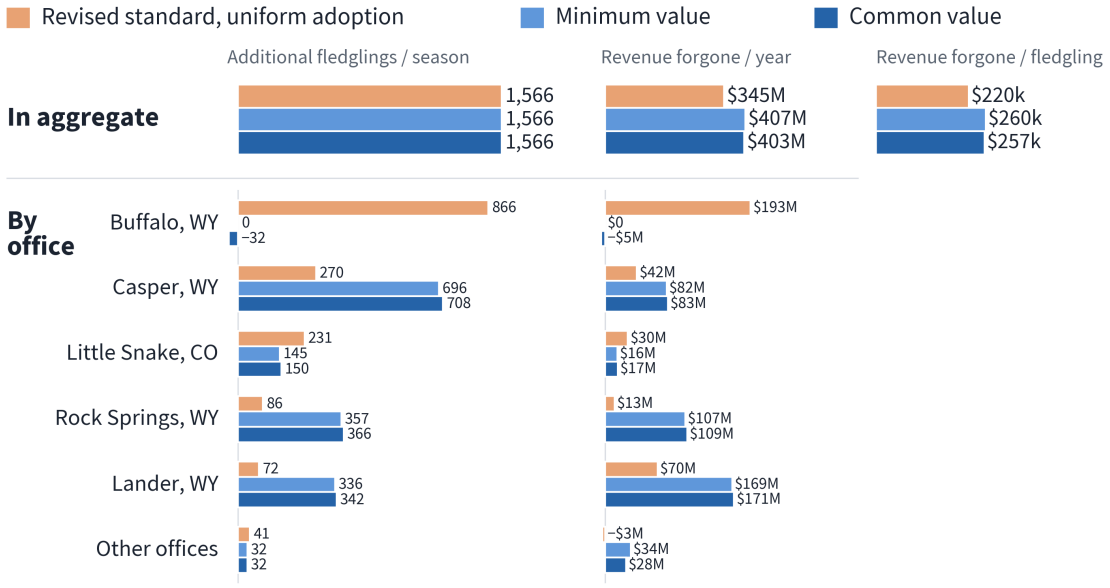
Notes: Figure plots the share of protective decisions that close the feature rather than restrict it under a stipulation by field office. The black tick is the realized share. The filled gray marker is the full fitted model's predicted probability of closing conditional on being protected. The hollow marker predicts the probability with the fitted model but with each office's closure fixed effect $\gamma_i(\text{closed})$ set to zero. At the right is the office's closure fixed effect in log-odds with its standard error (Table 1).

Figure A11: Revising the Standard under Heterogeneous and Uniform Adoption



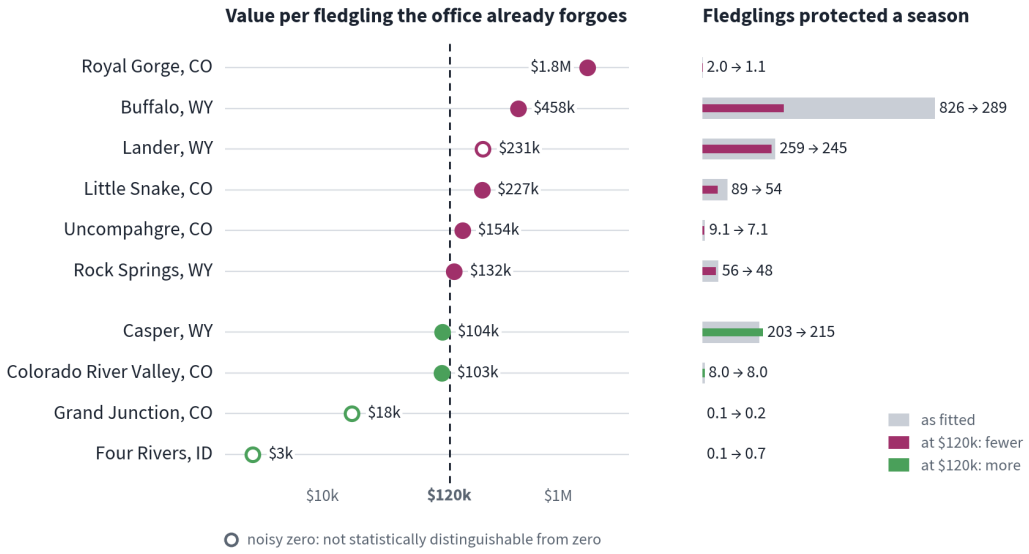
Notes: Figure plots changes in baby birds (fledglings) protected and revenue forgone, relative to the baseline, under heterogeneous and uniform adoption of the revised conservation standard. Dark orange bars show the standard revision under *heterogeneous adoption*, in which all offices respond to a common one-mile recommendation but retain office- and nest-specific unobserved heterogeneity in the conservation standards they ultimately follow. Light orange bars show the revision under *uniform adoption*, in which all offices respond to the same one-mile conservation standard without retaining this unobserved heterogeneity. The top panel reports aggregate changes, while the bottom panel breaks them out by office. The first column shows changes in fledglings protected, the second changes in revenue forgone, and the third, appearing only in the top panel, revenue forgone per additional fledgling. The "Other offices" category includes Uncompahgre, Colorado River Valley, Royal Gorge, and Grand Junction in Colorado, and Four Rivers in Idaho.

Figure A12: Protecting the Same Number of Birds with a Revised Standard (Uniform Adoption) or Price-Like Instruments



Notes: Figure replicates Figure 6 for the uniform adoption implementation of the standard revision counterfactual. It plots changes in baby birds (fledglings) protected and revenue forgone under the standard revision and price-like instrument counterfactuals, relative to the baseline. Light orange bars show the standard revision under *uniform adoption*, in which all offices respond to the same one-mile conservation standard, without retaining office- and nest-specific unobserved heterogeneity in the standards they follow. Light and dark blue bars show two implementations of the price-like instrument, both calibrated to protect the same aggregate number of additional fledglings as the standard revision. The *minimum value* implementation (light blue) raises each office's value of avoiding wildlife disturbance to at least $284,000 per relevance-weighted acre, while the *common value* implementation (dark blue) sets every office's value to $295,000. The top panel reports aggregate changes, while the bottom panel breaks them out by office. The first column shows changes in fledglings protected, the second changes in revenue forgone, and the third, appearing only in the top panel, revenue forgone per additional fledgling. The "Other offices" category includes Uncompahgre, Colorado River Valley, Royal Gorge, and Grand Junction in Colorado, and Four Rivers in Idaho.

Figure A13: Protecting Birds under a Common Value of $120,000 per Fledgling



Notes: Figure plots each office's fitted value of a fledgling and the change in baby birds (fledglings) protected, relative to the baseline, when every office's value is set to $120,000 per fledgling, the Fish and Wildlife Service's cost of offsetting one golden eagle. The left panel shows the value per fledgling each office already forgoes, on a log scale: the office's fitted value per relevance-weighted acre of avoided wildlife disturbance, times the disturbed acres per well in its plan (the median over its nest decisions), divided by two, since a well beside a nest costs a brood of two. The dashed line marks $120,000. Hollow markers show the three offices whose wildlife value is not statistically distinguishable from zero. The right panel shows fledglings protected a season at the baseline (grey bars) and with every office set to $120,000 per fledgling (colored bars), with the legend distinguishing offices that protect fewer fledglings from those that protect more. The counterfactual prices each nest decision at its own acres per well rather than the office's median, so an office near the line can move in either direction: Casper's median value is below $120,000, yet it protects more. Nest decisions that expect no well have no price and are left at the baseline.